DCCREG turbine — technical drawings

The bundle of the design lock (2026-10-09, design state 09243c7, revised at the settlement, 2026-10-09; built at bbaf186): 60 sheets after this register. The ledger is docs/ledger/DCCREG-design-ledger.md (and .pdf); the drawings themselves stay in the repository at the paths below, with the scripts that redraw them.

sheet	title	what it shows	file	generator
Part A — the design of record (locked)				
1	The locked machine: what drives what	the drive and the two counter-rotating bodies, both pumps, the hub and its two fields; where the belt's power goes; the reference	docs/ledger/figures/architecture.png	docs/ledger/make_ledger_figures.py
2	Rotor circuits: reluctance pump and electrostatic pump	(a) the magnetic dual doubler driving the AH pair (the 22 mF bypass dashed, PROPOSED); (b) the de Queiroz diode doubler, the clamps and the rings' two mirrored chains; parts and operating point	docs/schematic-rotor-circuits.svg	docs/make_schematic_rotor.py
3	The rings' DC supply, the symmetric pair	both Cockcroft-Walton chains stage by stage, each capacitor's DC and each diode's reverse peak; the hub with its AH cores at REF	docs/schematic-rings-supply.svg	docs/make_rings_supply_schematic.py
4	DCCREG-UTR-101: the wound utron's SiFe half-core	manufacturing drawing, Rev A draft for quotation: lamination profile 2:1, views B and C, GD&T, hole table, stack data, 3-D view	docs/drawings/DCCREG-UTR-101.pdf	docs/make_core_drawing.py
5	The wound utron against a bridge	the 200-turn coil on the split U-core, the 3.0 mm NiFe neck, the bonded slot cover and studs; front view, two sections, the 2-D flux, the data	docs/figures/utron-core-detail.png	docs/make_utron_drawing.py
6	The air vane stack, 6 mm gaps, 6 + 6 vanes	the electrostatic varicaps C1 / C2: stator vane (REF), rotor vane (node 1 / 4), both at minimum C, side A's half-section, the vane cell, the rim field (panel d still draws the placeholder hub)	docs/figures/air-vane-stack-6mm.png	docs/make_air_vane_drawing.py
7	DCCREG-HUB-201: rings A and B, copper, beaded	manufacturing drawing, Rev A draft: the half-section 2:1, the polar and equatorial beads in their grooves 5:1, one gore flat 4:1, the feature table and build notes	docs/drawings/DCCREG-HUB-201.pdf	docs/make_rings_drawing.py
8	The rings' field at the null	field drawings: the section's |E|, equipotentials and field lines; the field and the pressure along the axis and across the equator; ring B's beads	docs/figures/hub-rings-field.png	docs/make_rings_field_figure.py
9	The machine of record: section through the shaft	the record in solids, 940 mm: the air stack of record (6 + 6 vanes, 6 mm gaps) with Ca / Cb on the rotor, the reluctance sections and the locked hub; side A below	docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50-section.png	sim/tube_geometry.py --record
10	The machine of record: quarter cutaway	the same solids in 3-D, cut away	docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50-3d-cutaway.png	tools/step-viewer/shoot.py
11	The machine of record: half section	the same solids, half-sectioned	docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50-3d-half.png	tools/step-viewer/shoot.py
12	The locked hub in solids	the vessel, rings A / B with their beads, the gel, the PEEK retainer, the G10 coupler, the AH cores, formers and coils, the flanges	docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50-hub.png	sim/tube_geometry.py --record
13	Reluctance sections A and B, plan cuts	the three wound utrons per side on the rotor and the six bridges per side on the counter-rotor (B offset 30°)	docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50-reluctance-plan.png	sim/tube_geometry.py --record
14	Utron A1 with bridge A1, exploded	the parts of one wound utron and its bridge	docs/geometry/tube/tube-r150-n8-wound-g0p5-6br-3d-exploded.png	tools/step-viewer/shoot.py
15	Reluctance section A, quarter cut	utrons on their G10 carrier discs inside the bridge ring	docs/geometry/tube/tube-r150-n8-wound-g0p5-6br-3d-reluctance.png	tools/step-viewer/shoot.py
16	Reluctance section A, plan at mid-stack	the 0.5 mm gaps at the aligned position	docs/geometry/tube/tube-r150-n8-wound-g0p5-6br-3d-plan.png	tools/step-viewer/shoot.py
Part B — supporting drawings of the record				
17	The lock-down study of the hub	the hub as locked (50 mm sphere, AH cores, flanges) and the lock-down's pick of rings, its 3-stage 20–53° bands superseded by DCCREG-HUB-201	docs/figures/hub-locked.png	docs/make_hub_locked_figure.py
18	The rings as built: retainer, edges, stages	the PEEK retainer with its gel pocket and the beads to scale; the field against the DC by supply family and rating; the polar bead against the AH; the retainer's permittivity	docs/figures/hub-rings-build.png	docs/make_hub_rings_build_figure.py
19	What the bench test should see	start-up, the 120 Hz ripple (nodes, rings, field), the 6 h drift by material, the test's phases	docs/figures/hub-bench-predictions.png	docs/make_hub_bench_figure.py
20	One revolution of the rotor	the pump's phases, its nodes, the chains, and the field at the null: steady from B to A, no sign change	docs/figures/hub-rings-revolution.png	docs/make_hub_revolution_figure.py
21	Utron size against gain: the pick	6 / 12 bridges × 600 / 1200 rpm at 0.5 / 1.0 mm, with the corrected mean turn	docs/figures/utron-size-vs-gain.png	docs/make_variants_chart.py
22	The vane matrix: what the radius buys under the 6 + 6 cap	power, gain, stack length and aluminium against the vanes' outer radius	docs/figures/vane-matrix-radius.png	docs/make_vane_matrix_figures.py
23	The vane matrix: rim corona margins	the rim's corona margin against gap and vane thickness	docs/figures/vane-matrix-corona.png	docs/make_vane_matrix_figures.py
24	The AH's null in the locked hub	the cusp's field, the null against the coils' imbalance and over the 120 Hz cycle with and without the bypass, the rods' flux, the shaft's permeability	docs/figures/ah-null.png	sim/ah_null.py
25	The reversing gear and the belt drive (PROPOSED)	the first cut: a bevel reverser at side A, its carrier held by the frame, three POM-C pinions; the HTD belt and the motor	docs/figures/drive-gear-concept.svg	sim/drive_sizing.py --figure
26	The utrons in 3-D: their ends, their coupling and the pump	κ in 3-D against the record's 2-D section, where the flux goes, the coils' coupling round the machine, and the pump with each utron set (model check 34)	docs/figures/utron-3d.png	sim/utron_3d.py
27	The neck in a nonlinear field solve	Ψ(NI) of the built utron, where the iron saturates, and the pump with the field map's law (model check 31)	docs/figures/neck-nonlinear.png	sim/neck_nonlinear.py
28	Real diodes in both pumps	the kick's threshold by rectifier class, the electrostatic gain against the seed, and the rings with real HV sticks (model check 32)	docs/figures/diodes-real.png	sim/diodes_real.py
29	The tube's strays by a field solve	the strays per node and where they come from, the varicap with its rims, and the pump as built (model check 33)	docs/figures/tube-strays.png	sim/tube_strays.py
30	The multiplier stacked into the doubler (PROPOSED, the redesign of 2026-10-10)	the designer's ladder C3-C6 / D5-D8 run bipolar with C1 / C2 in phase, each chain one stage past its varicap node to its ring; every node's swing, reverse peak and DC as built; against the record's supply	docs/schematic-integrated-multiplier.svg	docs/make_integrated_multiplier_schematic.py
31	The multiplier laid out in the machine (PROPOSED, 2026-10-11)	the half-section without the hub-face bearings; side A's central bay with its gaps; deck A in plan (the sticks, C3 / C6 / C8, Csa and the clamp string round the grounded can); the parts, the interfaces, the strays, the clamp trim and the shafts	docs/figures/multiplier-layout.png	docs/make_multiplier_layout.py
32	The layout in solids: the section and the decks (PROPOSED, 2026-10-11)	the layout's 386 solids cut through the shaft, side A's central bay enlarged, both decks cut in plan at the sticks' height, and side A's end with its bridge-ring bearing: every part of the study's plans, the can, the column, the vanes with C2 in phase	docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50-layout-section.png	sim/tube_geometry.py --layout
33	The layout in 3-D, its circuit at 25.5 deg of the cycle (PROPOSED, 2026-10-11)	side A's central bay, quarter cut: every conductor by its potential (blue -15 kV, grey REF, red +15 kV), the diodes and the clamp lit while they conduct, insulators faded; the circuit of sim/multiplier_waveforms.py	docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50-layout-3d-bay-potential.png	tools/step-viewer/shoot.py (sim.json from tools/step-viewer/make_layout_data.py)
34	The bridge-ring bearings: the shafts' modes with them as drawn (SKF 6205-2Z, PROPOSED, 2026-10-11)	the first three modes of the shaft and the counter-rotor against the seats' stiffness at d 25 and d 30, with the 60 Hz criterion and the pumps' 120 / 240 Hz; the mode shapes at d 25 on G10 spiders	docs/figures/bridge-ring-bearings.png	sim/bridge_ring_bearings.py
35	Side A's end in 3-D: the end bearing and the bridge-ring bearing (PROPOSED, 2026-10-11)	quarter cut: the frame's end bearing on its spider; the bridge-ring bearing in its stainless cartridge on an 8 mm G10 spider, the 6 mm ring to bridge ring A, utron A1's end turns 9 mm above	docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50-layout-3d-ringA.png	tools/step-viewer/shoot.py

Register, continued

sheet	title	what it shows	file	generator
Part C — earlier phases, kept for the record (superseded)				
36	The machine as modelled before the record: section	superseded by the record in solids: the old vacuum 8 + 8 stack, Ca / Cb on the counter-rotor, the 120 mm placeholder hub (802 mm tube); its reluctance sections are the record's	docs/geometry/tube/tube-r150-n8-wound-g0p5-6br-section.png	sim/tube_geometry.py --rel wound
37	The machine as modelled before the record: quarter cutaway	as the section	docs/geometry/tube/tube-r150-n8-wound-g0p5-6br-3d-cutaway.png	tools/step-viewer/shoot.py
38	The machine as modelled before the record: half section	as the section	docs/geometry/tube/tube-r150-n8-wound-g0p5-6br-3d-half.png	tools/step-viewer/shoot.py
39	Hub drive: the first two-pump schematic (2026-10-07)	superseded by the rotor circuits: 300 rpm each way, 3 × 1150-turn toothed C-EM iron, the doubler on the stator vanes driving the bicone, a brush	docs/schematic-hub-drive.svg	docs/make_schematic_hub.py
40	First pole pick: pole-pair flux at 0.5 mm	superseded geometry (r_g 165, slot 50 × 60, stack 70, bridges 90 × 20), before the mean-turn correction	docs/figures/pole-pair-flux-g0p5.png	docs/make_pole_drawing.py
41	First pole pick: pole-pair flux at 1.0 mm	as the 0.5 mm sheet	docs/figures/pole-pair-flux-g1p0.png	docs/make_pole_drawing.py
42	The DC pair inside the vacuum (not chosen)	Rogowski electrodes on the AH null in the placeholder hub, 65.7 kV/cm; superseded by the rings outside the glass	docs/figures/core-null-field.png	docs/make_core_null_figure.py
43	Rings on the placeholder vessel, AC-coupled	the swinging-rings study through 1 nF: field, strays, gain	docs/figures/core-rings.png	docs/make_core_rings_figure.py
44	The floating cones' swing	cone electrodes on coupling capacitors swinging −4.4 / +2.7 kV at 120 Hz	docs/figures/core-swing-waveforms.png	docs/make_core_swing_figure.py
45	Earlier tube build: section	the spark-gap tube with clocking decks, C-EMs and the bicone, 1263 mm	docs/geometry/tube/tube-r150-n8-section.png	sim/tube_geometry.py
46	Earlier tube build: C-EM and utron plan	the designer's C-EM / utron sections	docs/geometry/tube/tube-r150-n8-reluctance-plan.png	sim/tube_geometry.py
47	Earlier tube build: clocking plan	the spark-gap clocking deck	docs/geometry/tube/tube-r150-n8-clocking-plan.png	sim/tube_geometry.py
48	Switchless C-EMs on the diode core	leg against bus placement of the C-EMs	docs/schematic-diode-core-switchless.svg	docs/make_schematic_switchless.py
49	Diode core with a DC bus	marked SUPERSEDED in the file: a DC bus with a switched C-EM drive	docs/schematic-diode-core-dcbus.svg	
50	C-EMs in the discharge path	the C-EMs in series with the rail gaps SG1 / SG2	docs/schematic-cem-in-discharge-path.svg	
51	C-EM motor placement	the C-EMs in the netlist of record; a buffer and angle-gated fire proposal	docs/schematic-cem-motor-placement.svg	
52	Disc machine: the KiCad schematic of record	the designer's disc machine: 43 parts, 8 spark gaps (the source is the .kicad_sch beside it)	docs/kicad/DCCREG_Turbine_circuit.svg	designer export (KiCad)
53	Disc machine: Ca / Cb simplification proposal	43 → 31 parts	docs/kicad/schematic_simplification.png	docs/kicad/render_simplification.py
54	Disc machine: reference cross-section	the disc's reference radii and named features from the r0.15 DXF	tools/cross-section.svg	tools/gen_artifacts.py
55	Disc machine: the boomerang C-EM block cap	plan and section	docs/boomerang-cap.png	
56	Disc machine: the placed C-EM motor	plan and radial section of the motor on the interleaved N = 2 disc	docs/geometry/motor/motor-il2f-6563b90d-rc40.png	
57	Disc machine: field cut at an SG1 station	meridional |E| before firing, v4 topology	docs/geometry/rt/slices/machine-SG1-station-before-firing.png	sim/field_slice.py
58	Disc machine: field cut at an SG1 station, zoom	as the station cut, magnified	docs/geometry/rt/slices/machine-SG1-zoom-before-firing.png	sim/field_slice.py
59	Disc motor: field cut at ±15 kV	the motor model at rotor 15°	docs/geometry/rt/slices/motor-A-station-pm15kV.png	sim/field_slice.py
60	Disc motor: field cut, unit drive	the motor model at rotor 15°, unit drive on C_A	docs/geometry/rt/slices/motor-A-station-unit-cA.png	sim/field_slice.py

CAD and DXF files (not placed on sheets)

part	file	what
A	docs/drawings/DCCREG-UTR-101_half-core_A.step	the utron half-core stack, hand A (hand B mirrored)
A	docs/drawings/DCCREG-UTR-101_lamination.dxf	the as-cut lamination profile (R12), tip face R130.20
A	docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50.step	the record in solids (153 solids, 940 mm)
A	docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50.glb	the same, for tools/step-viewer/
A	docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50.parts.json	its parts and layout list
B	docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50-layout.step	the redesign's layout in solids, with its bridge-ring bearings (386 solids, 1014 mm; PROPOSED)
B	docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50-layout.glb	the same, for tools/step-viewer/layout.html (the circuit running on it)
B	docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50-layout.parts.json	its parts, layout list and each conductor's net
C	docs/geometry/tube/tube-r150-n8-wound-g0p5-6br.step	the machine as modelled before the record (152 solids)
C	docs/geometry/tube/tube-r150-n8-wound-g0p5-6br.glb	the same, for tools/step-viewer/
C	docs/geometry/tube/tube-r150-n8-wound-g0p5-6br.parts.json	its parts and layout list
C	docs/geometry/tube/tube-r150-n8.step	the earlier spark-gap tube build (307 solids, 1263 mm)
C	docs/geometry/freeze-v010-CaCb.step	the disc's v0.10 freeze with Ca / Cb
C	docs/geometry/floor-56b6cb83.step	round-trip floor build (disc)
C	docs/geometry/opt-c9ac780b.step	round-trip opt build (disc)
C	docs/geometry/il2-ce1a9380.step	interleaved N = 2 disc build
C	docs/geometry/il2f-6563b90d.step	final interleaved N = 2 disc build
C	docs/geometry/il2f-6563b90d-rc40-FULL-pump+motor.step	the disc pump with its C-EM motor
C	docs/geometry/motor/C-em_and_motor_coil_export.step	the designer's C-EM and utron source (Fusion 360)
C	docs/varcap-nodeanalysis-template-r0.15_TMD_layout.dxf	the designer's disc layout, r0.15
C	docs/varcap-nodeanalysis-template-r0_6_TMD_layout.dxf	the designer's disc layout, r0.6
C	docs/kicad/DCCREG_Turbine_circuit.kicad_sch	the designer's KiCad source of the disc schematic

Analysis plots of the earlier phases (the repository root's *.png and sim/*.png) are data, not drawings; they stay with their findings.

The locked machine: what drives what

the drive and the two counter-rotating bodies, both pumps, the hub and its two fields; where the belt's power goes; the reference

DCCREG turbine, the locked design (2026-10-09): what drives what

Two counter-rotating bodies on one shaft; on each shaft half a magnetic pump and an electrostatic pump; both pumps hold a static field at the hub's centre. Side A below, side B above, mirror images.

DRIVE (the frame)
drive motor and belt
reversing gear 1 : −1
end bearings, housings
(a first cut, PROPOSED)

1200 rpm relative

ROTOR, +600 rpm
shaft (REF), sleeve, hub
utrons, rotor vanes
Ca / Cb, D1–D4, Z1 / Z4
both pumps' circuits

four inner bearings

COUNTER-ROTOR, −600 rpm
stator cage, stator vanes
(REF via one inner bearing)
SiFe bridges, no wires

THE MAGNETIC PUMP: a reluctance machine, the planar dual of the electrostatic doubler

Utrons × bridges
3 wound utrons per side (rotor)
6 SiFe bridges per side (counter)
gap 0.5 mm, 120 Hz
L 81 ↔ 9.4 mH (3-D: 83 ↔ 12.4)

→

Magnetic dual doubler
groups A / B in antiphase
La / Lb 0.146 H, D1*–D4*
clamp: the NiFe neck saturates
kick start; z 1.14–1.21

→

AH coil pair
160 turns, one per branch
22 mF bypass each (PROPOSED)
300 A-t ±3 % (best est. 249–272)
belt 18.1 W + 1.25 W iron

THE ELECTROSTATIC PUMP: the de Queiroz diode doubler, HV side on the rotor

Air vane stack C1 / C2
6 + 6 vanes per side, Al 3 mm
6 mm gaps, R1.5 full rounds
rotor vanes = nodes 1 / 4
stator vanes = REF
55 ↔ 410 pF (128 ↔ 451 as built)

→

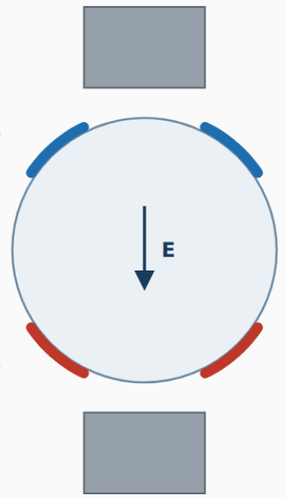
Diode doubler
Ca / Cb 451 pF on the rotor
D1–D4; as built a 0.12–3.7 kV seed
clamps Z1 / Z4 at 13.1 kV
nodes 1 / 4: −13.2 ↔ −5.7 kV
z 1.31 (1.06 as built); 2.14 W

→

Two mirrored chains
Cockcroft-Walton, 2 stages
each, from the shaft
node 1 → ring A −15.0 kV
node 4 → ring B +15.0 kV
8 diodes, 8 × 100 pF

THE HUB (rotor)

AH cores (MnZn) on the axis
ring B +15.0 kV (blue), A −15.0 kV (red)



50 mm borosilicate sphere, vacuum,
1.5 mm wall; PEEK retainer + 0.5 mm gel,
G10 coupler; flanges at |z| 72–80 mm

at the null:
B: the AH's steady cusp (the null)
E: 7.62 kV/cm, B to A, 2.57 Pa (8.2 settled)
(ripple 0.19 %, no sign change)

WHERE THE BELT'S POWER GOES (at 1200 rpm relative)

magnetic pump	18.1 W belt + 1.25 W iron: utron copper 13.5 W, AH coils 2.0 W, diodes 2.0 W, La / Lb 0.7 W; the bypass 8 mW; the best estimate (3-D utrons, the neck's field map) 12.7–15.0 W
electrostatic pump	2.14 W belt, all into the clamps Z1 / Z4 (1.07 W each), 0.65 W as built with the tube's strays solved; the rings' leakage 4.5 mW at 100 GΩ per ring
the mechanics	windage about 12 W and the bearings 5 W, so about 39 W in all from the belt; the gear puts the same torque on both bodies, the frame takes twice it
the products	two static fields at the hub's centre: the AH's cusp (300 A-turns per coil; best estimate 249–272) and the DC field 7.62 kV/cm (8.2 settled); no electrical output is drawn
the reference	the shaft (REF); the counter-rotor's stator vanes reach it through one inner bearing (0.41 mA rms, pure AC)
not yet built in	the gear and belt, La / Lb and the HV parts are not in the 3-D model; the record's stacks, hub, AH and rings are (see the ledger's register)

Sources: docs/ledger/DCCREG-design-ledger.md · generator: docs/ledger/make_ledger_figures.py

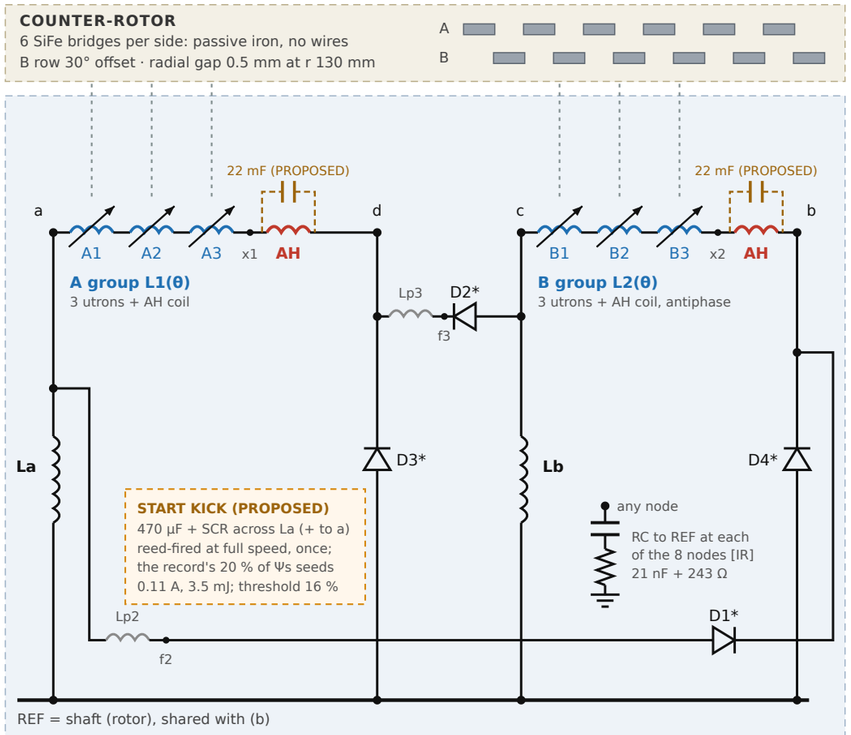
Rotor circuits: reluctance pump and electrostatic pump

(a) the magnetic dual doubler driving the AH pair (the 22 mF bypass dashed, PROPOSED); (b) the de Queiroz diode doubler, the clamps and the rings' two mirrored chains; parts and operating point

DCCREG turbine: the rotor's two circuits as built (tube, wound utrons; rotor and counter-rotor geared 1 : −1)

(a) the reluctance pump drives the AH pair · (b) the electrostatic pump's rings hold a DC field on the AH null; only C1 / C2 straddle the bodies · operating point g 0.5 / 6 bridges / 1200 rpm relative · only diodes switch

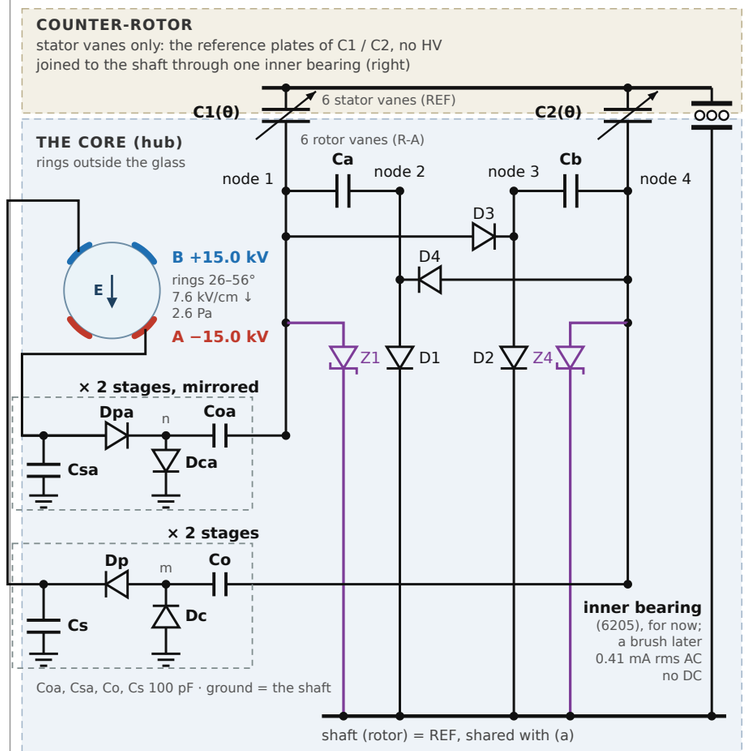
(a) RELUCTANCE: magnetic dual doubler → AH pair



A1–A3, B1–B3	wound utrons at 0 / 120 / 240°, 200 t of Ø1.55 mm Cu: 81.0 / 9.4 mH (aligned / unaligned), 0.58 Ω each
group	L̂ 0.243 H, ratio κ 8.6, τ = L/R 0.140 s; Ψs 0.134 Wb-t (the 3.0 mm NiFe neck)
	in 3-D: 83.3 / 12.4 mH, κ 6.70; 5.96–6.44 with the coils coupled (sim/utron-3d-findings.md)
bridges	a passing bridge closes the utron's flux path: L(θ) 6 × per rev; B row half a pitch on
AH	anti-Helmholtz pair at the hub, one coil per branch: 160 t, 1.01 mH, 0.27 Ω each [RH]
	bypass 22 mF across each coil, ESR 10 mΩ [RH] (PROPOSED)
La, Lb	0.146 H DC chokes: 0.87–1.15 A, ≤ 110 V, 0.29 Ω in the deck (τ 0.5 s [RH])
	first cut (PROPOSED): EI-84 × 35 mm, 150 t of Ø1.40 mm, 0.126 mm gap, 0.276 Ω, 1.66 kg each
Lp2, Lp3	4.4 mH wiring stray, as modelled (dual of Cpar) [IR]
D1*–D4*	0.54 V at 1 A in the deck; 1.9 / 2.3 A peak; reverse 113 / 104 / 58 / 60 V with the start-up
	first cut (PROPOSED): Schottky, 200 V (D1* / D2*) and 150 V (D3* / D4*), ≥ 3 A

At 1200 rpm relative (120 Hz): 0.87–2.81 A per branch, 101 V peak per group; AH 139–449 A-turns unbypassed.
With the bypass (PROPOSED): 290–308 A-turns per coil (300 mean), top and bottom within 17.
Belt 17.6 W = utron Cu 6 × 2.15 + La / Lb 0.66 + AH 2.00 + diodes 2.01 W; + iron 1.25 W = 18.8 W.
With the bypass the belt reads 18.1 W + iron 1.25 W (sim/ah-steady-cusp-findings.md).
3-D utrons: AH 221–245 A-t mean with the bypass; with the neck's field map 249–272 (sim/neck-nonlinear-findings.md).

(b) ELECTROSTATIC: de Queiroz diode doubler → DC field on the null



C1, C2	6 + 6 vanes, 6 × 22° / 22°, 6 mm air gaps, 3 mm full rounds: 55–410 pF, κ 7.5
	as built (3-D, the rims in): 128–451 pF, κ 3.51; Ca 480 pF; strays 66.2 / 24.5 pF
	z 1.058 bare, 1.031 with the chains; 0.65 W (sim/tube-strays-findings.md)
Ca, Cb	451 pF (1.1 C_max): full-annulus plates, now on the rotor
D1–D4	HV stacks, reverse peaks 6.1 / 6.1 / 13.2 / 13.2 kV; parts not chosen
Z1, Z4	avalanche strings, BV 13.1 kV (e.g. 66 × 200 V), on the rotor: 1.07 W, 0.95 mA peak each
core	two rings outside the 50 mm glass: Cu foil bands 26.2–55.7°, 29.9 mm apart along it
	beaded edges Ø3 / Ø2 mm (polar / equatorial) in gel, under the PEEK retainer
supply	A: 2 CW stages on node 1, mirrored, −15.0 kV · B: 2 CW stages on node 4, +15.0 kV
	100 pF storage; Dca, Dpa, Dc, Dp ≤ 7.5 kV reverse; 4 mW leakage at 100 GΩ/ring [RH]
	29.9 kV across: 7.6 kV/cm, 2.6 Pa at the null (as connected; 8.2 settled)
strays	≈ 20 pF per node [RH]; the rings 5.1 pF each to REF, 1.1 pF between
bearing	one inner bearing, for now: 0.41 mA rms AC (0.91 mA pk), no DC

At 1200 rpm relative (120 Hz): belt 2.14 W, into Z1 + Z4; z at start 1.191 (bare 1.310).
As built, real HV sticks: rings −13.38 / +13.40 kV, 6.82 kV/cm (sim/diodes-real-findings.md §6);
no self-start: a seed of 122 V on nodes 1 / 4 (1.0 kV at maximum leakage, 3.65 kV hot).

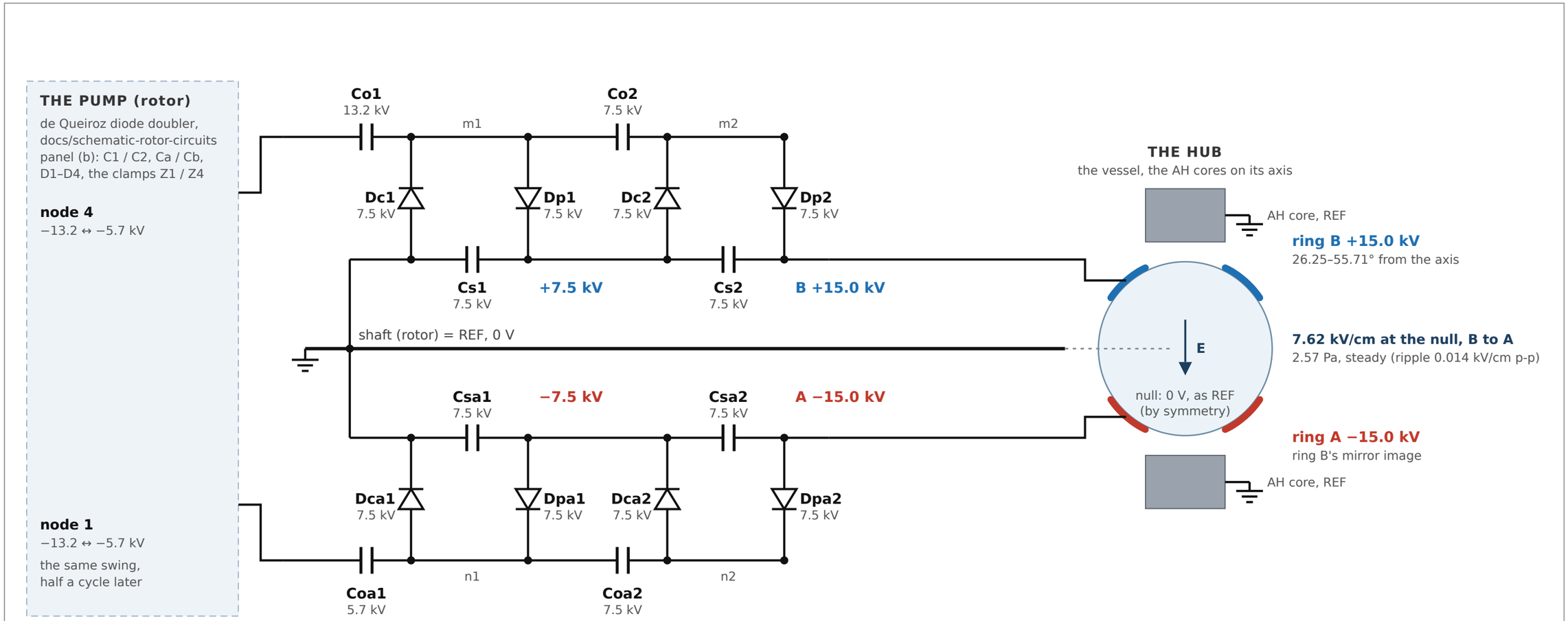
Bodies. Rotor: shaft, sleeve, rotor vanes (nodes 1 / 4), Ca / Cb plates, the 6 utrons on their carrier discs, the hub (the glass vessel with the rings outside it, the AH pair, retainer and coupler).
Counter-rotor: stator cage, stator vanes (REF), bridges. Frame: end bearings, the gear.
Not in the solids yet: La, Lb, D1*–D4*, the RC snubbers, the kick source, D1–D4, Z1 / Z4, the two CW chains, the rings (all rotor);
the record's solids (sim/tube_geometry.py --record) carry Ca / Cb on the rotor, the hub and the rings. Each pump's reaction torque goes into the gear.
(a) is the planar dual of (b) [OC]: C → L, V → I, Q → Ψ, Ca / Cb → La / Lb, Cpar → Lp2 / Lp3, D1–D4 → D1*–D4*, the clamps Z1 / Z4 → the NiFe neck's saturation.
A and B swap every half cycle: one group generates (L falling) while the other motors. Each branch carries its own side's AH coil (A below, B above; sim/magnetic_doubler.py's AHt / AHb date from when A was on top).
The AH coils: wound alike, start to x1 / x2, finish to d / b, mounted end over end with the start leads toward the vessel: anti-Helmholtz (sim/ah_winding.py).
Polarity: (b) runs negative; ring A on its own chain from node 1, negative, and ring B on the mirror chain from node 4, positive (the symmetric supply): E at the null is steady, B to A.
de Queiroz's Fig. 1 draws the core positive, all four diodes the other way (sim/queiroz_fig1_check.py).
Numbers: sim/pole_design_variants_op.json (g 0.5 / 6 bridges / 1200 rpm), sim/utron_profile.py, sim/rotor_parts_duty_results.json, sim/core_field_results.json, sim/hub_rings_build_results.json (b: the air build's capped stack)
Netlists: sim/magnetic_doubler.py, sim/core_field.py · findings: sim/pole-design-findings.md, sim/hub-rings-build-findings.md · generator: docs/make_schematic_rotor.py

The rings' DC supply, the symmetric pair

both Cockcroft-Walton chains stage by stage, each capacitor's DC and each diode's reverse peak; the hub with its AH cores at REF

DCCREG turbine: the rings' DC supply of record, the symmetric pair (each ring on its own chain from the shaft)

Ring B on 2 positive Cockcroft-Walton stages driven by the pump's node 4, ring A on the mirror image driven by node 1; both chains stand on the shaft (REF), which the AH and the null share. 100 pF storage, 100 GΩ per ring.

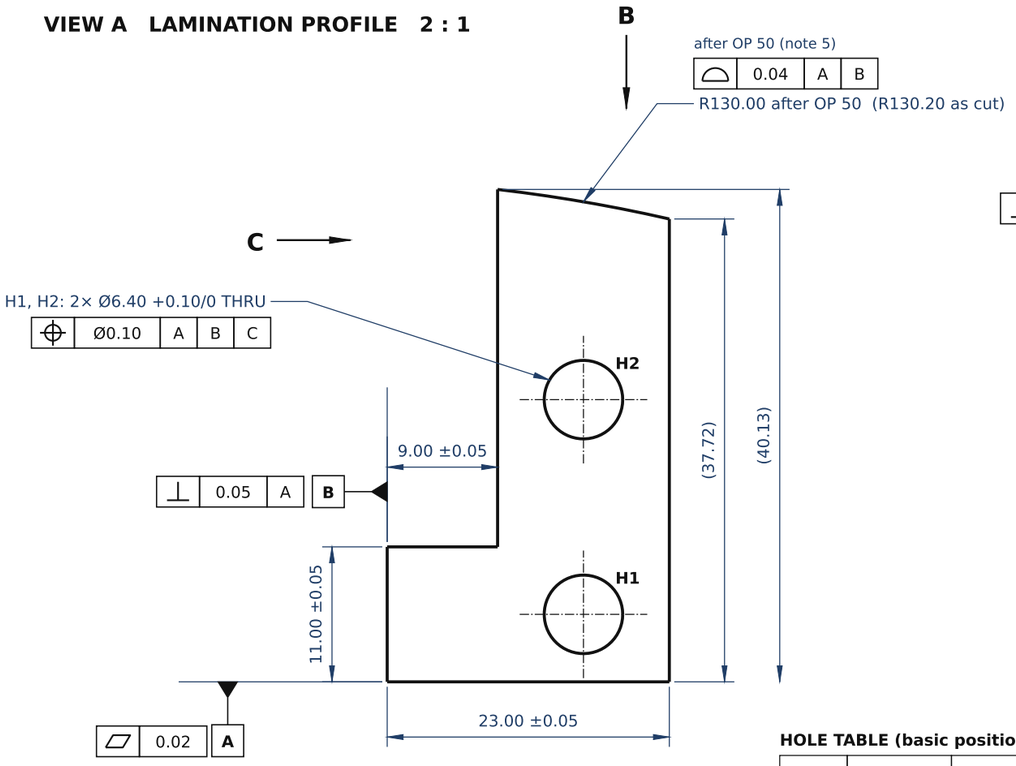


chains	2 stages each; ring B's from node 4 (Co, Dc, Dp, Cs), ring A's from node 1 (Coa, Dca, Dpa, Csa), the diodes reversed
levels	each stage adds node 4's / node 1's swing: 7.5 kV a stage at 100 GΩ per ring; 29.9 kV across the rings
diodes	8 HV stacks, reverse ≤ 7.5 kV each (20 kV stacks, derated 1.5×)
capacitors	8 × 100 pF / 30 kV; each holds about one stage's 7.5 kV DC, but Co1 13.2 and Coa1 5.7 kV: nodes 4 and 1 sit at −8.6 kV mean
ripple	ring A 34 V, ring B 29 V p-p at 120 Hz; ring A tops up as node 1 bottoms, ring B as node 4 peaks, 0.124 of a cycle later: 57 V p-p across the gap, 0.2 % of the field; no swing
pump	belt 2.14 W into the clamps; leakage 4.5 mW; start-up gain z 1.191
rings	Cu foil bands, beads Ø3 / Ø2 mm (polar / equatorial) in gel; strays 5.1 pF each to REF, 1.1 pF between

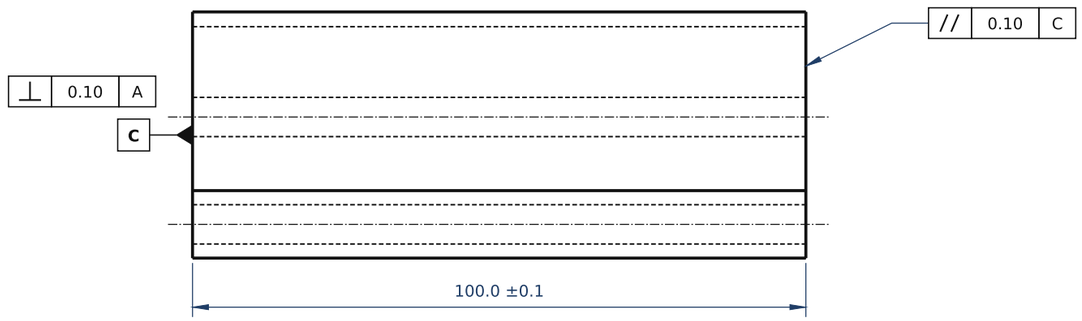
Numbers: sim/hub_rings_build_results.json (record, record_supply), sim/hub_drift_results.json (swing), sim/core_field.py (the deck) · findings: sim/hub-rings-build-findings.md · generator: docs/make_rings_supply_schematic.py

REVISIONS		
Rev	Date	Description
A	2026-10-08	First issue, draft for quotation

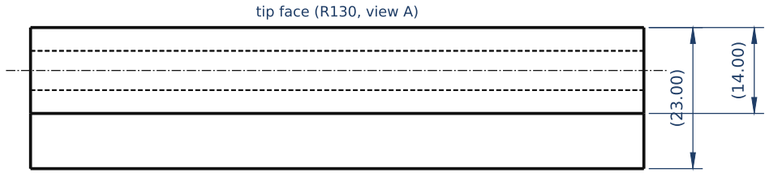
VIEW A LAMINATION PROFILE 2 : 1



VIEW C 1 : 1



VIEW B 1 : 1



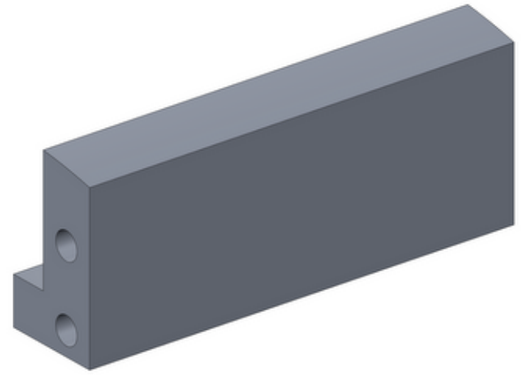
HOLE TABLE (basic positions)

HOLE	X from B	Y from A	SIZE
H1	16.00	5.50	Ø6.40 +0.10/0 THRU
H2	16.00	23.00	Ø6.40 +0.10/0 THRU

NOTES

1 Material: non-oriented electrical steel M235-35A to EN 10106, 0.35 mm, fully processed. Insulation coating on both sides: C-5 (EN 10342) for an epoxy-bonded stack, or the supplier's self-bonding varnish for a varnish-bonded stack.
2 Laminations: laser or wire-EDM cut to the DXF DCCREG-UTR-101_lamination.dxf (as-cut profile, tip face R130.20). Burr ≤ 0.02 mm, all burrs to one side within a stack. No interlocks, rivets or welds: the two Ø6.40 holes are the only through-features.
3 Stack: 100.0 ±0.1 mm at a stacking factor ≥ 0.95; laminations aligned on datums A and B and the two holes; bonded (self-bonding varnish cured to the supplier's cycle, or vacuum epoxy). Hand B is stacked from laminations turned over.
4 Stress-relief anneal after cutting (optional, to agree): 750 °C for 2 h in a non-oxidising atmosphere, before bonding. Epoxy route only: a self-bonding varnish does not survive the anneal.
5 Tip face: supplied as cut to R130.20 (0.20 mm stock). OP 50 (by DCCREG, on the assembled rotor) finishes it to R130.00: wire EDM or a light grind, then deburr and re-varnish the face. The profile tolerance on the tip face applies after OP 50. The centre of R130 is the machine axis: basic 6.00 from B (away from the tip) and 89.00 below A.
6 Face A seats the NiFe neck strip (a lap joint in the magnetic circuit): keep A flat, clean and free of burrs.
7 Marking: hand (A / B) and stack number on the tip's outer face (x' = 23) by paint or laser; never on A, B, C or the tip face.
8 Inspection: profile and holes on the bonded stack (CMM or optical); stack height; interlaminar insulation by sample (Franklin test, IEC 60404-11).
9 General tolerances ISO 2768-mK. Dimensions in mm. Break sharp edges 0.1 max, except the tip-face edges (keep sharp).
10 Views by reference arrows (ISO 128-30). Boxed values are basic dimensions. Do not scale from this drawing.

LAMINATION / STACK DATA	
lamination	one part for both hands (hand B = laminations turned over)
thickness, material	0.35 mm, M235-35A (EN 10106), coated both sides
stack	100.0 mm, ≈ 271 laminations at a stacking factor ≥ 0.95
section, mass	5.81 cm² net, 0.42 kg per stack
per machine	6 stacks hand A + 6 stacks hand B (3 utrons per side, 2 sides)
mates with	NiFe neck strip on A, G10 spacer on B, G10 cheeks on C and its opposite end



3-D VIEW, hand A: the L-profile end face, the outer face and the tip face (axonometric, not to scale)

DCCREG turbine · wound utron, magnetic doubler (rotor)

HALF-CORE, LAMINATED (hand A; hand B mirror)

Drawing no. **DCCREG-UTR-101**

Material: M235-35A, 0.35 mm (EN 10106), bonded stack

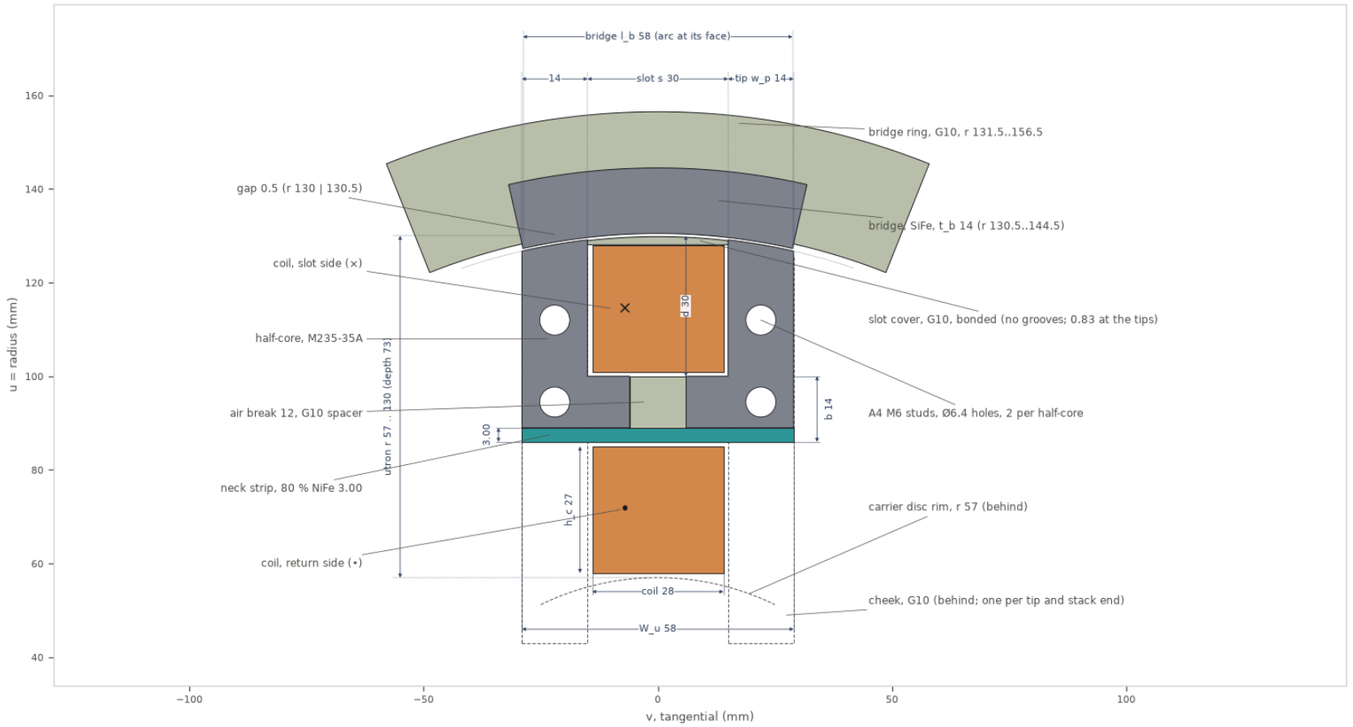
Mass: 0.42 kg	Units: mm	Rev A (draft) Sheet 1 / 1 A3
Scale: 2:1 (A), 1:1 (B, C)	Tolerances: ISO 2768-mK	Date: 2026-10-08
Drawn: generated from sim/utron_profile.py	Checked: — Approved: —	**DRAFT FOR QUOTATION**

The wound utron against a bridge

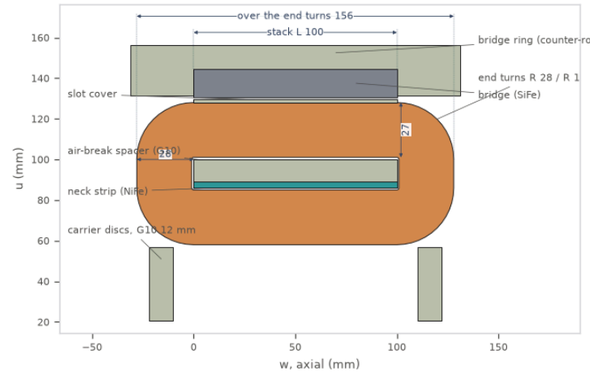
the 200-turn coil on the split U-core, the 3.0 mm NiFe neck, the bonded slot cover and studs; front view, two sections, the 2-D flux, the data

Wound utron, detail: 200-turn yoke coil on a split U-core with a NiFe neck, against a passive bridge (all dimensions in mm)

A. front view, the plane of rotation (the stack runs into the page); the aligned bridge above



B. side section at v = 0 (the coil's centre): the window holds the neck strip and the break



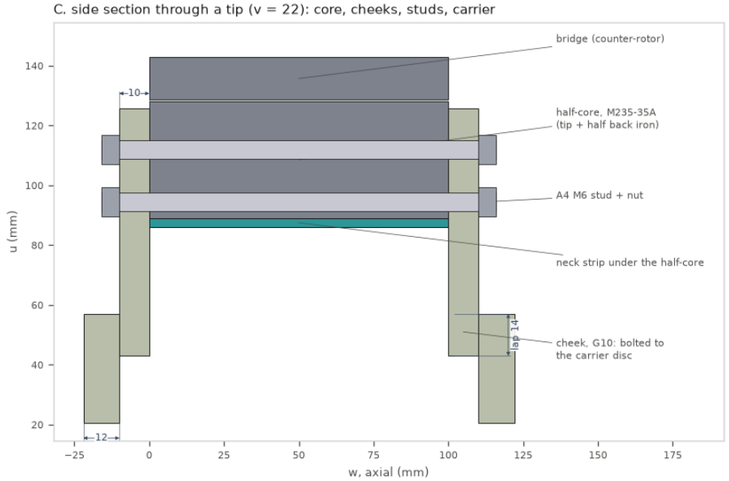
E. data (sim/pole_design_variants_op.json, sim/utron_profile.py)

operating point	g 0.5 / 6 bridges / 1200 rpm
section (2-D model)	r_g 130, tips 14, slot 30 x 30, back iron 14, gap 0.5
stack	100 mm of M235-35A (0.35 mm); over the end turns 156 mm
winding	200 turns, 1.89 mm² (Ø1.55 bare), fill 50%, mean turn 319 mm
per coil	R 0.581 Ω, L 81.0 / 9.4 mH (aligned / unaligned)
ratio, time constant	κ 8.6, τ = L/R 0.140 s; f 120 Hz, f·τ 16.7
group (3 in series)	L 0.243 H, I 2.81 A pk / 1.91 A rms, V 101 V pk
saturation	Ψs 0.1336 Wb-t (group) = 0.223 mWb per utron
neck	NiFe 3.0 x 100 mm (30 x 0.1) at 0.75 T; Φs +1.1 % (op: 1.48 mm SiFe at 1.5 T); body 0.16 T
losses	Cu 2.15 W per coil; machine 18.8 W (iron 1.25 W)
coil temperature	46 °C in air (forced), 63 °C in vacuum (radiation)
start	kick ≥ 20 % of I_pk (38 mJ) with Si diodes
mass per utron	2.23 kg: SiFe 0.89, NiFe 0.15, Cu 1.07, G10 0.12
counter-rotor, per side	6 bridges x 0.65 kg (SiFe), G10 ring r 131.5..156.5
rotor, per side	3 utrons at 120°, 2 carrier discs (0.20 kg each); B bridges offset 30°

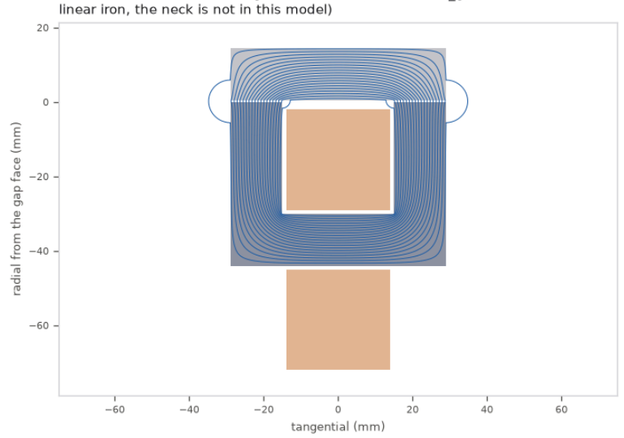
Assembly: wind the coil on its 1 mm G10 former; slide the NiFe strip and the G10 spacer through the window; insert the two half-cores from each side (they sit on the strip); lay in the slot cover; studs + cheeks; bolt the cheeks to the carrier discs; vacuum-impregnate. Grind the tip faces to r_g and bore the bridge faces in their ring after assembly: the 0.5 mm gap needs the rotor and counter-rotor runout well below it.

[OC] the section and its L(θ) are the solved 2-D model; [IR] the build details; [RH] the neck (B_sat, lap joints, the break's knee), the slot cover and the stud sizes are first-cut estimates: a nonlinear field check of the neck is the next step.

C. side section through a tip (v = 22): core, cheeks, studs, carrier



D. the solved 2-D section, aligned: flux lines (unrolled at r_g; linear iron, the neck is not in this model)

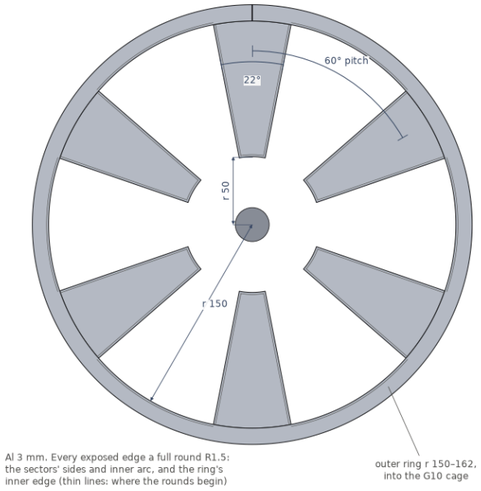


The air vane stack, 6 mm gaps, 6 + 6 vanes

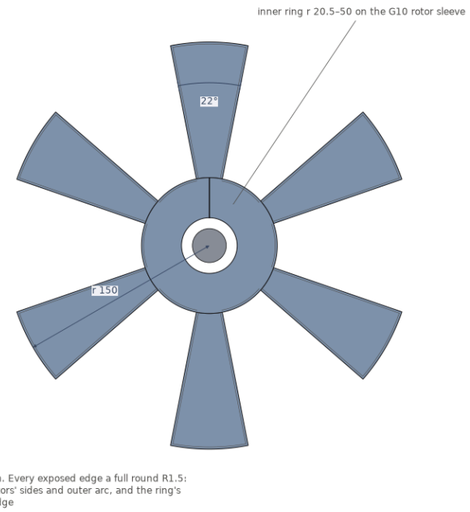
the electrostatic varicaps C1 / C2: stator vane (REF), rotor vane (node 1 / 4), both at minimum C, side A's half-section, the vane cell, the rim field (panel d still draws the placeholder hub)

Electrostatic vane stack for AIR: 6 mm gaps, 6 sectors of 22° (stator) / 22° (rotor), 6 + 6 vanes per varicap per side; vanes and Ca / Cb plates Al 3 mm with full-round edges R1.5 (dimensions in mm)

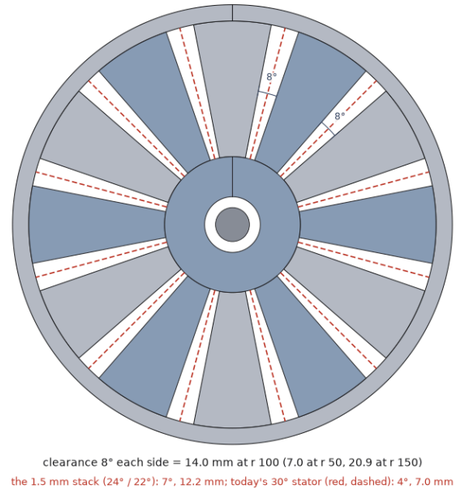
(a) stator vane (counter-rotor, REF of C1 / C2): 6 × 22° on a 60° pitch



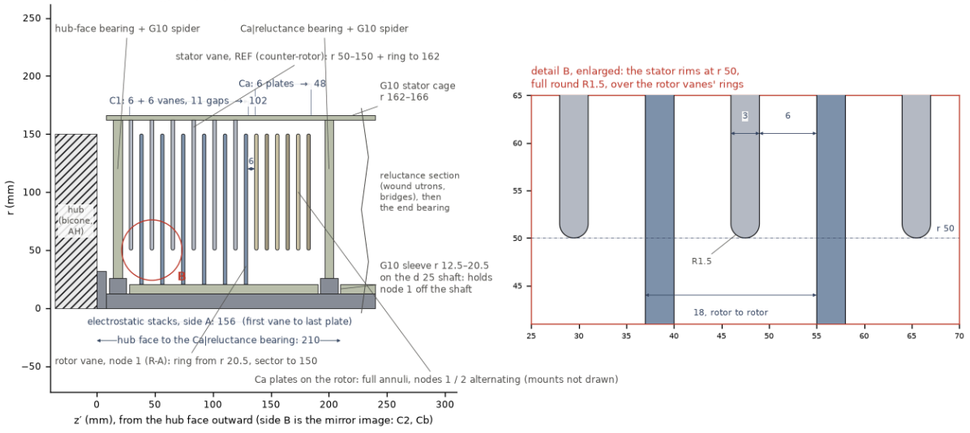
(b) rotor vane (rotor, node 1 / node 4): 6 × 22° on a 60° pitch



(c) at minimum C: the rotor sectors centred between the stator sectors

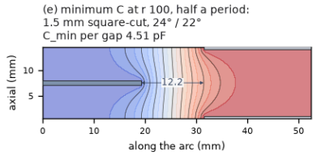


(d) side A, half-section in the plane of a sector at alignment (stack_sizing.layout):
hub-face bearing, C1, Ca, Ca|reluctance bearing

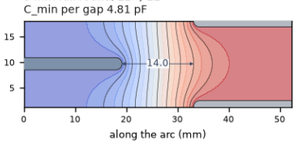


detail B, enlarged: the stator rims at r 50,
full round R1.5, over the rotor vanes' rings

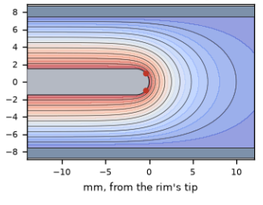
(e) minimum C at r 100, half a period:
1.5 mm square-cut, 24° / 22°
C_min per gap 4.51 pF



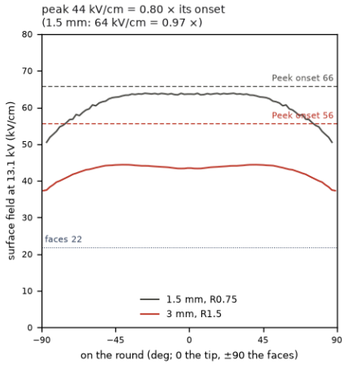
3 mm full round, 22° / 22°
C_min per gap 4.81 pF



(f) a rim at 13.1 kV, 6 mm to the
faces; equipotentials every 0.1, ● the peak



peak 44 kV/cm = 0.80 × its onset
(1.5 mm: 64 kV/cm = 0.97 ×)



(g) data (sim/air_stack_sizing_results.json: thick_vanes)

air, 1 atm	gap 6 mm: breakdown 19.7 kV (uniform field), operating 13.1 kV (÷ 1.5)
vanes	6 stator + 6 rotor per varicap per side, Al 3 mm, full-round edges R1.5: 11 gaps
sectors	6 on a 60° pitch: stator 22°, rotor 22° (searched for 4 mm at the C_max rule)
clearance at min C	8° each side, 14.0 mm at r 100 (1.5 mm stack: 7°, 12.2 mm)
C per gap	4.81 – 37.3 pF (1.5 mm stack: 4.51 – 38.4)
C1 = C2	55 – 410 pF, κ 7.47 (16 + 16: 7.65; 1.5 mm stack 8.38; vacuum 15.7)
Ca = Cb	451 pF: 6 full-annulus plates, Al 3 mm, rims R1.5; 5 gaps of 6
gain z	1.309 per cycle (16 + 16: 1.380; 1.5 mm stack 1.402; vacuum 1.515)
pump	2.14 W clamped at 13.1 kV, 1200 rpm relative (120 Hz)
	(16 + 16: 6.96; 1.5 mm stack 7.43 W; today's stack in air 2.44 W at 7.3 kV)
stacks, side A	102 C1 + 6 + 48 Ca = 156 mm, first vane to last plate
	(16 + 16: 417; 1.5 mm stack 354; today's 105)
tube	916 mm (stack_sizing.layout; 16 + 16: 1438; the wound build 802)
aluminium	rotor: rotor vanes 2.9 + Ca / Cb plates 6.1 kg; counter-rotor: stator vanes 3.4 kg
HV side	on the rotor: the rotor vanes (nodes 1 / 4), Ca / Cb, D1–D4, Z1 / Z4
	the stator vanes are C1 / C2's REF (sim/core-field-findings.md)
rims at the peak	44.4 kV/cm (× 2.03 the faces' 21.9); onset at 16.5 kV smooth, 14.0 kV handled (m 0.85)
	(smooth / handled, kV: 1.5 mm 13.5 / 11.5, 2.5 mm 15.7 / 13.3, 4 mm 17.8 / 15.1)

The cap: 6 + 6 vanes give 11 working gaps per side and C_max 410 pF against the 1113 pF the ladder was sized for (Ca = Cb = 1.1 C_max scale with it). Power falls to 0.31 of the 16 + 16 stack's, the stacks to 156 mm per side and the tube to 916 mm. z falls 1.380 → 1.309, mostly because the fixed 20 pF stray weighs about 3× more against the smaller stack; with 4 mm vanes it drops below 1.3.

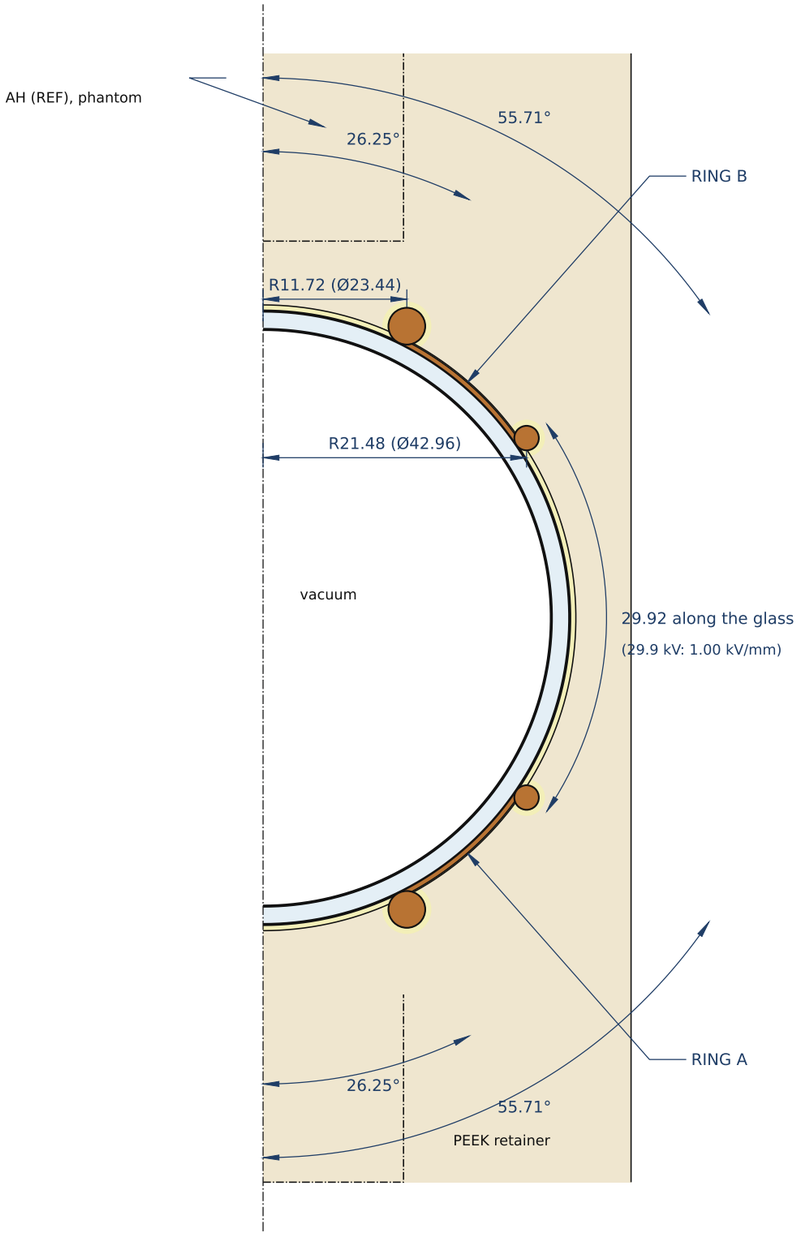
The thickness: thicker edges fringe more at minimum C (κ at the C_max rule on 22° / 22°: 2.5 mm 8.04; 3 mm 7.65; 4 mm 6.86) but hold more voltage at the rims; against the 13.1 kV peak, 3 mm reaches Peek's onset at 16.5 kV smooth and 14.0 kV handled, 1.5 mm at 13.5 / 11.5 kV.

Against today's stack in air (3 mm gaps, 8 + 8 × 1.5 mm, clamped near 7 kV): 2.44 W from 105 mm per side; this one gives 2.14 W at 13 kV from 156 mm.

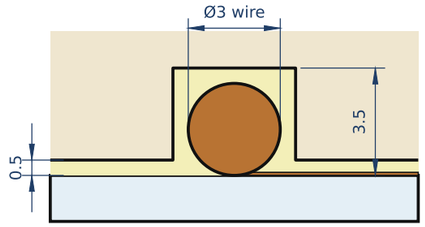
[OC] the Laplace solves (vane cell, rim), the core's z; Peek's law (empirical). [IR] the full round on the grid, the 2-D cell with its 2 pF rim floor and 0.25 mm grid (finer grids raise C_min by up to 7 %), the cosine C(θ), the sector widths (searched for 4 mm at the C_max rule, not for this count). [RH] the 1.5 margin on breakdown, the rim field read as a corona margin and the surface factor, the air's humidity and temperature.

A HALF-SECTION ON THE AXIS (2:1)

rings A and B mirror the equator; angles from the axis

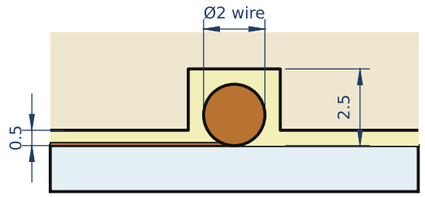


B POLAR BEAD, Ø3 (5:1)

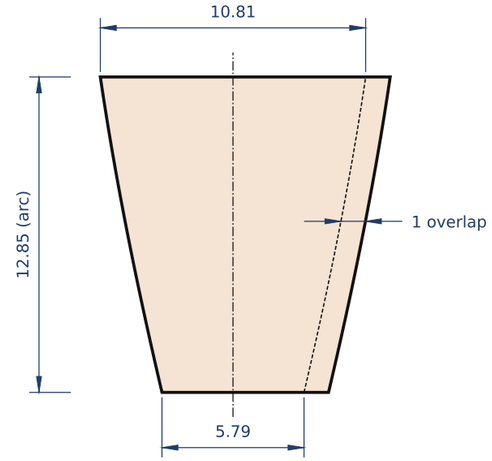


glass 1.5 foil 0.1, soldered to the bead

C EQUATORIAL BEAD, Ø2 (5:1)



glass 1.5 foil 0.1, soldered to the bead



D GORE, FLAT (4:1): 12 per band, 24 in all

width = 2π·25·sin θ / 12 along the arc; the edge θ at the narrow end

RINGS A AND B: band 12.85 along the glass, 13.1 cm² each; the gap 29.92

feature (each ring)	polar edge	equatorial edge
edge angle from the axis	26.25°	55.71°
bead's contact on the glass	r 11.06 / z ±22.42	r 20.65 / z ±14.08
bead's centre	r 11.72 / z ±23.77	r 21.48 / z ±14.65
bead: Cu wire, ring centre-line	Ø3 wire, Ø23.44	Ø2 wire, Ø42.96
length per ring, the centre line (note 3)	73.6	135.0
groove in the PEEK (depth)	3.5	2.5
peak field in the gel (record)	4.85 kV/mm	4.36 kV/mm

NOTES

1. Material: foil Cu-ETP (CW004A), annealed, 0.10 thick; bead rings Cu-ETP wire. Glass: the vessel as supplied (borosilicate, OD 50, wall 1.5), clean and degreased.
2. Cut 12 gores per band (D), deburr every edge; burnish them into a gel film (no adhesive) from the polar edge, each overlapping its neighbour by 1; fillet each lap's free edge with solder, >= 0.2 wide. No void under a lap.
3. Polar rings (Ø3) seamless: turned, or brazed and turned round. Equatorial (Ø2): butt joint dressed round within 0.03 over 1. Solder each along its foil's edge at the edge angle (A), touching the glass; no solder ball, no burr.
4. The leads: PTFE-insulated HV wire soldered to the equatorial bead, out through the PEEK and the G10 coupler.
5. The PEEK retainer: unfilled, annealed stock; the pocket 0.5 over the glass and the grooves (B, C) machined to size; degassed silicone gel vacuum-cast into the pocket and the grooves, void-free, above all at the contacts.
6. Ring A is ring B mirrored in the equator. Test before potting: continuity; after: hold-off per docs/bench-test-rings.md phase 1 (c).

DCCREG turbine · the hub: the field's rings on the vessel

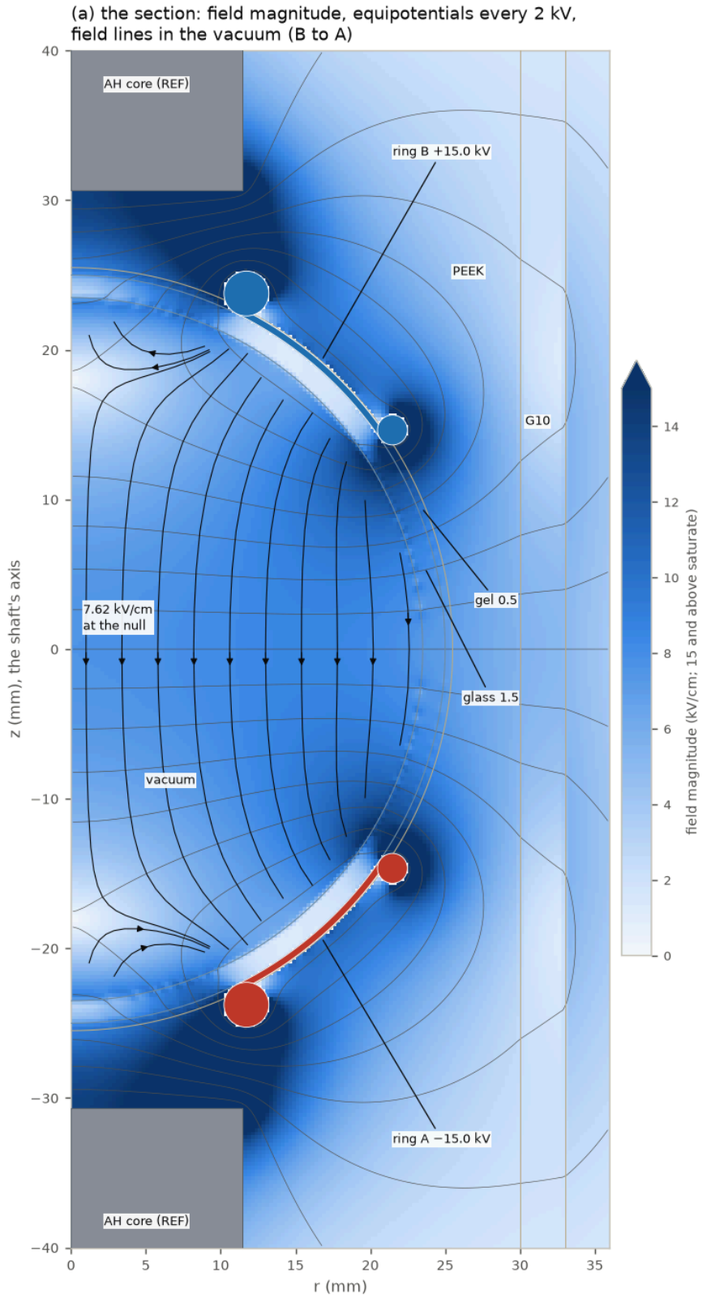
RINGS A AND B, COPPER, BEADED	Drawing no.
Cu-ETP foil 0.10 + wire; in gel under PEEK	**DCCREG-HUB-201**
Scale: 2:1 (A), 5:1 (B, C), 4:1 (D) Units: mm	Rev A (draft) Sheet 1/1 A3
Drawn: generated from sim/hub_rings_build.py	Date: 2026-10-09

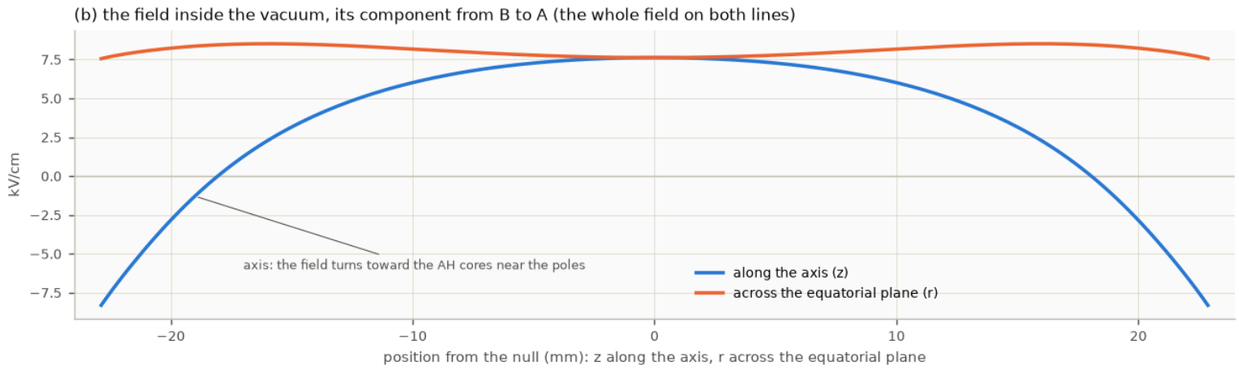
The rings' field at the null

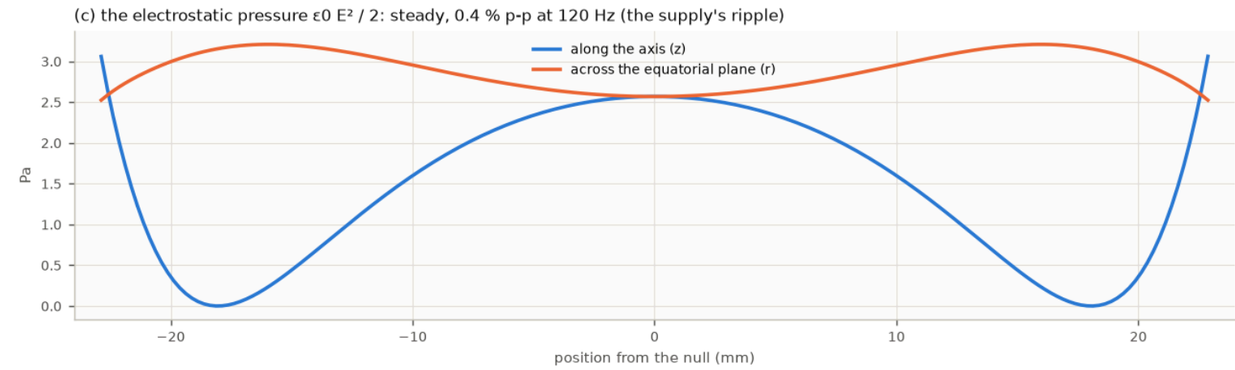
field drawings: the section's |E|, equipotentials and field lines; the field and the pressure along the axis and across the equator; ring B's beads

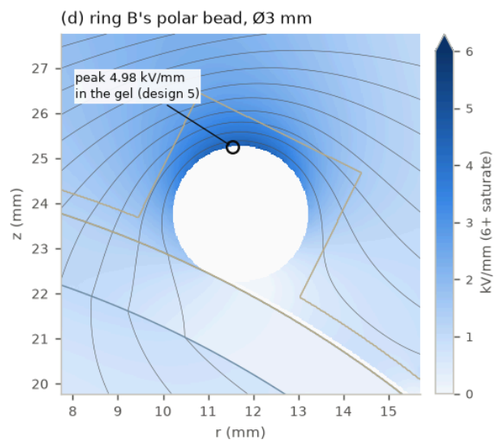
The rings' field, as connected: −15.0 / +15.0 kV on the symmetric supply, 7.62 kV/cm and 2.57 Pa at the null, steady

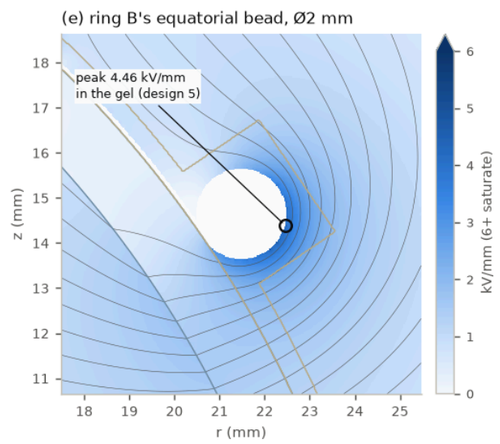
Ring A on its own negative chain from the shaft (node 1), ring B on the positive one (node 4), two stages each; the bands on the glass from the polar to the equatorial bead. Finite volumes, 0.25 mm cells; the beads by local solves, 0.025 mm cells.









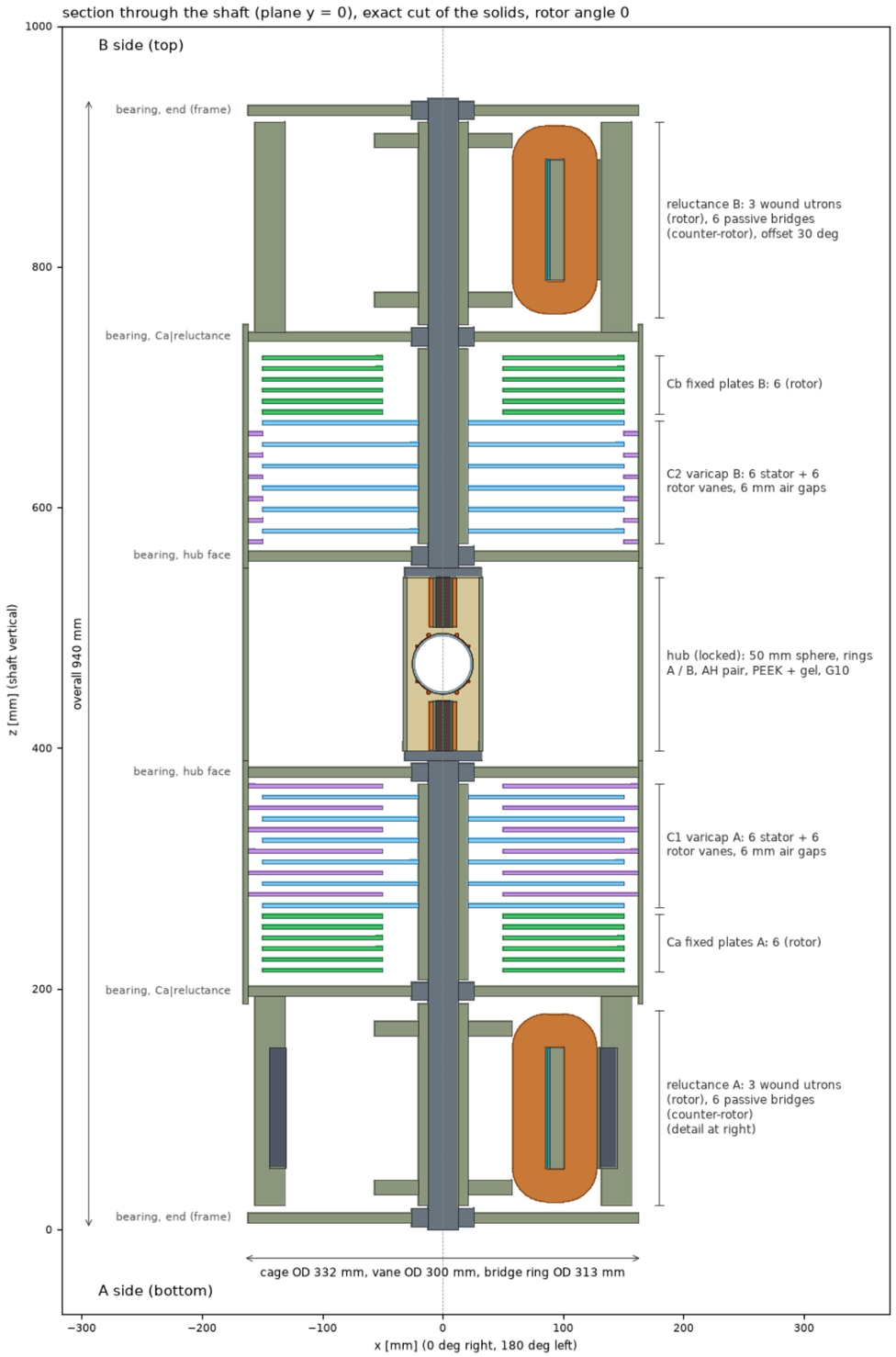


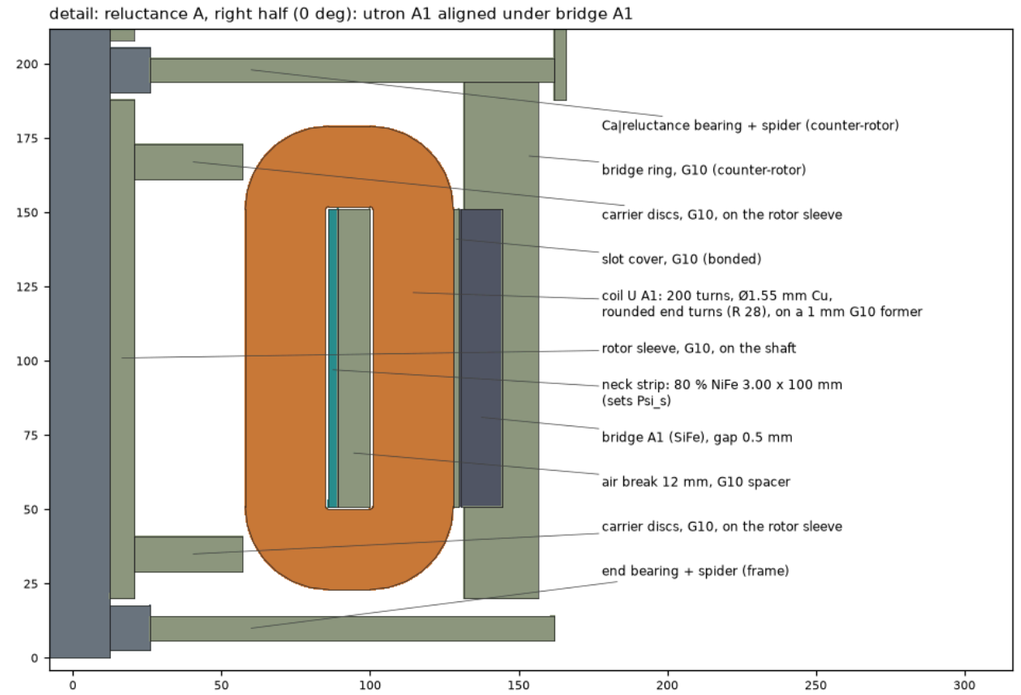
Ring A −15.0 kV, ring B +15.0 kV: 29.9 kV across, the null at the shaft's potential (0 V) by symmetry. Bands 26.25–55.71° from the axis, 29.9 mm apart along the glass (1.00 kV/mm). At the null 7.62 kV/cm, 2.57 Pa; within ±5 mm it varies 4.4 % along the axis and 2.0 % across the equator. The supply's ripple moves it 0.014 kV/cm p-p (sim/hub_drift.py swing): the field does not swing. With the beads in their grooves as drawn (the bands alone give 7.10 kV/cm). Beads in the gel: polar 4.98, equatorial 4.46 kV/mm (design 5); in the glass 1.46 / 3.01. Each ring to the AH cores: 1.81 kV/mm on average. Settled (the insulators' leakage, 25 °C): 8.20 kV/cm at the null (sim/hub_drift.py).

The machine of record: section through the shaft

the record in solids, 940 mm: the air stack of record (6 + 6 vanes, 6 mm gaps) with Ca / Cb on the rotor, the reluctance sections and the locked hub; side A below

tube-r150-n6-air6-wound-g0p5-6br-hub50: arrangement (g 0.5 / 6 bridges / 1200 rpm), 940 mm long





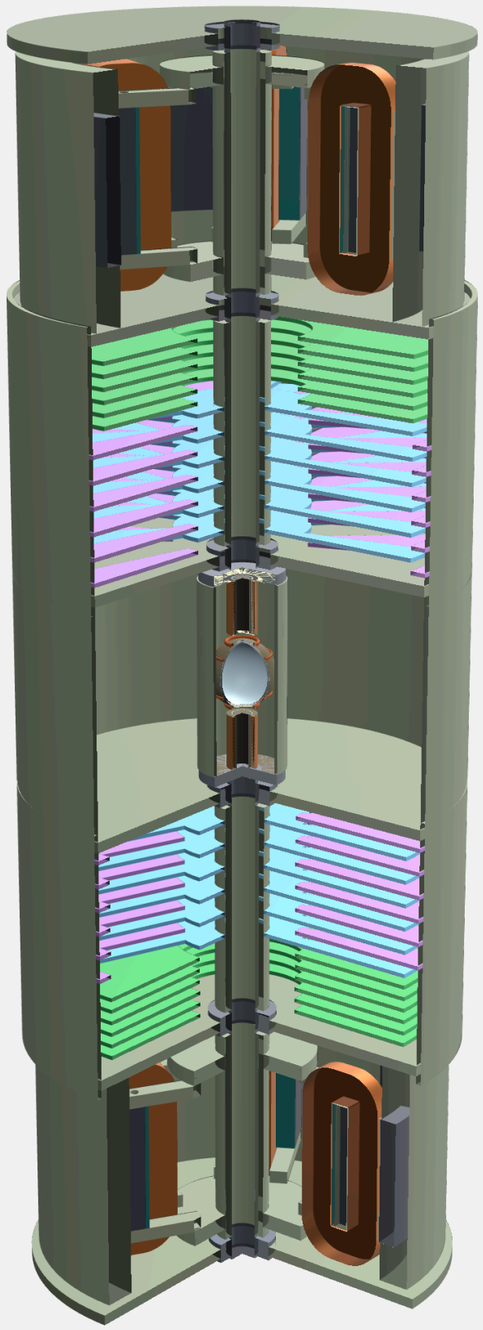
SiFe laminations (utron cores, bridges)
80 % NiFe neck strip
Cu windings
Al stator vane
Al rotor vane
Al Ca / Cb plate
G10 (sleeve, cage, spiders, carriers, ring)
steel (shaft, flanges, bearings)
borosilicate vessel
Cu rings + beads
silicone gel
PEEK retainer
MnZn AH cores

Bodies. Rotor: shaft halves, flanges, the hub (vessel, rings, gel, retainer, coupler, AH pair), rotor sleeve, rotor vanes (nodes 1 / 4), Ca / Cb plates (mounts not drawn), utrons + carrier discs. Counter-rotor, geared 1 : -1 (gear not drawn): stator cage (one tube across the hub), stator vanes (REF), the hub-face and Ca|reluctance spiders, bridges + bridge rings. Frame: the two end bearings and their spiders. The inner bearings run at the relative speed. At rotor angle 0: utron A1 is aligned (gap 0.5 mm), utron B1 sits between two B bridges (antiphase); C1 is at its maximum and C2, its stator vanes half a pitch on, at its minimum. This plane (v = 0) cuts the coil, the neck strip and the air-break spacer; the SiFe half-cores and the cheeks lie at |v| >= s / 2 (see the plan cuts and the utron detail drawing).

The machine of record: quarter cutaway

the same solids in 3-D, cut away

Quarter cutaway at rotor angle 0 (rotor + counter-rotor geared 1 : −1; frame: the end bearings)

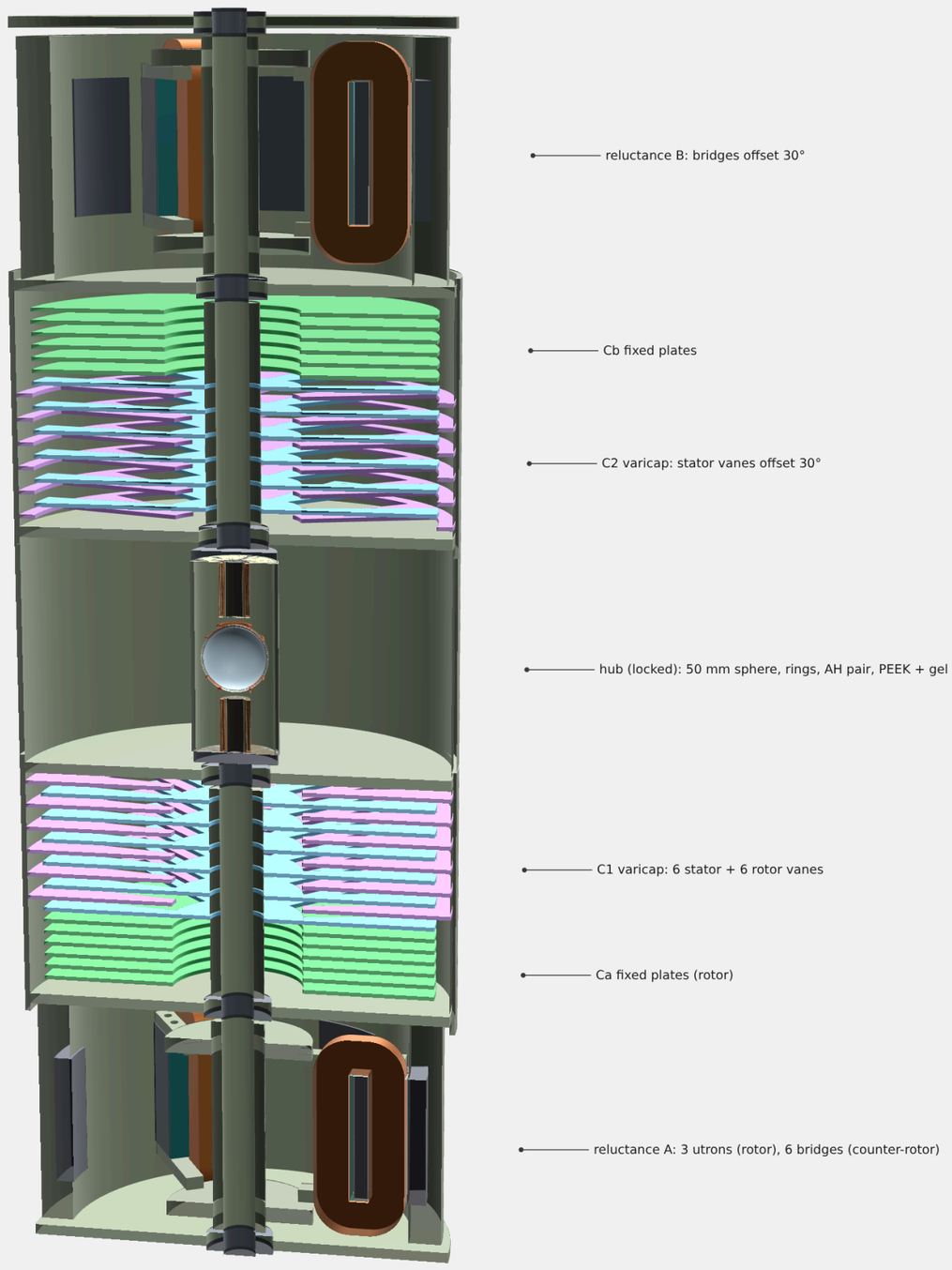


docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50.step, rendered from the STEP (sim/step_to_glb.py + three.js)

The machine of record: half section

the same solids, half-sectioned

Half section (the near half removed): the machine from the front

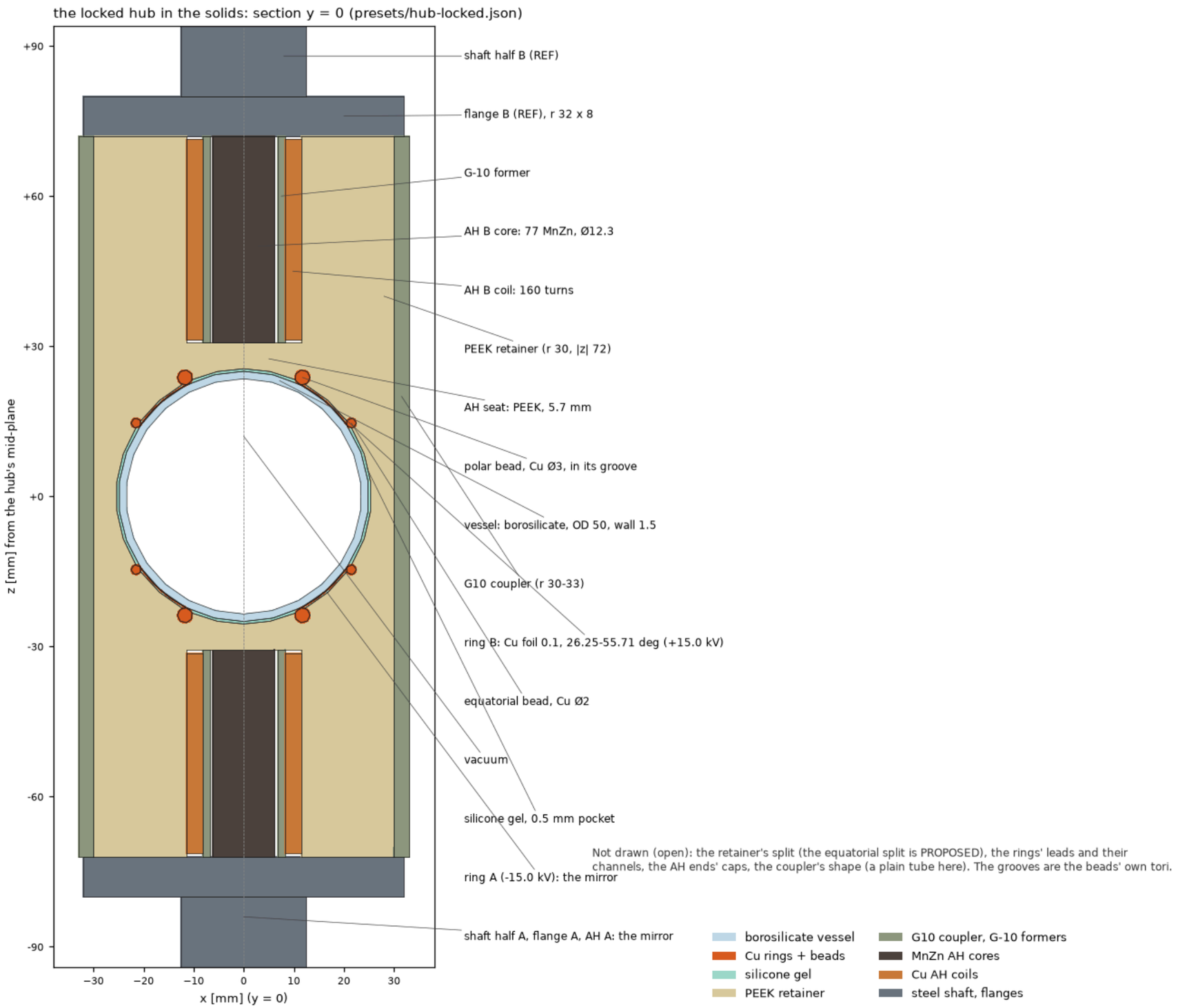


docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50.step, rendered from the STEP (sim/step_to_glb.py + three.js)

The locked hub in solids

the vessel, rings A / B with their beads, the gel, the PEEK retainer, the G10 coupler, the AH cores, formers and coils, the flanges

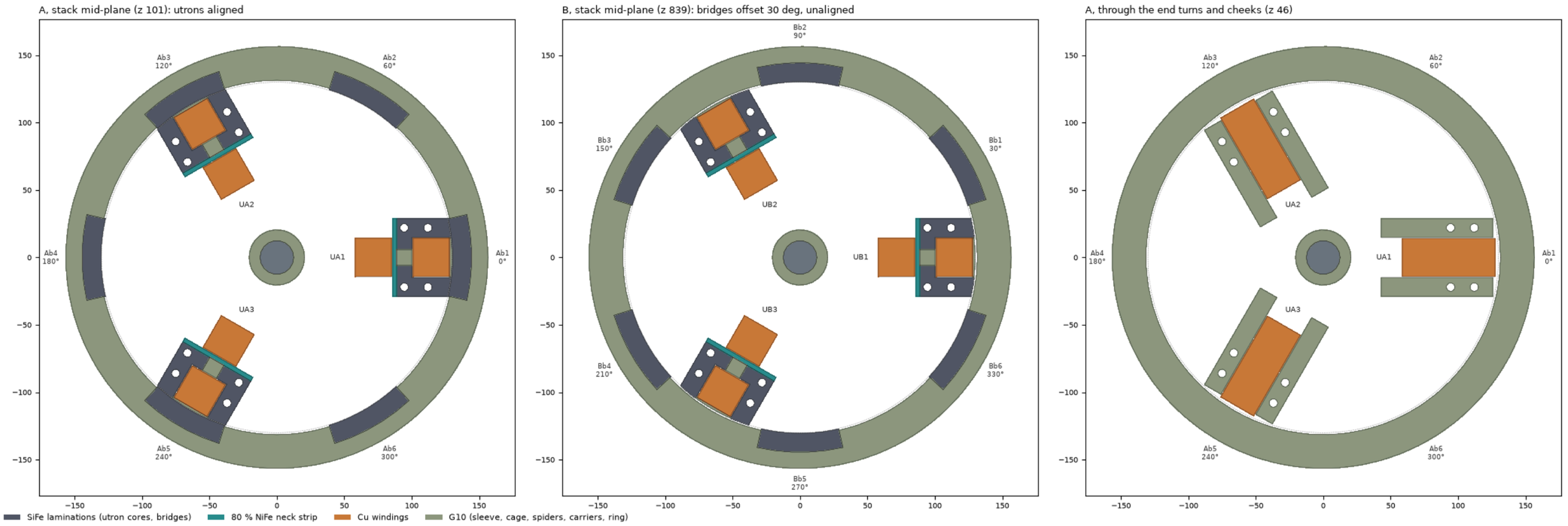
tube-r150-n6-air6-wound-g0p5-6br-hub50: the hub



Reluctance sections A and B, plan cuts

the three wound utrons per side on the rotor and the six bridges per side on the counter-rotor (B offset 30°)

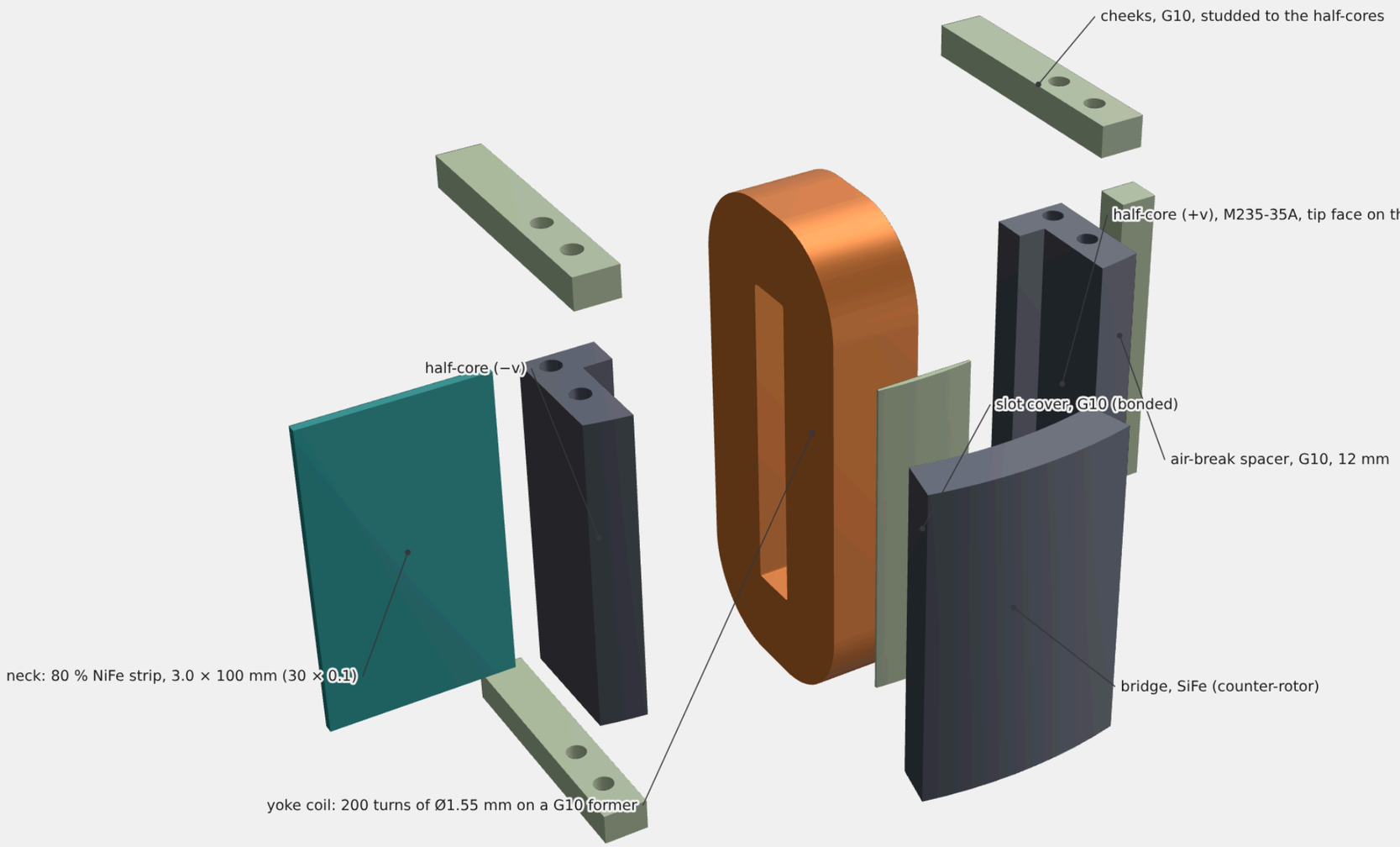
tube-r150-n6-air6-wound-g0p5-6br-hub50: plan cuts through the reluctance sections, rotor angle 0 (dashed: the gap radius r_g 130 mm, gap 0.5 mm). 3 utrons per side at 120 deg on the rotor, 6 bridges per side at 60 deg on the counter-rotor.



Utron A1 with bridge A1, exploded

the parts of one wound utron and its bridge

Utron A1 exploded along its assembly directions, with bridge A1

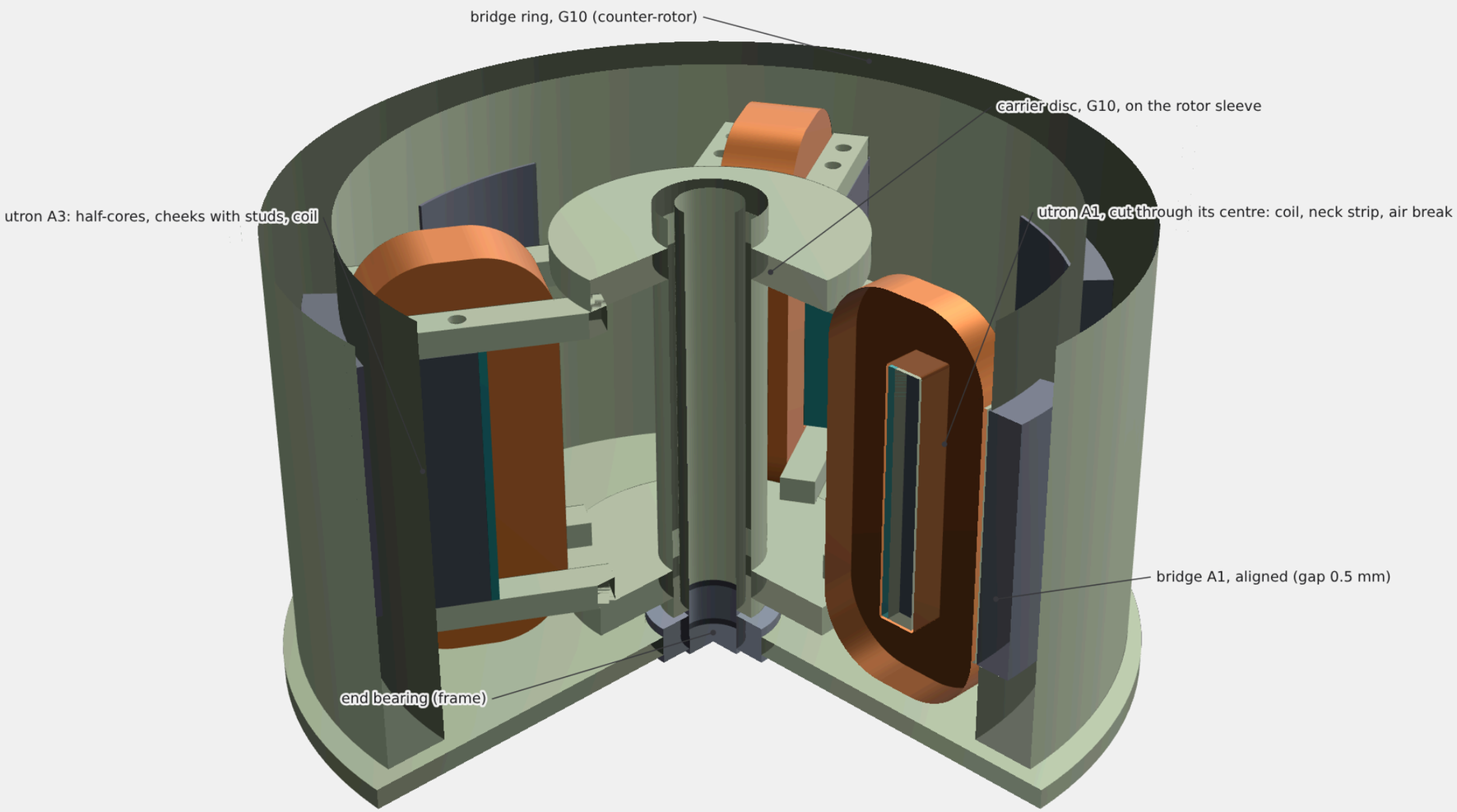


docs/geometry/tube/tube-r150-n8-wound-g0p5-6br.step, rendered from the STEP (sim/step_to_glb.py + three.js)

Reluctance section A, quarter cut

utrons on their G10 carrier discs inside the bridge ring

Reluctance section A, quarter cut: 3 wound utrons on the rotor, 6 passive bridges in the G10 ring

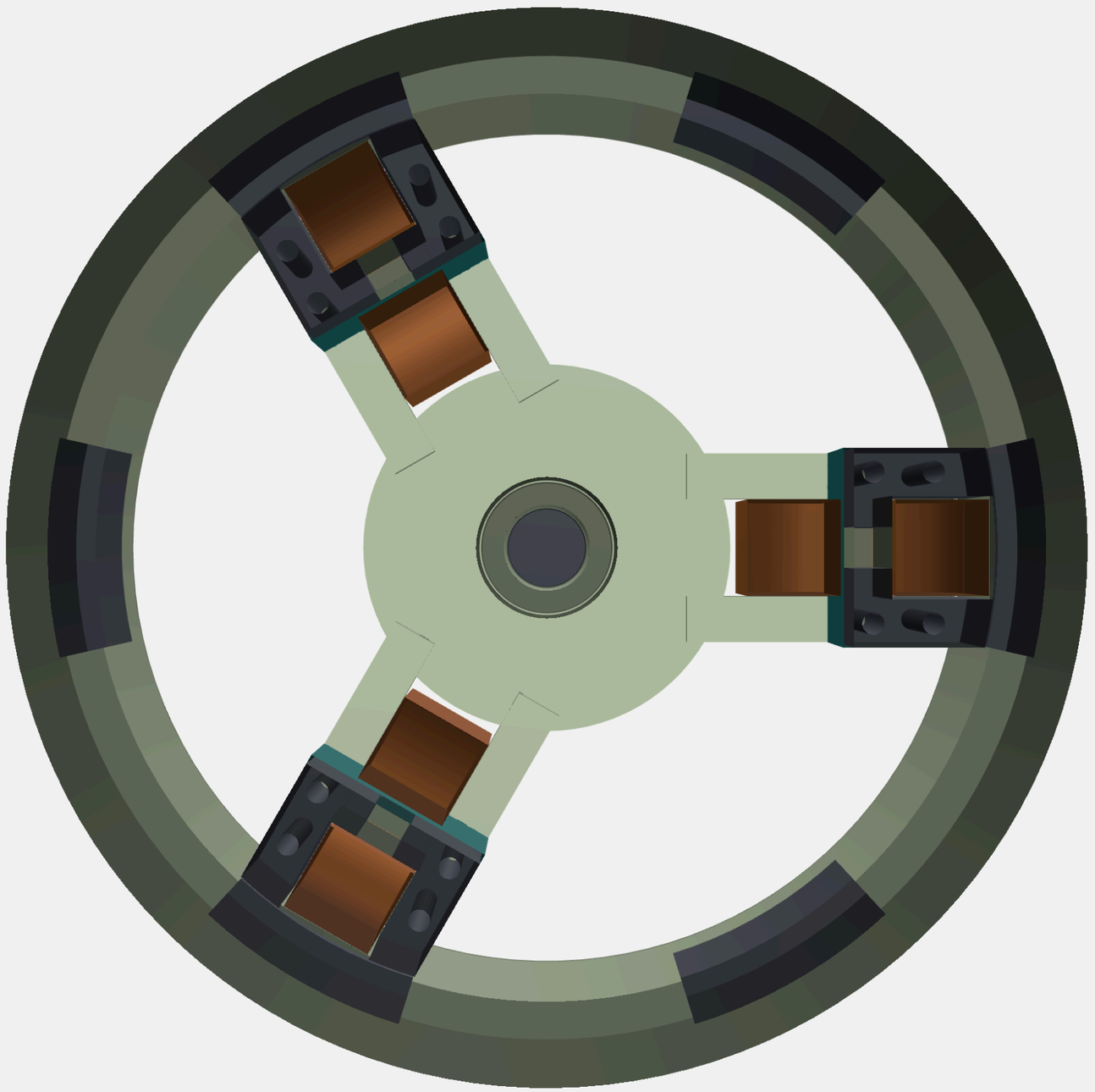


docs/geometry/tube/tube-r150-n8-wound-g0p5-6br.step, rendered from the STEP (sim/step_to_glb.py + three.js)

Reluctance section A, plan at mid-stack

the 0.5 mm gaps at the aligned position

Plan view of reluctance A, cut at mid-stack: utrons A1 to A3 aligned under bridges (B is unaligned)



docs/geometry/tube/tube-r150-n8-wound-g0p5-6br.step, rendered from the STEP (sim/step_to_glb.py + three.js)

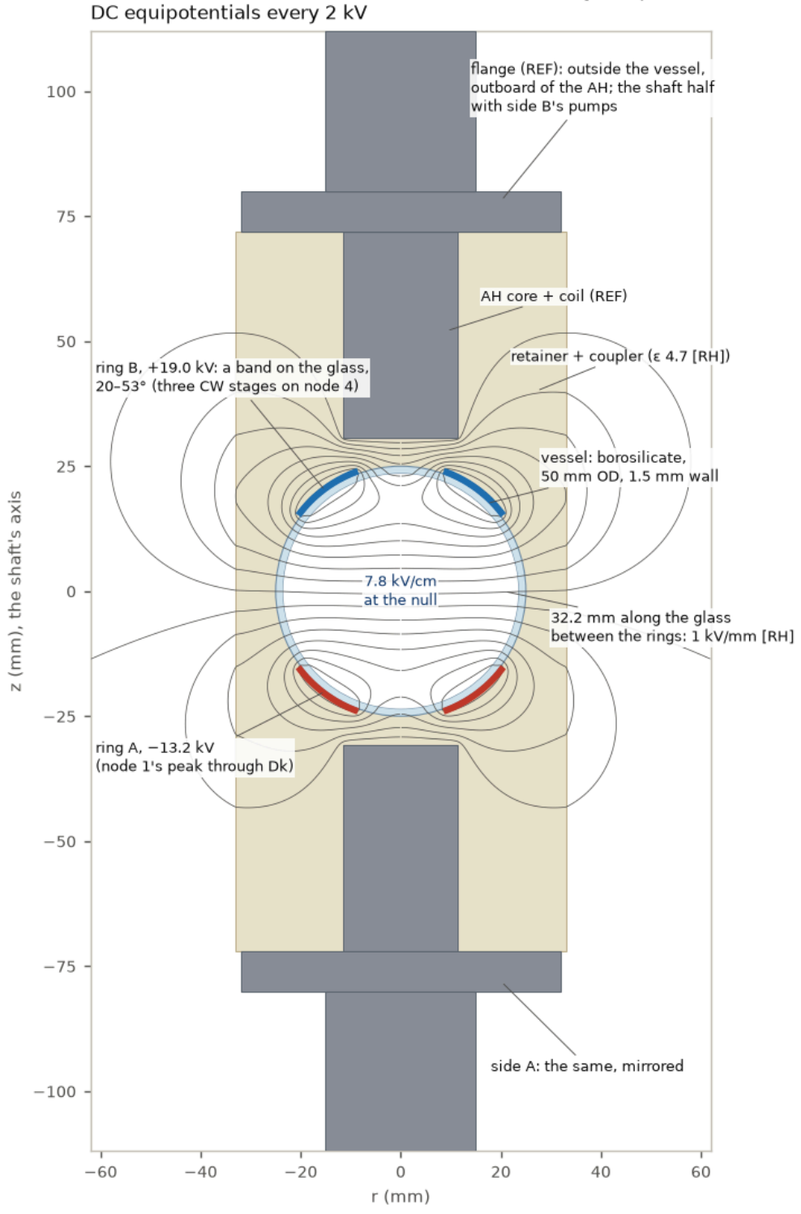
The lock-down study of the hub

the hub as locked (50 mm sphere, AH cores, flanges) and the lock-down's pick of rings, its 3-stage 20–53° bands superseded by DCCREG-HUB-201

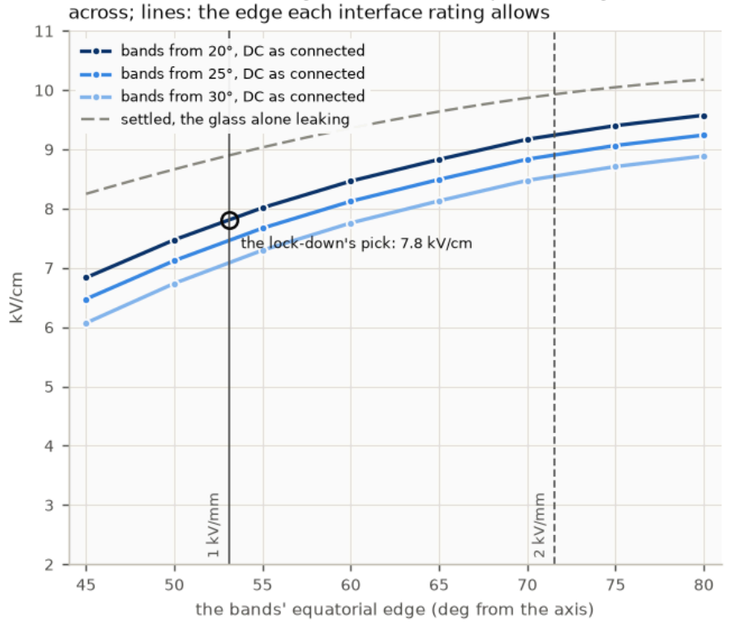
The locked hub, as the lock-down study sized its rings (2026-10-08; the rings superseded by the record, docs/rings-design.md)

Designer's stack-up and choices (presets/hub-locked.json): 50 mm borosilicate sphere, 1.5 mm wall; the rings outside it; the AH cores on the z axis, their flanges outside the vessel; retainer, shaft coupler, shaft halves with their pumps. 1200 rpm relative.

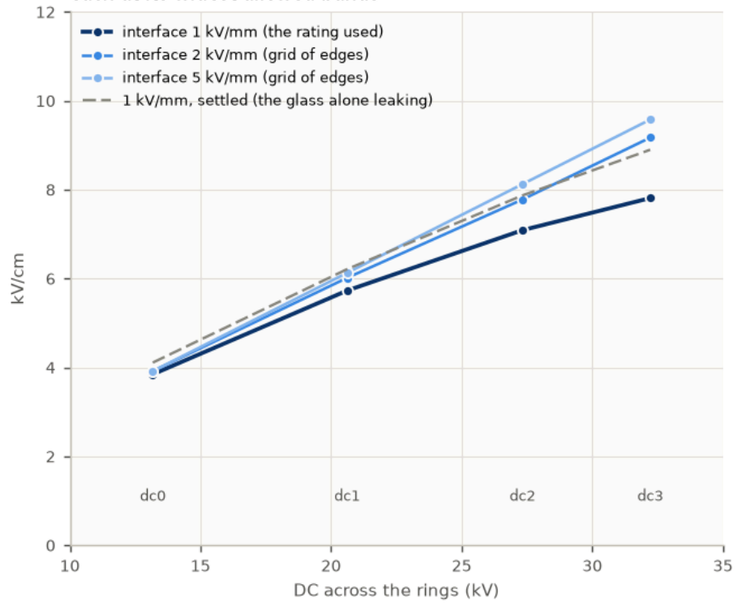
(a) the locked hub to scale, with the lock-down's rings (superseded); DC equipotentials every 2 kV



(b) the field at the null against the bands' equatorial edge, 32.2 kV across; lines: the edge each interface rating allows



(c) the field at the null against the DC across the rings (0–3 CW stages), each at its widest allowed bands



(d) the lock-down's pick and the alternatives (the record now: 2 + 2 stages, ±15.0 kV, 7.62 kV/cm with its beads)

option	across	gap	kV/cm	Pa	settled	start z	C to REF / across (pF)
dc3: bands 20–53° (its pick)	**32.2 kV**	**32.2 mm**	**7.8**	**2.70**	**8.9**	**1.178**	**7.4 / 1.3**
dc2: bands 20–59°	27.3 kV	27.3 mm	7.1	2.22	7.9	1.179	7.6 / 1.7
dc1: bands 20–66°	20.6 kV	20.6 mm	5.7	1.45	6.2	1.184	7.8 / 2.5
dc0: bands 20–75°	13.2 kV	13.2 mm	3.8	0.65	4.1	1.237	8.0 / 3.9
the swing: 8 mm bands at 50°	±7.4 kV		±1.5	0–0.10	AC	1.286	5.8 / 1.6
not chosen: the Rogowski pair inside	20.6 kV	3.14 mm	65.7	191	steady	1.184	1.0 / 0.7

The supply limits the rings, not the vacuum: at 7.8 kV/cm the vacuum is far below its rule. More DC widens the gap the rings need between them (1 kV/mm along the glass under the retainer [RH]), yet still gains: three stages give 7.8 kV/cm, 1.4× one stage. A better-rated interface lets the bands reach the equator: 9.2 kV/cm at 2 kV/mm, 9.6 at 5.
The multiplier sags at 100 pF: its third stage adds 4.9 kV, not 7.4 (the 10 GΩ placeholder leakage per ring). Each ring's average field to the AH cores is 2.6 kV/mm through the retainer and the seats (5 kV/mm derated garolite [IR]).
DC through glass drifts: borosilicate leaks (τ ≈ 7 min), so the glass beyond each ring follows it and the field rises toward the settled value; charge on the vacuum side of the wall can screen it. The swing does neither, at a fifth of the field.

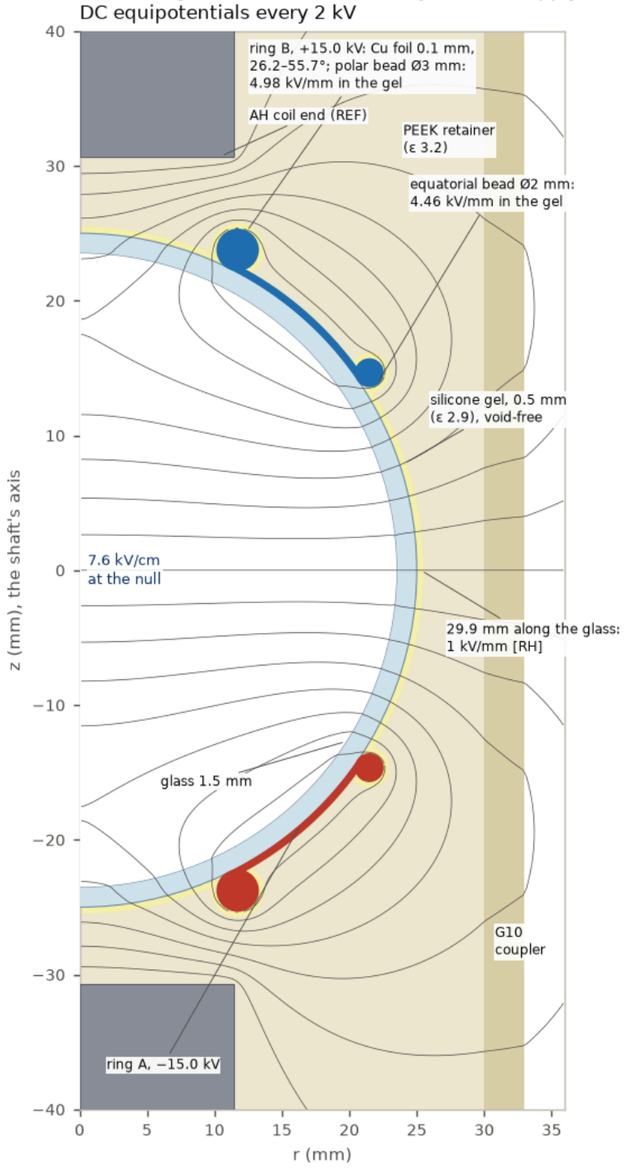
The rings as built: retainer, edges, stages

the PEEK retainer with its gel pocket and the beads to scale; the field against the DC by supply family and rating; the polar bead against the AH; the retainer's permittivity

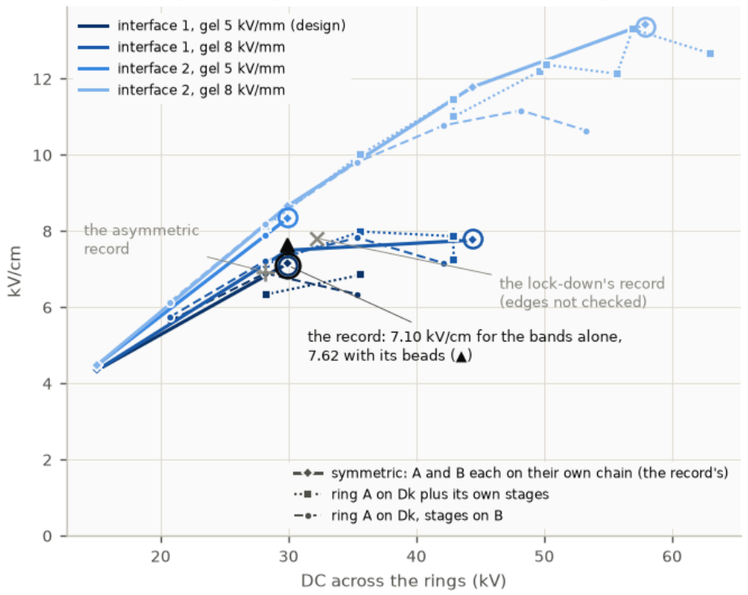
The rings as built: the edges set the stages — rings A and B on 2 stages each, symmetric, 29.9 kV across, 7.6 kV/cm at the null

The retainer: unfilled PEEK with a 0.5 mm pocket over the glass, filled void-free with silicone gel; G10 coupler outside. The rings: 0.1 mm copper foil, beaded edges. Leakage 100 GΩ per ring (estimate). Limits: the gap along the glass, and both beads in the gel.

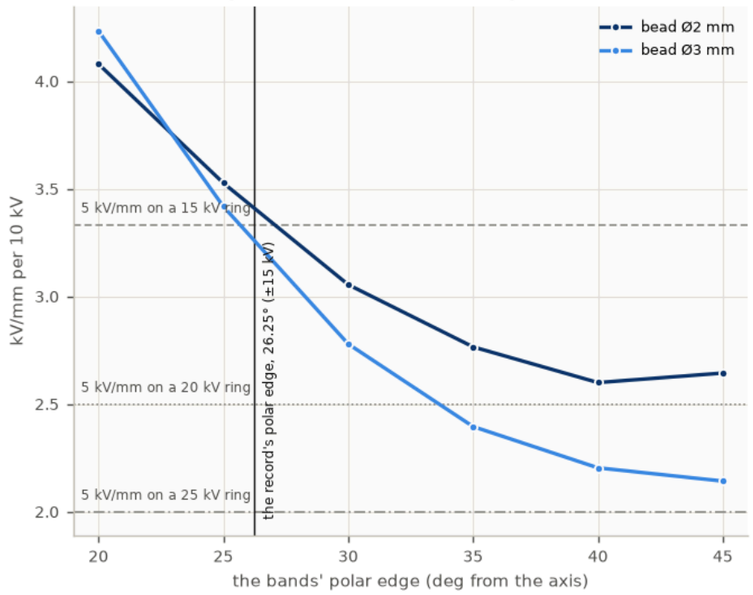
(a) the rings as built, to scale (the symmetric supply); DC equipotentials every 2 kV



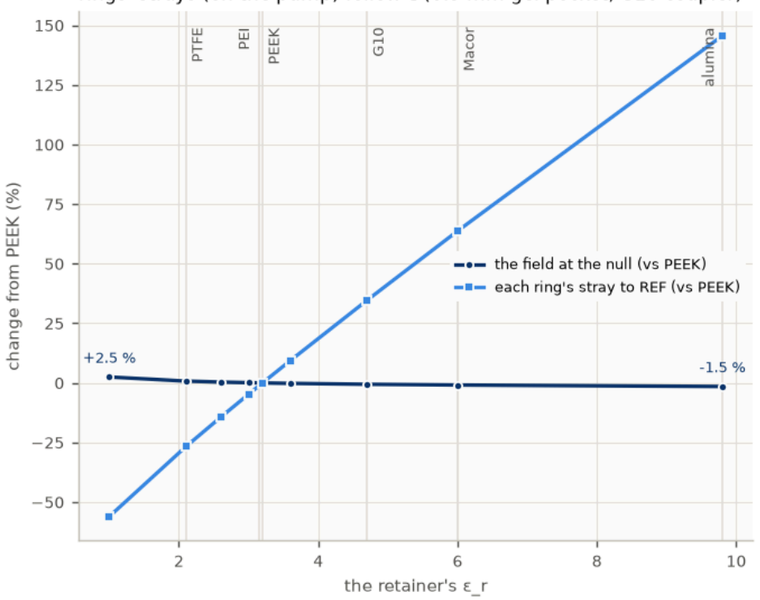
(b) the field at the null against the DC across the rings (each supply at its edge-limited bands; colour: the ratings, line: the family)



(c) the polar bead faces the AH coil's end: its peak field in the gel per 10 kV on the ring (bands to 50°; the other ring adds little)



(d) the retainer's permittivity: the field at the null barely moves; the rings' strays (on the pump) follow ε (0.5 mm gel pocket, G10 coupler)



(e) the best symmetric design at each pair of ratings, solved directly (the bands alone; with the beads as drawn, about 7 % more at the null)

ratings (kV/mm)	stages, rings A / B	bands, beads	gel, polar / eq. bead	at the null	stacked family's best
glass 1 gel 5	**2 + 2 ±15.0 kV**	**26.25–55.7° Ø3 / Ø2 mm**	**4.85 / 4.36**	**7.10 kV/cm 2.23 Pa**	**6.93 kV/cm B 2**
glass 1 gel 8	3 + 3 ±22.2 kV	24.00–39.2° Ø3 / Ø2 mm	7.95 / 7.02	7.80 kV/cm 2.69 Pa	7.97 kV/cm A 1 + B 2
glass 2 gel 5	2 + 2 ±15.0 kV	25.50–72.9° Ø3 / Ø4 mm	4.97 / 4.85	8.35 kV/cm 3.09 Pa	7.91 kV/cm B 2
glass 2 gel 8	4 + 4 ±28.9 kV	30.75–56.8° Ø3 / Ø3 mm	7.89 / 6.90	13.38 kV/cm 7.92 Pa	13.25 kV/cm A 2 + B 4

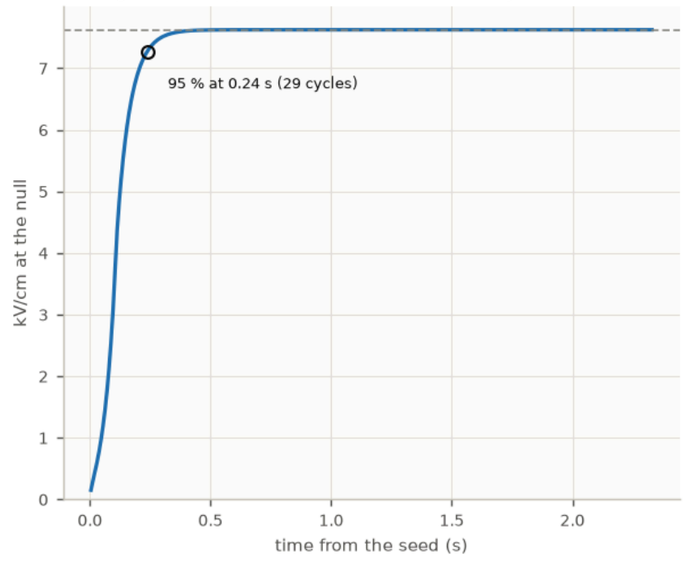
The rings' separation is not the only limit: each ring's polar bead faces the AH coil's end (REF) across the PEEK seat, and holds 5 kV/mm in the gel only away from the pole; the equatorial bead's field grows with the DC even at a fixed field along the glass. On the symmetric supply each ring has its own chain from the shaft (A negative on node 1, B positive on node 4): each holds half the DC, and the null sits at the shaft's potential.
Bold: the design ratings (1 kV/mm along the glass [RH]; 5 kV/mm in the gel at a bead [RH]). The others need the bench test's qualification (docs/bench-test-rings.md).

start-up, the 120 Hz ripple (nodes, rings, field), the 6 h drift by material, the test's phases

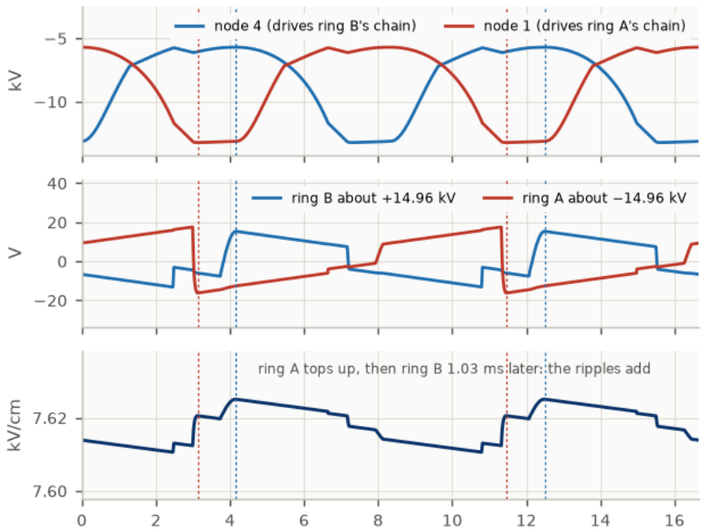
The bench test of the rings: what the field at the null should do (the symmetric supply, 2 + 2 stages, 29.9 kV across, bands 26.2–55.7°)

sim/hub_drift.py: the pump's supply in ngspice (100 pF storage, 100 GΩ per ring), the hub's finite volumes for the drift (C dV/dt + G V = b; conductivities datasheet-class [IR]). The field at the null as connected follows the DC across the rings; the leaking insulators then move it.

(a) the start-up: the pump self-excites and charges the rings through the multiplier (the lowest field each cycle)

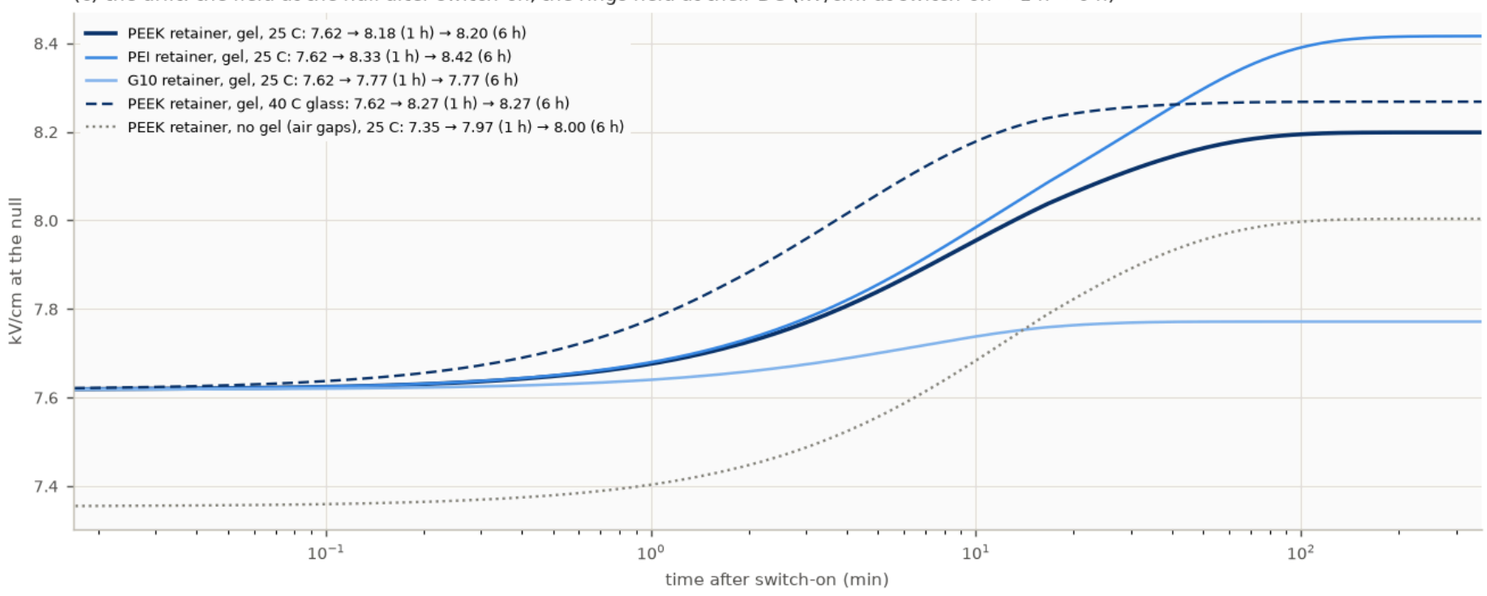


(b) the 120 Hz ripple: 0.014 kV/cm p-p on 7.62 (0.2 %), no swing from A to B



the AH's ampere-turns ripple 5.9 % p-p (22 mF across each coil)

(c) the drift: the field at the null after switch-on, the rings held at their DC (kV/cm: at switch-on → 1 h → 6 h)



(d) the bench test (docs/bench-test-rings.md)

1. hold-off

how: coupons, then the sphere: ramp the DC to flashover / breakdown; partial discharge at the design voltage

expect: sets the ratings: 1 kV/mm along the glass and 5 kV/mm in the gel (the record), 2 / 8 to qualify more stages

2. leakage

how: each ring from a lab supply; the current to REF on an electrometer; then the multiplier's parts

expect: ≥ 100 GΩ per ring (the ledger: 158 GΩ); the diodes and capacitors dominate

3. DC drift

how: rings on lab supplies at the record's DC; the field at the null on the probe for 6 h

expect: 7.62 kV/cm at switch-on, 7.95 at 10 min, 8.18 at 1 h, 8.20 at 6 h (PEEK, gel, 25 °C)

4. the pump

how: the rings on the rotor's supply: start-up, ripple, then the drift again

expect: 95 % in 0.24 s; 0.014 kV/cm p-p at 120 Hz on 7.62: a steady field from B to A, no swing

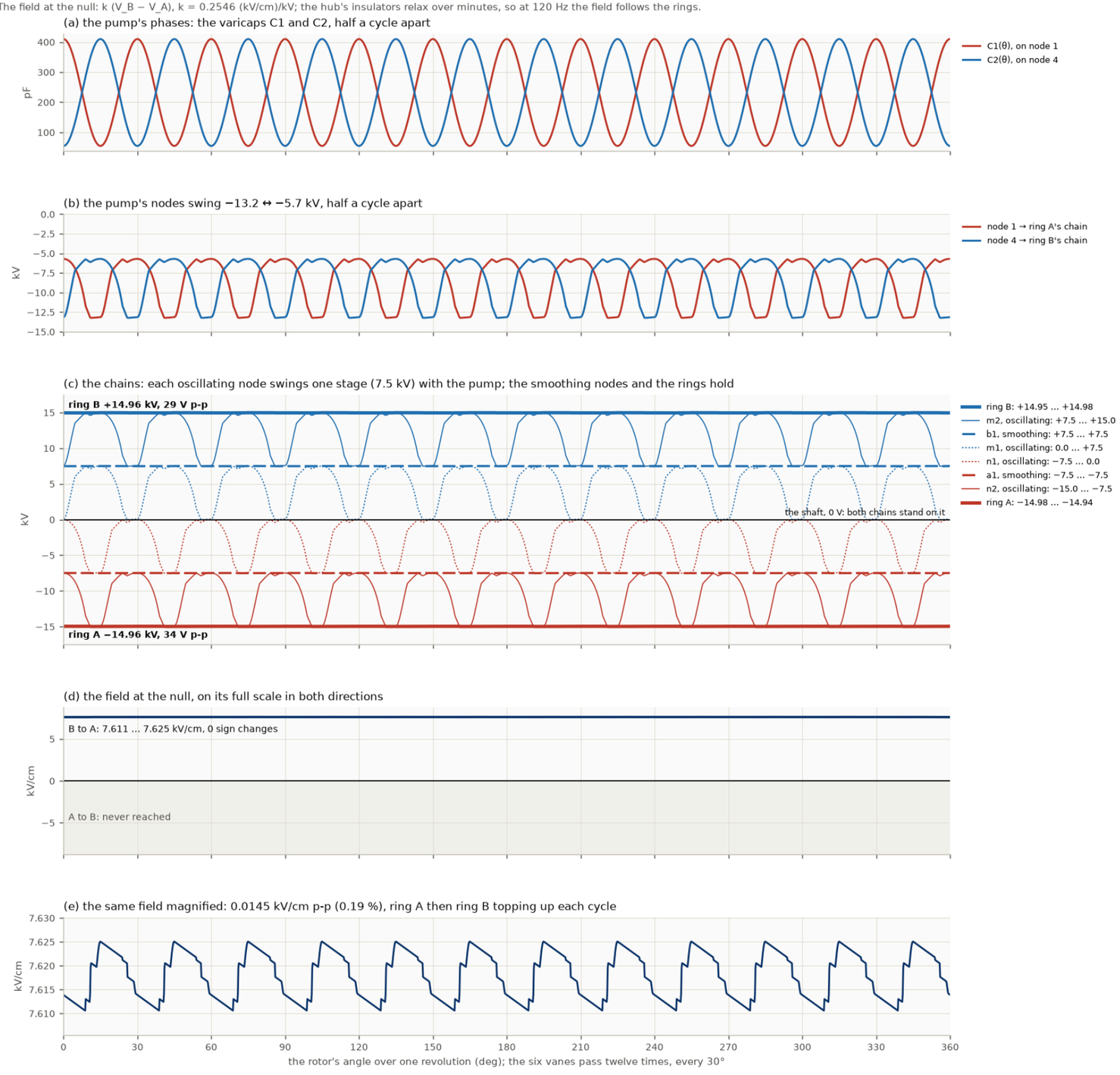
The probe: an electro-optic (Pockels) BGO sensor on a fibre, entering through the pumping tube at one pole along the axis, where it barely disturbs the axial field. The bench's vessel must be evacuated: air inside would carry its own ions and screen the DC in minutes.
The drift is the leakage redistributing the potential along the glass and through the gel and retainer: the field moves from the electrostatic value at switch-on toward the conduction-settled value. Fit E(t) with one or two exponentials; the time constants check the conductivities, the end value their ratios.

One revolution of the rotor

the pump's phases, its nodes, the chains, and the field at the null: steady from B to A, no sign change

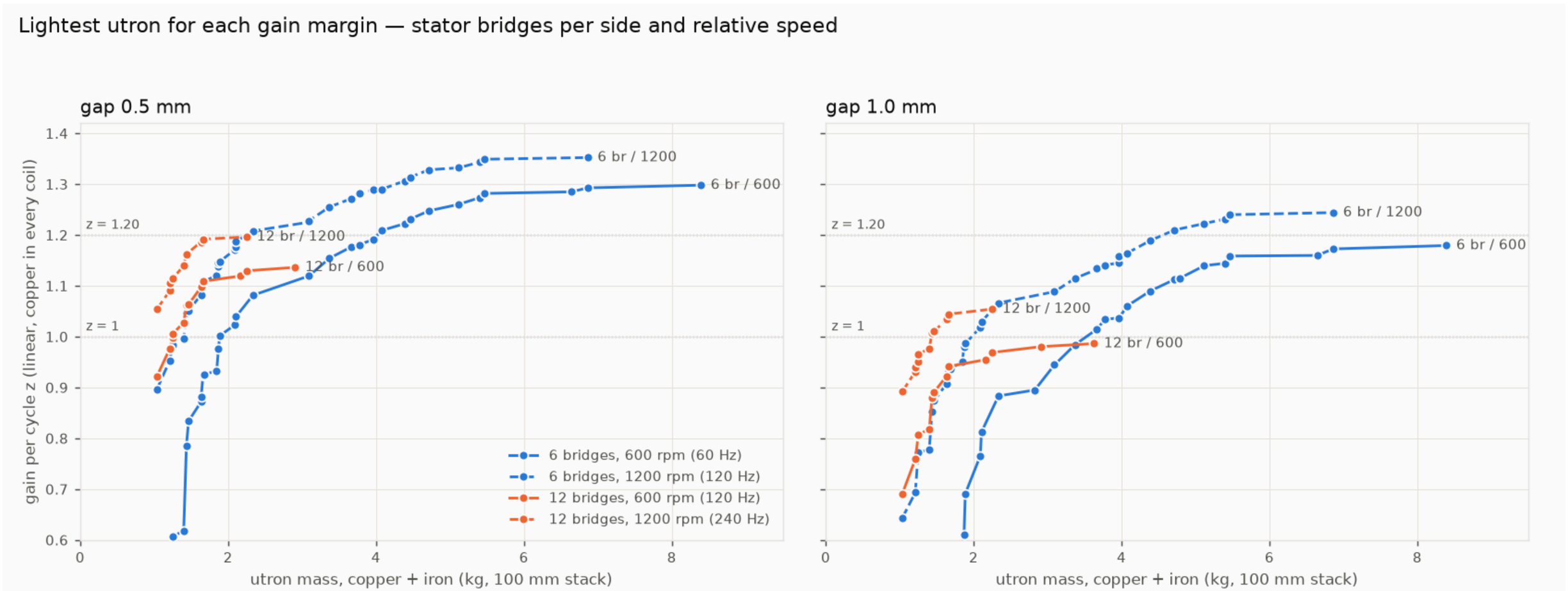
One revolution of the rotor: the pump goes through its phases 12 times, and the field at the null stays 7.62 kV/cm from B to A

sim/hub_revolution.py: the supply of record in ngspice (the symmetric pair, 2 + 2 stages, 100 pF, 100 GΩ per ring), settled over 280 cycles, then one revolution at 600 rpm each way (12 × 120 Hz cycles, 100 ms).
The field at the null: k (V_B − V_A), k = 0.2546 (kV/cm)/kV; the hub's insulators relax over minutes, so at 120 Hz the field follows the rings.



Utron size against gain: the pick

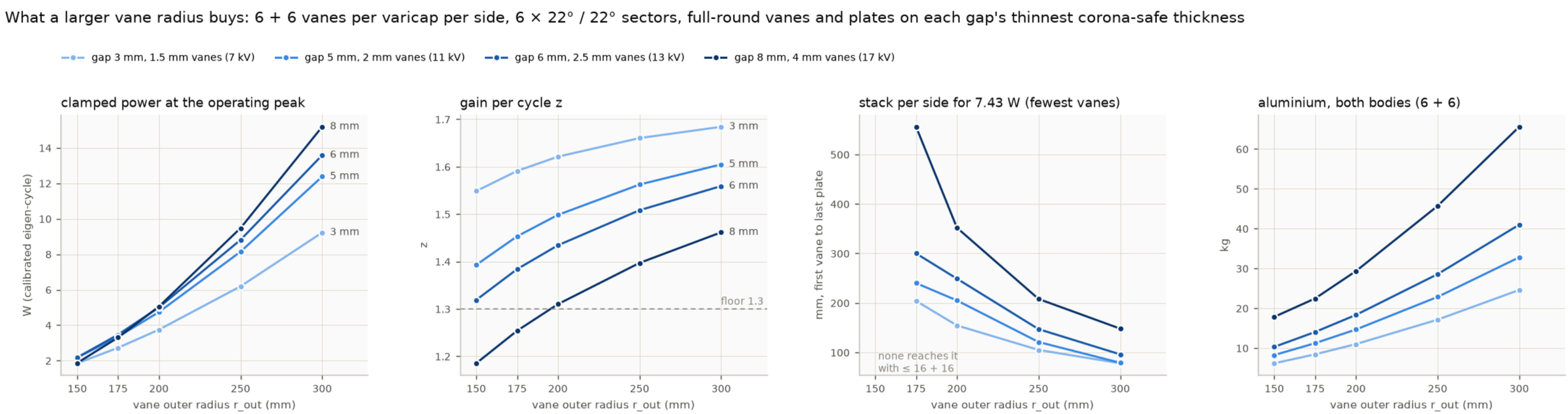
6 / 12 bridges × 600 / 1200 rpm at 0.5 / 1.0 mm, with the corrected mean turn



Each line is a Pareto front: every point is the lightest screened utron reaching that gain. Hue = bridges per side, line style = speed (solid 600 rpm, dashed 1200 rpm relative).
Dotted: z = 1 (growth threshold) and z = 1.20 (healthy margin). Below z 0.6: off the axis. Copper with the corrected mean turn. Source: sim/pole_design_variants.json (pole_design.py variants / variants_ext / turnfix).

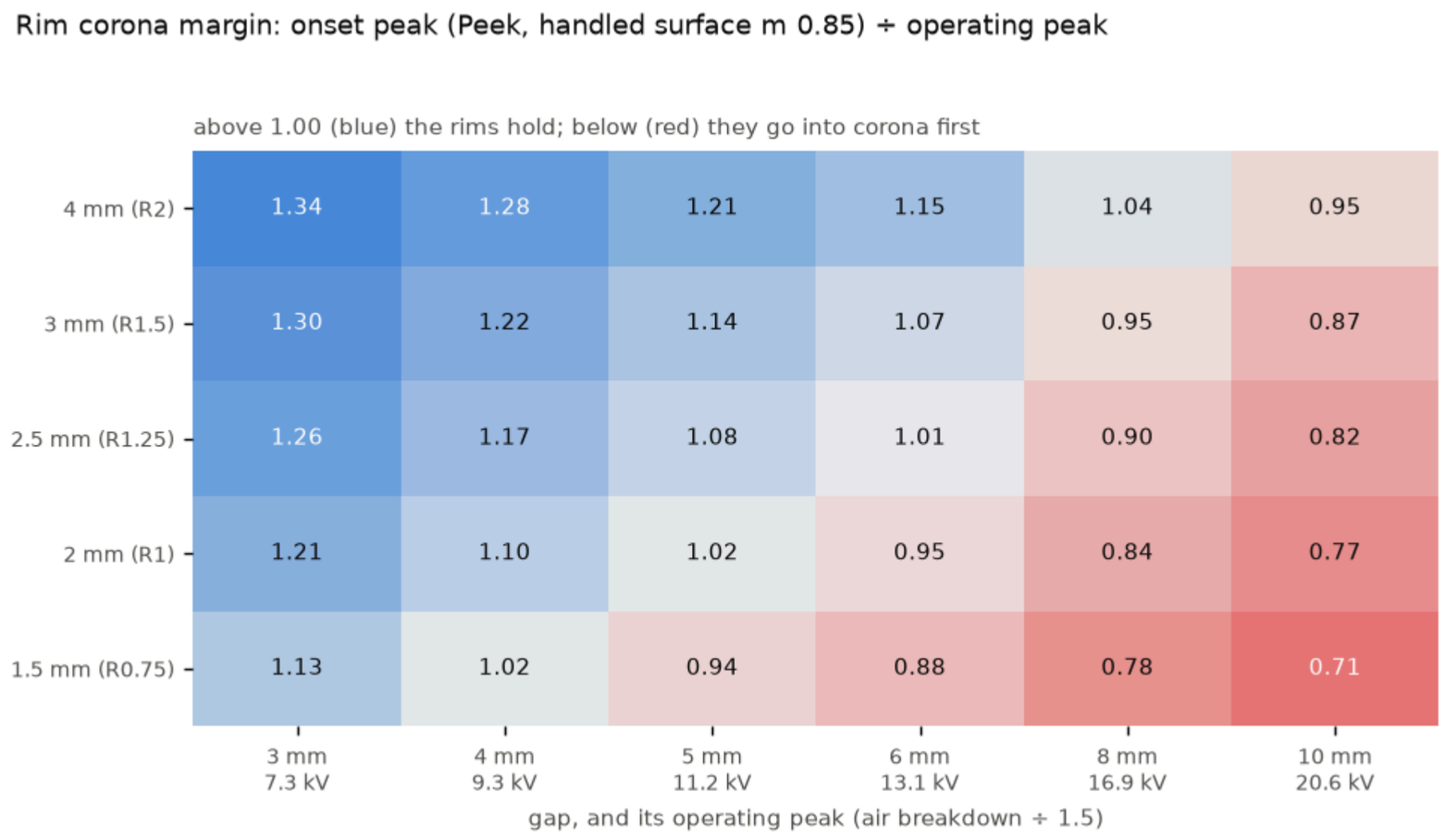
The vane matrix: what the radius buys under the 6 + 6 cap

power, gain, stack length and aluminium against the vanes' outer radius



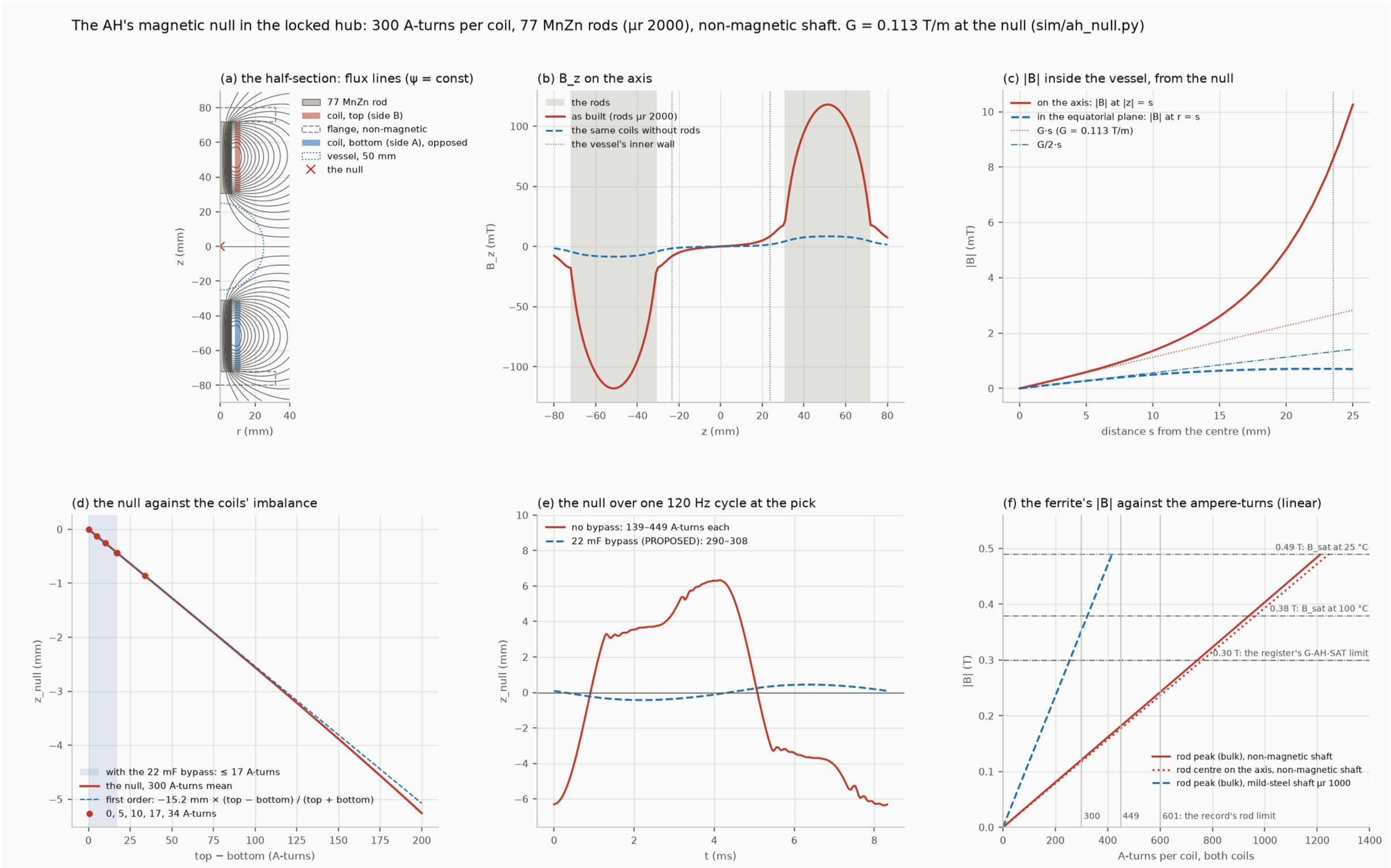
The vane matrix: rim corona margins

the rim's corona margin against gap and vane thickness



The AH's null in the locked hub

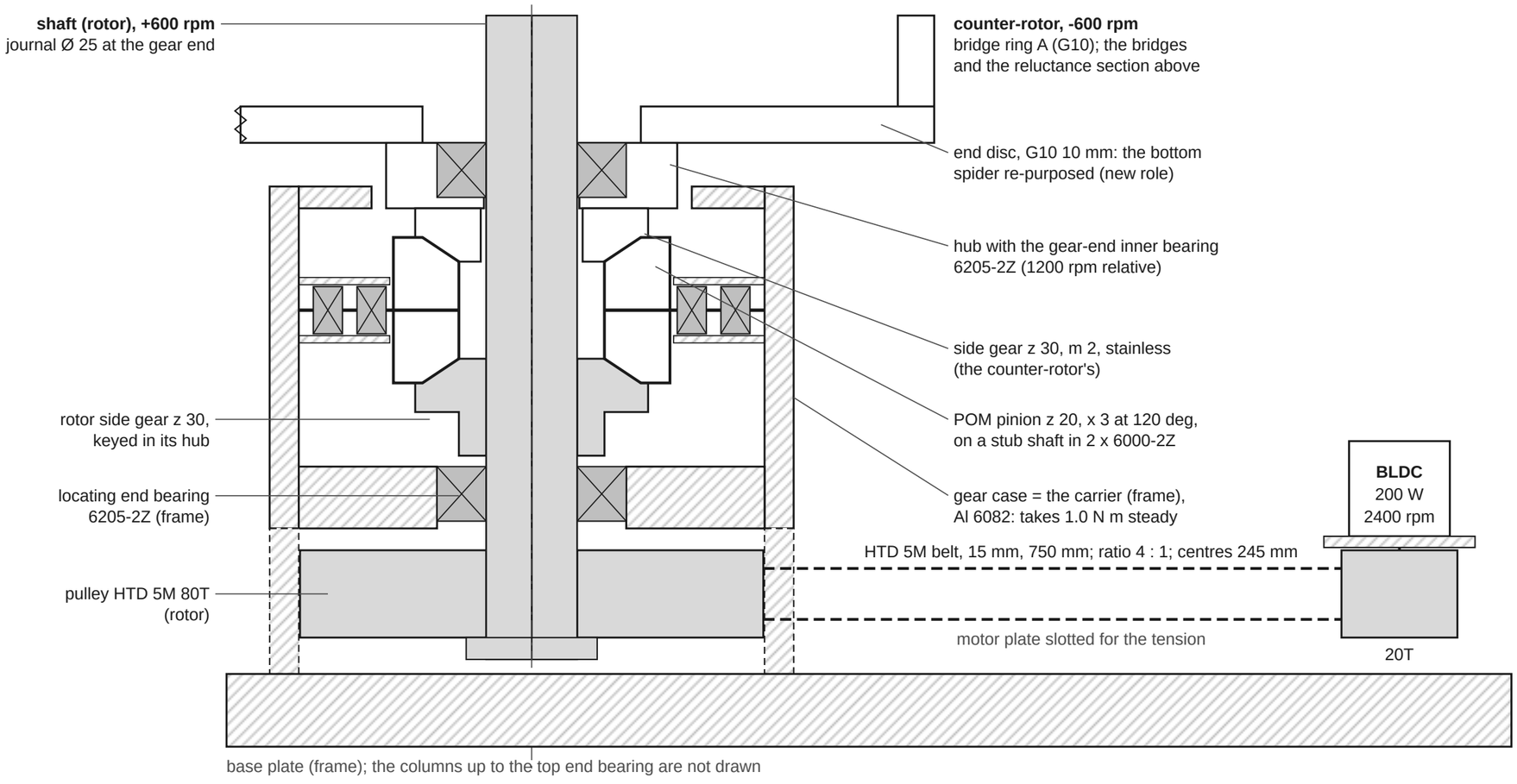
the cusp's field, the null against the coils' imbalance and over the 120 Hz cycle with and without the bypass, the rods' flux, the shaft's permeability



the first cut: a bevel reverser at side A, its carrier held by the frame, three POM-C pinions; the HTD belt and the motor

The 1 : -1 bevel reverser at the frame end (side A, below), and the belt drive

PROPOSED. Kinematic section through the axis, drawn by sim/drive_sizing.py --figure. Heights to scale; radii to scale out to r 80 mm and compressed beyond (right half); the left half is broken at r 80.
Fills: rotor light grey; counter-rotor white; frame hatched; bearings crossed. The motor's offset is drawn shortened.



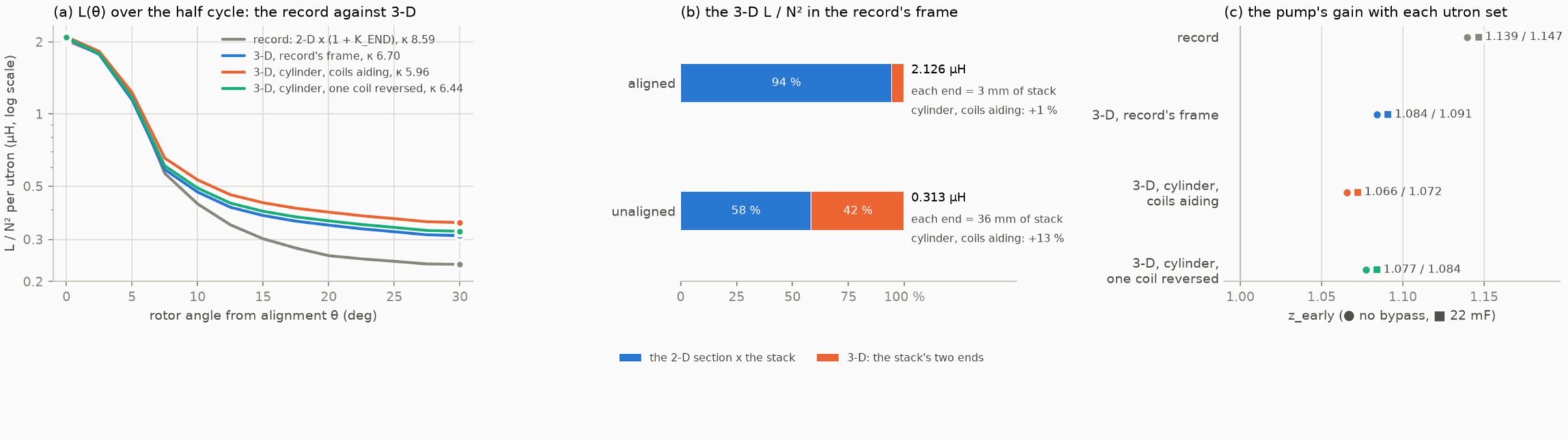
Steady, 600 rpm each way
pumps between the bodies 0.171 N m (1200 rpm relative)
swinging -0.71 .. 1.45 N m at 120 / 240 Hz
with bearings and 25 W of windage 0.40 N m
gear on each body 0.48 N m; frame (carrier) 1.00 N m

Drive
belt at the shaft 0.94 N m, 59 W; motor 61 W
run-up to 600 rpm 32 s at the motor's rated torque
gear peak 1.6 N m (run-up); ripple 0.06 N m p-p
the ripple stays in the bodies: the gear-path mode is 20 Hz

The utrons in 3-D: their ends, their coupling and the pump

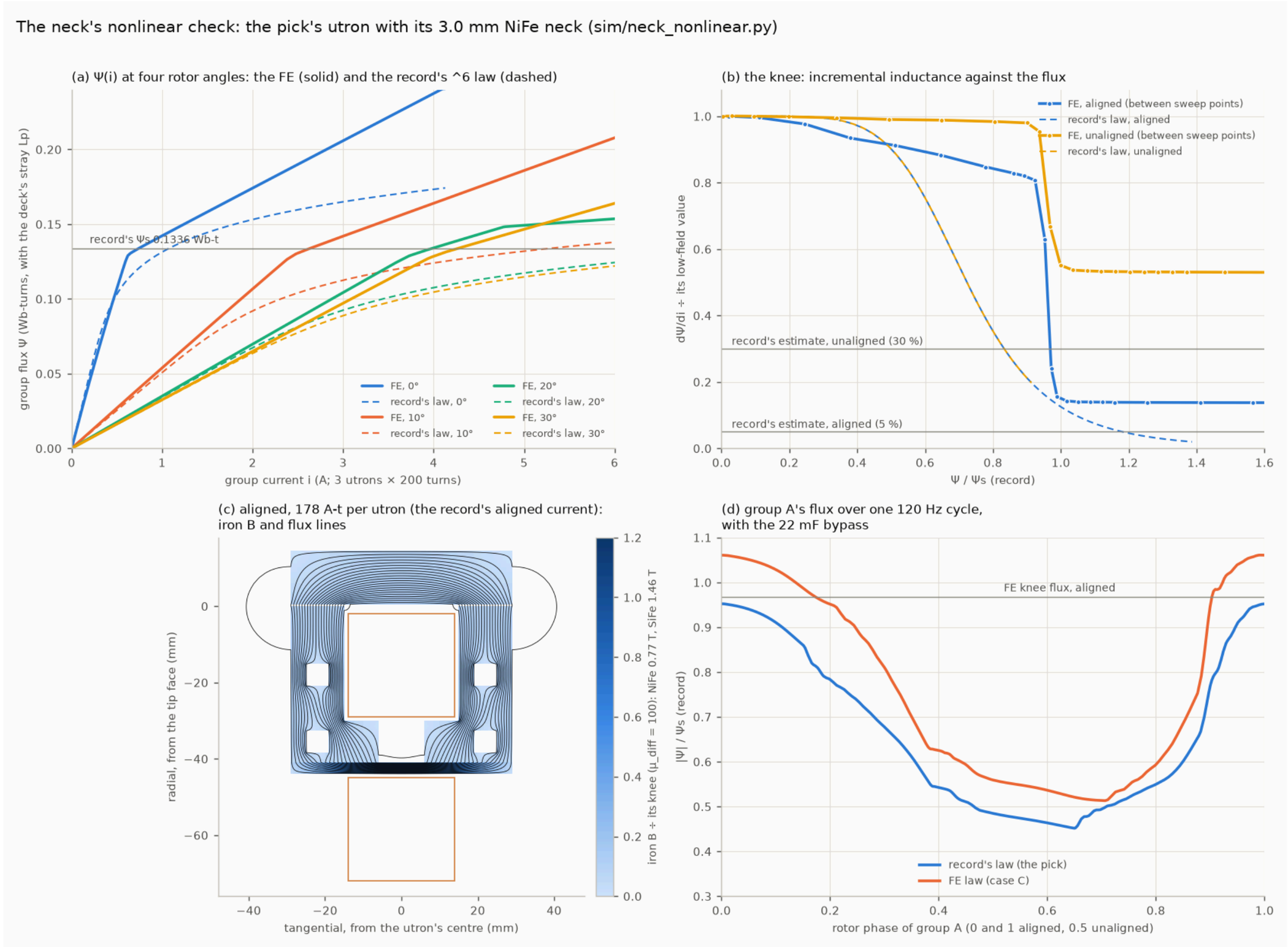
κ in 3-D against the record's 2-D section, where the flux goes, the coils' coupling round the machine, and the pump with each utron set (model check 34)

The pick's utron in 3-D (sim/utron_3d.py): kappa, where the flux goes, and the pump



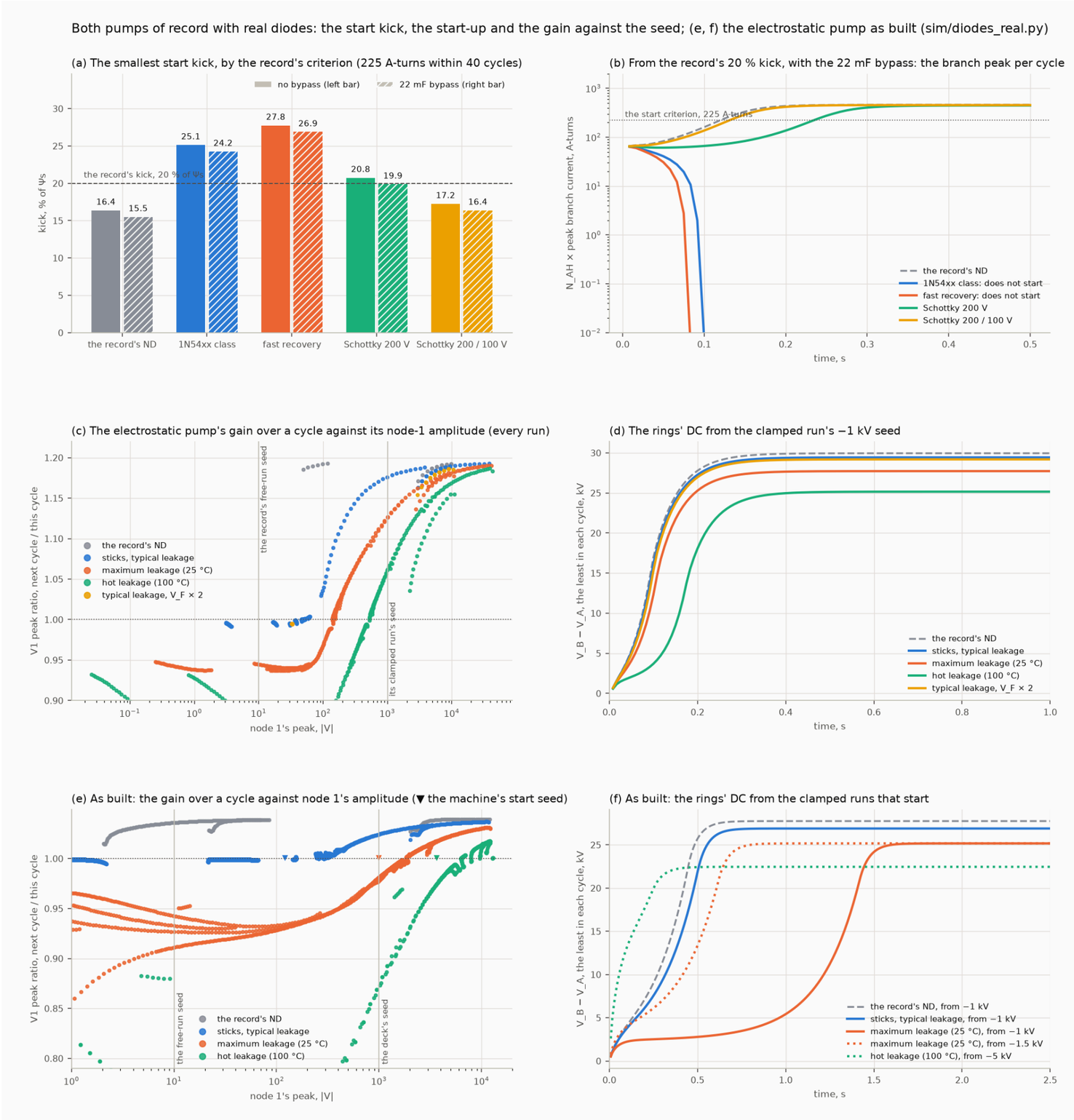
The neck in a nonlinear field solve

Ψ(NI) of the built utron, where the iron saturates, and the pump with the field map's law (model check 31)



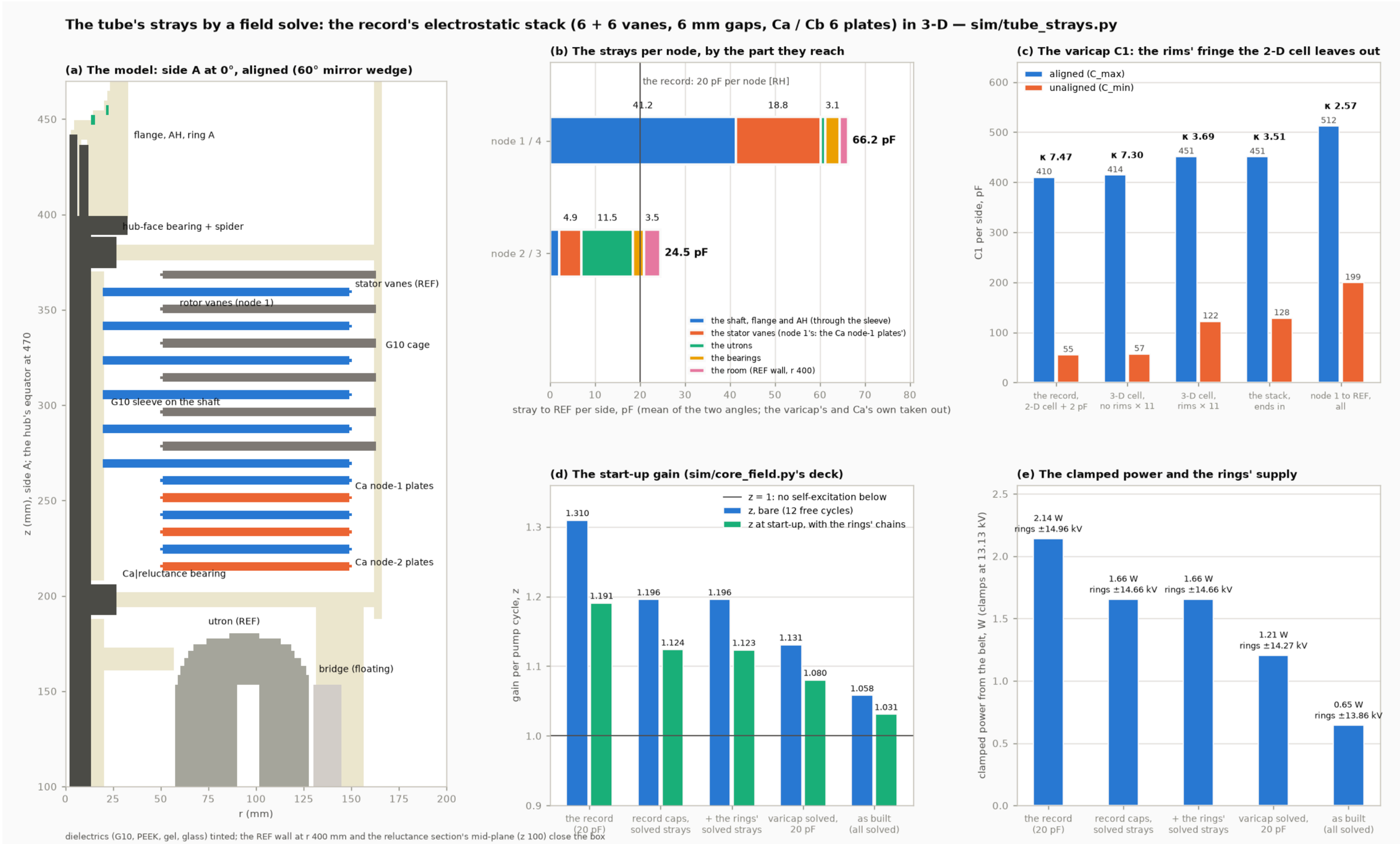
Real diodes in both pumps

the kick's threshold by rectifier class, the electrostatic gain against the seed, and the rings with real HV sticks (model check 32)



The tube's strays by a field solve

the strays per node and where they come from, the varicap with its rims, and the pump as built (model check 33)

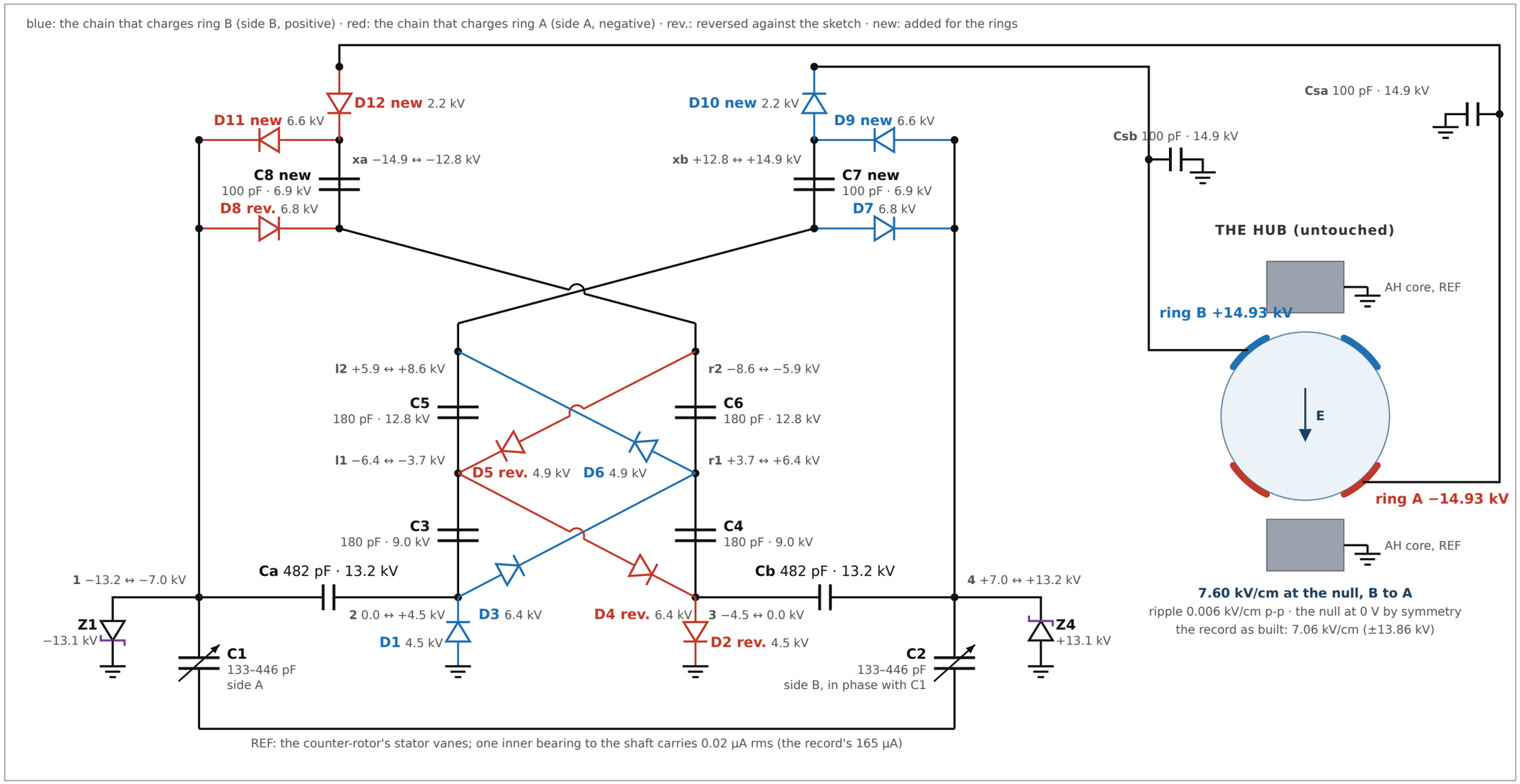


The multiplier stacked into the doubler (PROPOSED, the redesign of 2026-10-10)

the designer's ladder C3-C6 / D5-D8 run bipolar with C1 / C2 in phase, each chain one stage past its varicap node to its ring; every node's swing, reverse peak and DC as built; against the record's supply

DCCREG turbine: the multiplier stacked into the doubler, run bipolar, feeding the rings — PROPOSED (2026-10-10)

The designer's ladder C3–C6 / D5–D8 as sketched; chain 2 reversed (D2, D4, D5, D8): side A negative, side B positive; C1 / C2 in phase; one stage past each varicap node to its ring. As built, ideal diodes.



circuit	the designer's ladder (C3–C6 180 pF, D5–D8), bipolar: chain 2 reversed; C1 / C2 in phase (in antiphase it does not pump); each chain one stage past its varicap node to its ring
levels	nodes 1 / 4 at the clamps' −13.2 / +13.2 kV (V_op of record); the rings −14.93 / +14.93 kV, 7.60 kV/cm at the null (the record as built ±13.86 kV, 7.06 kV/cm)
diodes	12 HV sticks, one a position, reverse ≤ 6.8 kV (the record: 14, its D3 / D4 13.2 kV on two each)
capacitors	Ca / Cb now hold 13.2 kV (the record's 6.9); C5 / C6 12.8, C3 / C4 9.0, C7 / C8 6.9, Csa / Csb 14.9 kV
pump	z 1.0142 as built (the record with its chains 1.0322); the clamps 630 mW; 95 % in 1.45 s; the counter-rotor can float (Q1 = −Q2)
real sticks	seeds 51 V / 947 V / 4870 V (typical / maximum / hot; the record 122 V / 1.0 kV / 3.65 kV); the rings ±14.67 / ±14.40 / ±13.95 kV, 7.47 / 7.33 / 7.10 kV/cm
open	the new nodes' strays (24.5 pF each assumed) set the rings: with none they reach ±16.3 kV; trim the clamp strings on the bench before the hub sees it

Numbers: sim/integrated_multiplier_results.json (pick, as_built, real) · findings: sim/integrated-multiplier-findings.md · generator: docs/make_integrated_multiplier_schematic.py

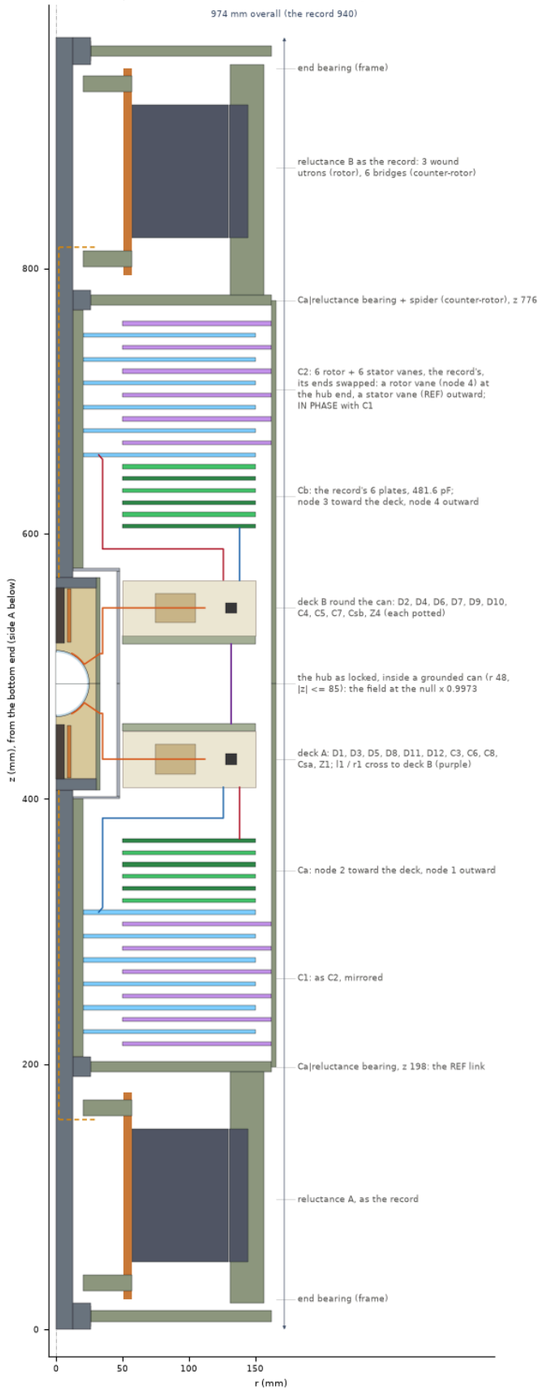
The multiplier laid out in the machine (PROPOSED, 2026-10-11)

the half-section without the hub-face bearings; side A's central bay with its gaps; deck A in plan (the sticks, C3 / C6 / C8, Csa and the clamp string round the grounded can); the parts, the interfaces, the strays, the clamp trim and the shafts

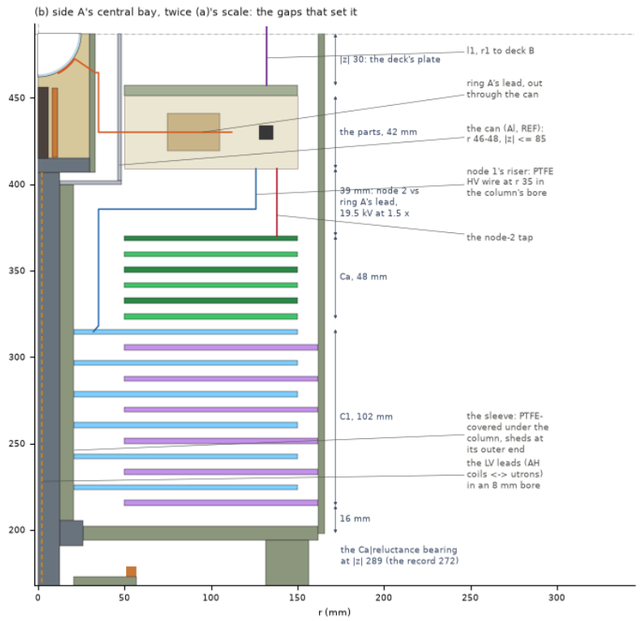
The multiplier stacked into the doubler: the mechanical layout (PROPOSED, 2026-10-11) — Ca / Cb in each side's column, every diode and the multiplier's capacitors on a potted deck round a grounded can over the hub

the circuit: sim/integrated_multiplier.py's proposal at its pick (C1 / C2 in phase, chain 2 reversed, one ring stage per chain, C3-C6 180 pF); the layout and its checks: sim/multiplier_layout.py; the record's palette (sim/tube_geometry.py)

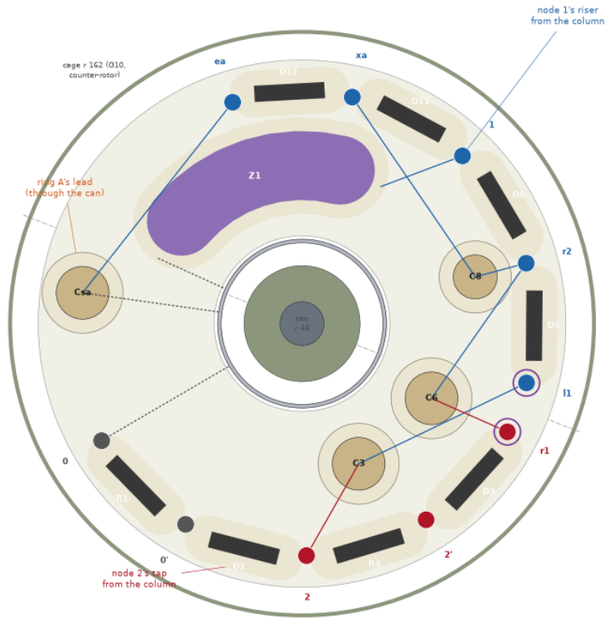
(a) the machine through the shaft, side B above: no hub-face bearings, the counter-rotor on the two Ca|reluctance bearings; amber dashed: the magnetic pump's LV leads in the shaft



(b) side A's central bay, twice (a)'s scale: the gaps that set it



(c) deck A in plan: 13 parts potted, 78 pairs checked: all clear bare and potted; 1.85 kg, its centroid 13 mm off the axis (balanced with the rotor). Junctions coloured by sign (blue −, red +)



deck A from the hub's side; deck B is its mirror about the dash-dot line (its nets mirrored: 1 -> 4, 2 -> 3, l1 <-> r1, r2 -> l2, xa -> xb, ea -> eb), so l1 and r1 (purple rings) run straight across between the decks

(d) the parts, the interfaces and what the layout does (sim/multiplier_layout_results.json)

the parts, each potted (>= 8 mm of silicone)

deck A	deck B	part	duty (as built)	rating
D1	D2	HV stick, 2CL77 class	4.52 kV reverse	20 kV (x2 rule)
D3	D4	HV stick, 2CL77 class	6.41 kV reverse	20 kV (x2 rule)
D5	D6	HV stick, 2CL77 class	4.95 kV reverse	20 kV (x2 rule)
D8	D7	HV stick, 2CL77 class	6.77 kV reverse	20 kV (x2 rule)
D11	D9	HV stick, 2CL77 class	6.64 kV reverse	20 kV (x2 rule)
D12	D10	HV stick, 2CL77 class	2.16 kV reverse	20 kV (x2 rule)
C3	C4	180 pF, class 1 / film	9.0 kV DC	30 kV
C6	C5	180 pF, class 1 / film	12.8 kV DC	30 kV
C8	C7	100 pF, class 1 / film	6.9 kV DC	30 kV
Csa	Csb	100 pF to REF	14.9 kV DC	40 kV
Z1	Z4	60 x 1.5KE200A, potted	V_op 11.94 kV (66 in the record)	200 V per part
R1, R3	R2, R4	22 kOhm surge	6.6 kV step	>= 15 kV, 1 W

the interfaces (IEC 60664-1 as sim/parts_first_cut.py; clearance at 1.5 x the peak, creepage x 1.25)

interface	peak / rms kV	clearance	creepage PTFE / G10	has	
Ca's plates, node 1 / 2 (6 mm air)	13.2 / 12.3	25 mm	77 / 152 mm	6 mm	x 1.49 to breakdown
the vanes, node 1 / REF (6 mm)	13.2 / 9.9	25 mm	62 / 123 mm	6 mm	x 1.49 to breakdown
the column's node-2 plate <-> the can's end disc (REF)	4.5 / 3.1	6 mm	20 / 39 mm	30 mm	ok
the column's node-2 plate <-> the deck's parts (their potted outer f...	19.5 / 17.7	39 mm	110 / 221 mm	39 mm	ok
node 1's riser in the column's bore <-> the node-2 plates' inner rims	13.2 / 12.3	25 mm	77 / 152 mm	13 mm	insulated
the decks' outermost housings <-> the cage (r 162; O10 on the counte...	14.9 / 14.9	29 mm	93 / 187 mm	13 mm	ok
deck A <-> deck B, across the equator (ring stage to ring stage)	29.9 / 29.9	67 mm	187 / 373 mm	72 mm	ok
the cross nets l1 <-> r1	12.8 / 10.5	24 mm	66 / 131 mm	—	insulated
the cross nets <-> the can (REF)	6.4 / 5.2	10 mm	33 / 66 mm	82 mm	ok
a ring's lead <-> the can's wall (bushing)	14.9 / 14.9	29 mm	93 / 187 mm	—	insulated
ring A's lead <-> ring B's lead	29.9 / 29.9	67 mm	187 / 373 mm	—	insulated
the O10 sleeve, hub side: the first rotor vane's ring (node 1) <-> t...	13.2 / 9.9	25 mm	62 / 123 mm	84 mm	O10 no, PTFE ok
the O10 sleeve, outer side: the last rotor vane's ring (node 1) <-> ...	13.2 / 9.9	25 mm	62 / 123 mm	18 mm	O10 no, PTFE no

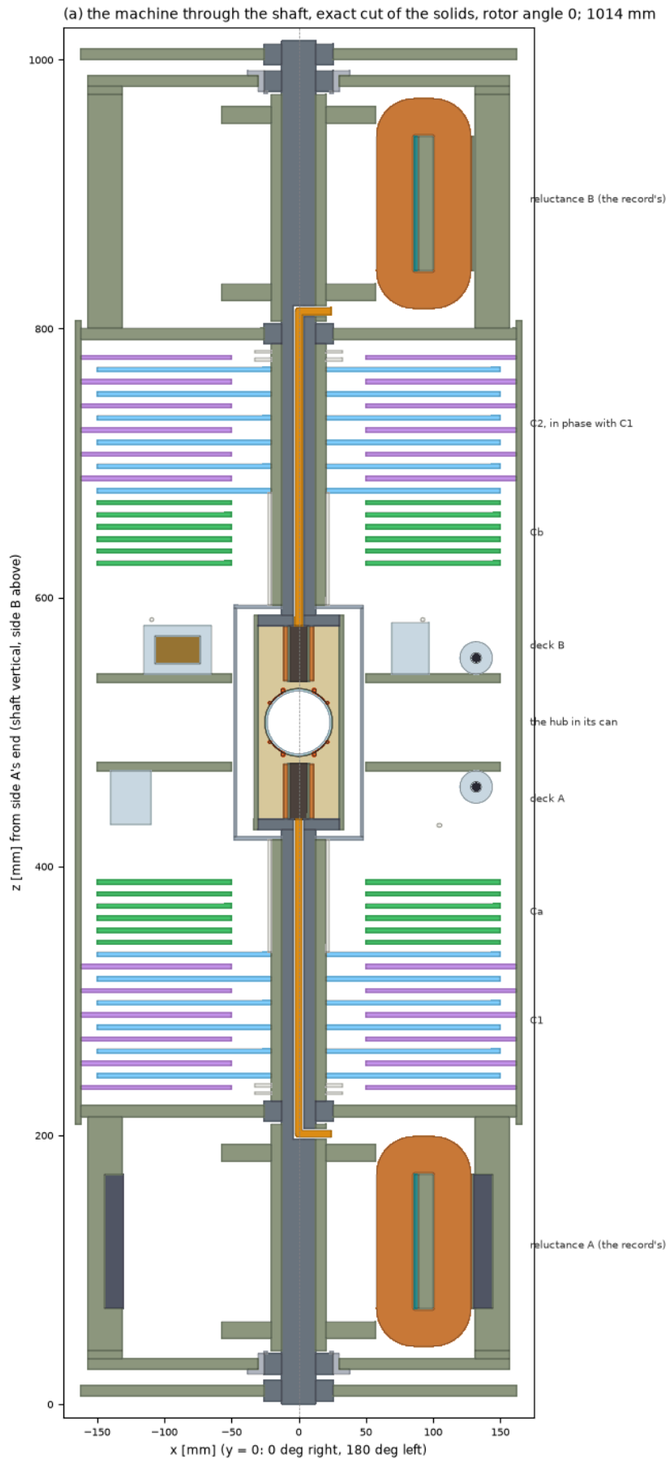
what it does

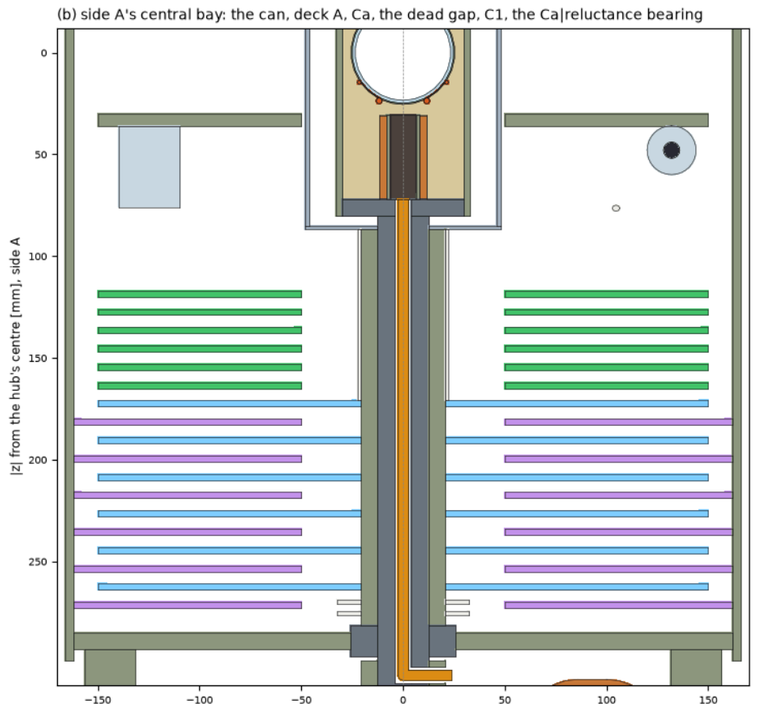
- length 974 mm (the record 940): +17 mm a side; the decks sit where the hub-face bearings were
- strays (estimated, [RH]): node 1 / 4 +4.6 pF, node 2 / 3 -10.8 pF, each new node 1.1-3.3 pF (the study assumed 24.5)
- so the free gain z 1.0164 against the study's 1.0142, and the rings rise to ±16.25 kV; C3-C6 at 100-180 pF gives ±16.08-16.26, C7 / C8 at 47-68 pF ±16.19-16.22
- so trim the clamps: strings of 60 parts (V_op 11.94 kV) hold the rings at ±14.77 kV, the most within the record's ±14.96
- shafts: without the hub-face bearings d 25 drops to 14 Hz (the record 29), d 40 to 26; with the bridge-ring bearings 89 Hz (stiff) / 85 Hz (realistic seats); the criterion 60 Hz
- the hub's field with the grounded can: x 0.9973; each ring's capacitance to REF 6.05 -> 7.21 pF
- the sleeve (red rows): PTFE over it under the column (84 mm against 62); at its outer end two 12 mm PTFE sheds (17.5 + 4 x 12 = 65.5 mm) — the record's own open point, moved there

The layout in solids: the section and the decks (PROPOSED, 2026-10-11)

the layout's 386 solids cut through the shaft, side A's central bay enlarged, both decks cut in plan at the sticks' height, and side A's end with its bridge-ring bearing: every part of the study's plans, the can, the column, the vanes with C2 in phase

tube-r150-n6-air6-wound-g0p5-6br-hub50-layout: the multiplier laid out in the machine

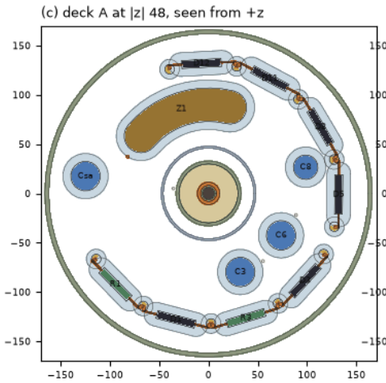


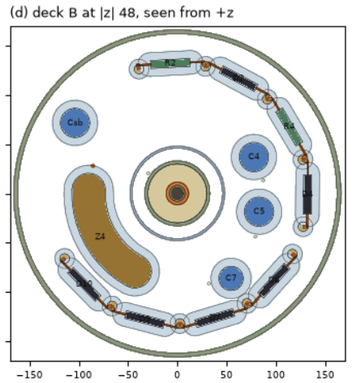


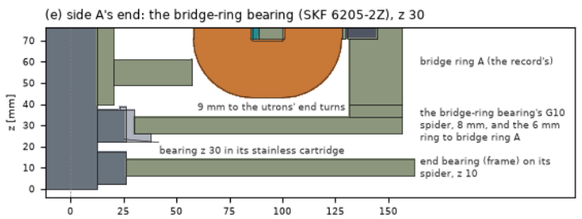


The layout of sim/multiplier-layout-findings.md (PROPOSED, 2026-10-11) in solids: every part of the decks from the study's plans, the record's hub, column plates, vanes and reluctance sections; the wiring as sim/tube_geometry.py LayoutMachine routes it; the bridge-ring bearings as sim/bridge_ring_bearings.py places them (PROPOSED).

Not drawn: Ca / Cb's rods and mounts, the decks' standoffs to the can, the plates' interconnections, the gear.



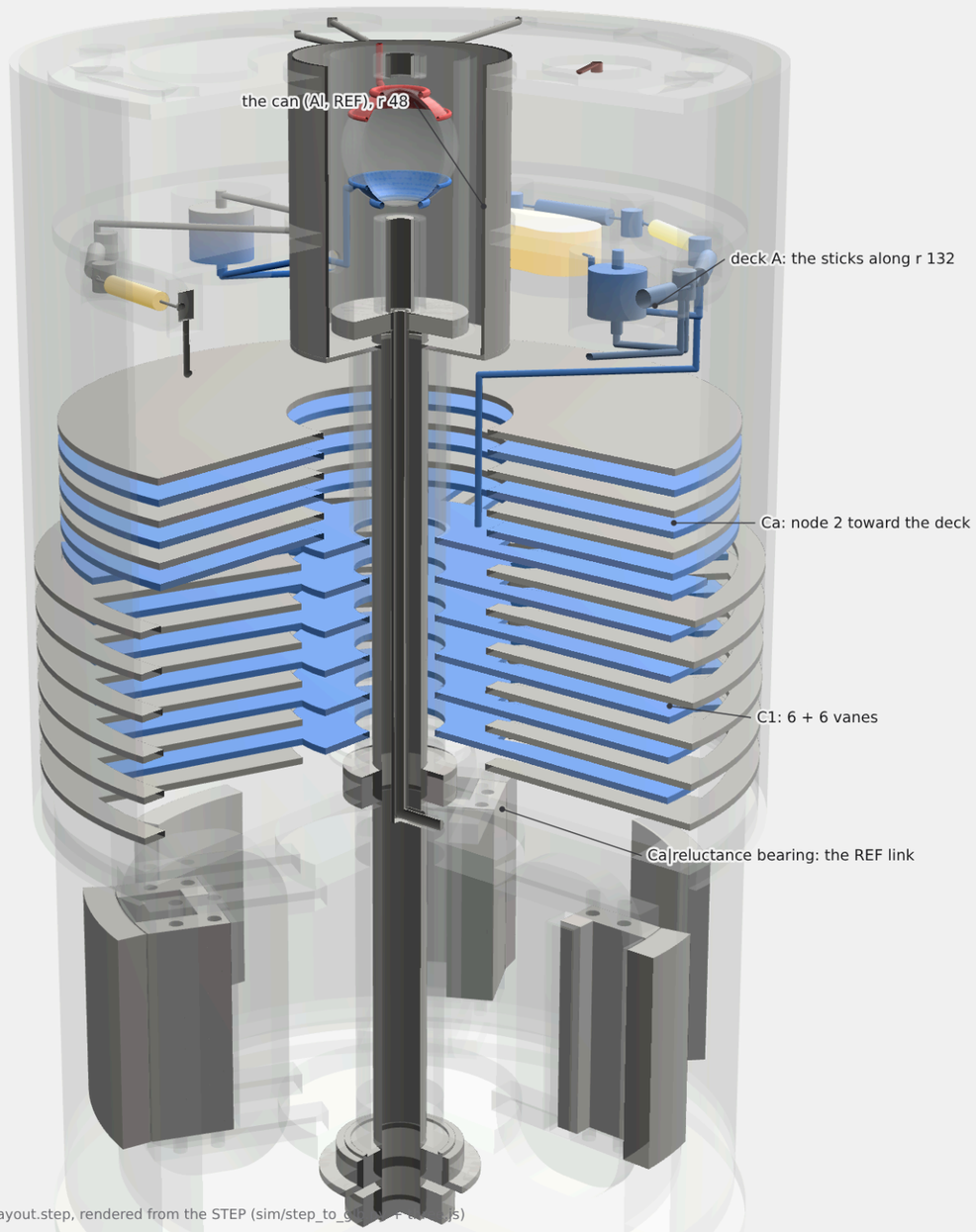




The layout in 3-D, its circuit at 25.5 deg of the cycle (PROPOSED, 2026-10-11)

side A's central bay, quarter cut: every conductor by its potential (blue -15 kV, grey REF, red +15 kV), the diodes and the clamp lit while they conduct, insulators faded; the circuit of sim/multiplier_waveforms.py

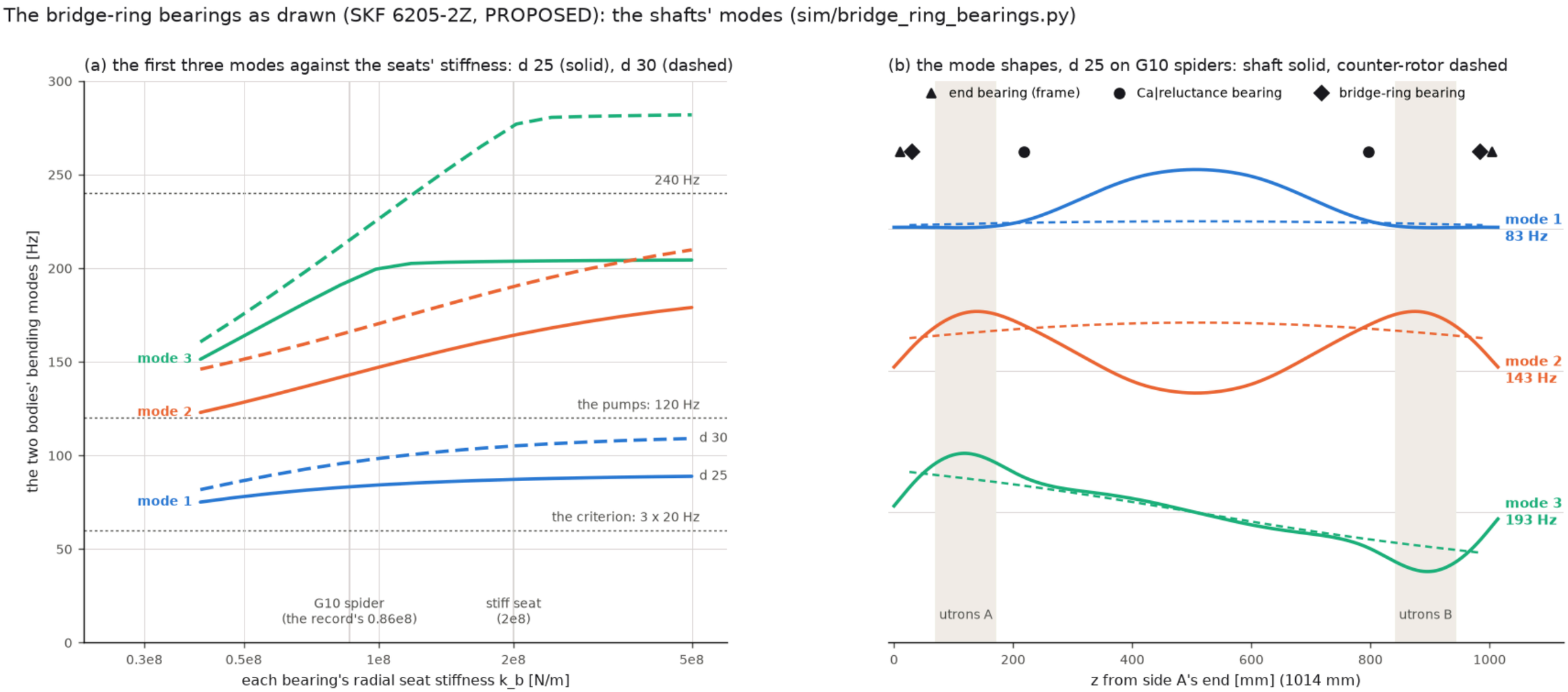
Side A's central bay, quarter cut: the hub in its can, deck A, Ca, the dead gap, C1, the Ca|reluctance bearing. the circuit at 25.5 deg of its cycle (sim/multiplier_waveforms.py): every conductor by its potential, blue -15 kV, grey REF, red +15 kV; lit amber: a diode or clamp conducting; insulators faded



docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50-layout.step, rendered from the STEP (sim/step_to_[illegible])

The bridge-ring bearings: the shafts' modes with them as drawn (SKF 6205-2Z, PROPOSED, 2026-10-11)

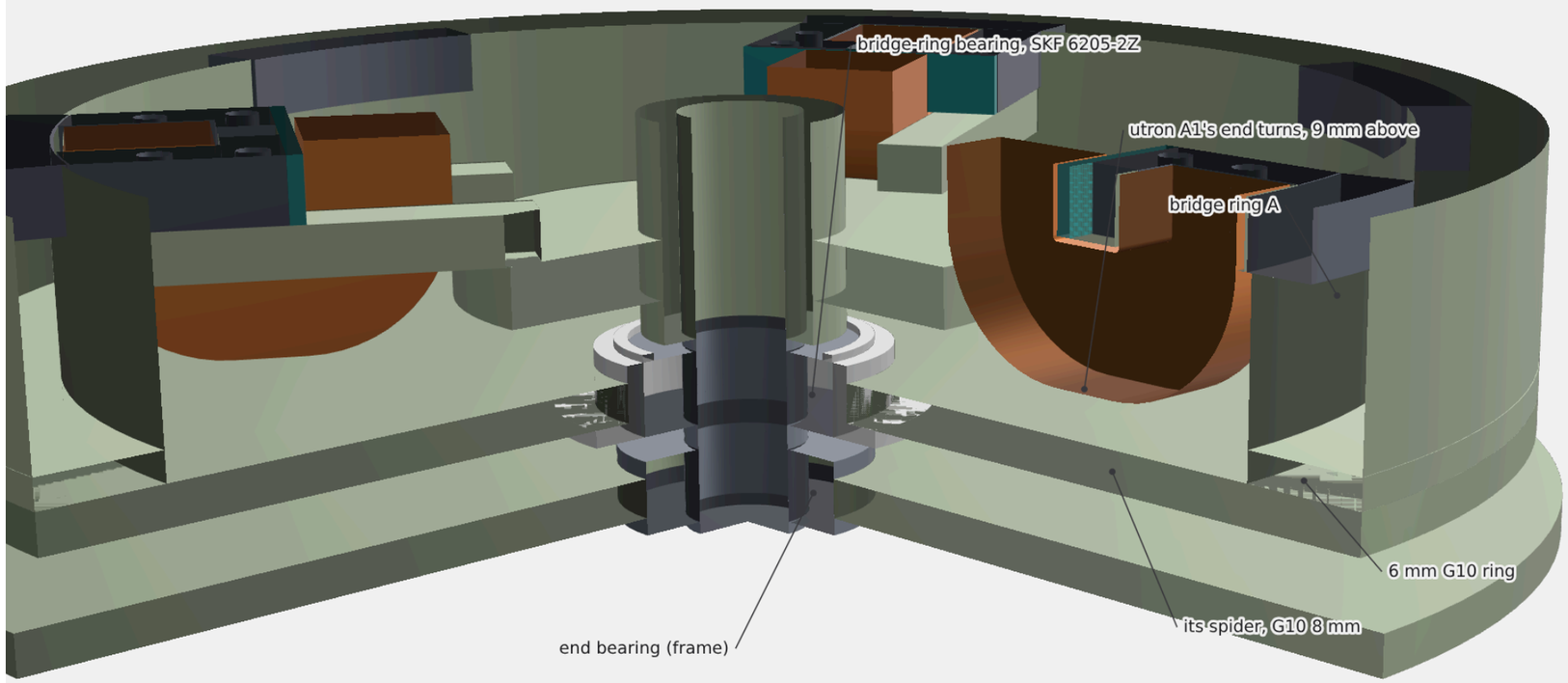
the first three modes of the shaft and the counter-rotor against the seats' stiffness at d 25 and d 30, with the 60 Hz criterion and the pumps' 120 / 240 Hz; the mode shapes at d 25 on G10 spiders



Side A's end in 3-D: the end bearing and the bridge-ring bearing (PROPOSED, 2026-10-11)

quarter cut: the frame's end bearing on its spider; the bridge-ring bearing in its stainless cartridge on an 8 mm G10 spider, the 6 mm ring to bridge ring A, utron A1's end turns 9 mm above

Side A's end, quarter cut: the end bearing (frame), and the bridge-ring bearing (SKF 6205-2Z, PROPOSED) in its cartridge on its spider under bridge ring A

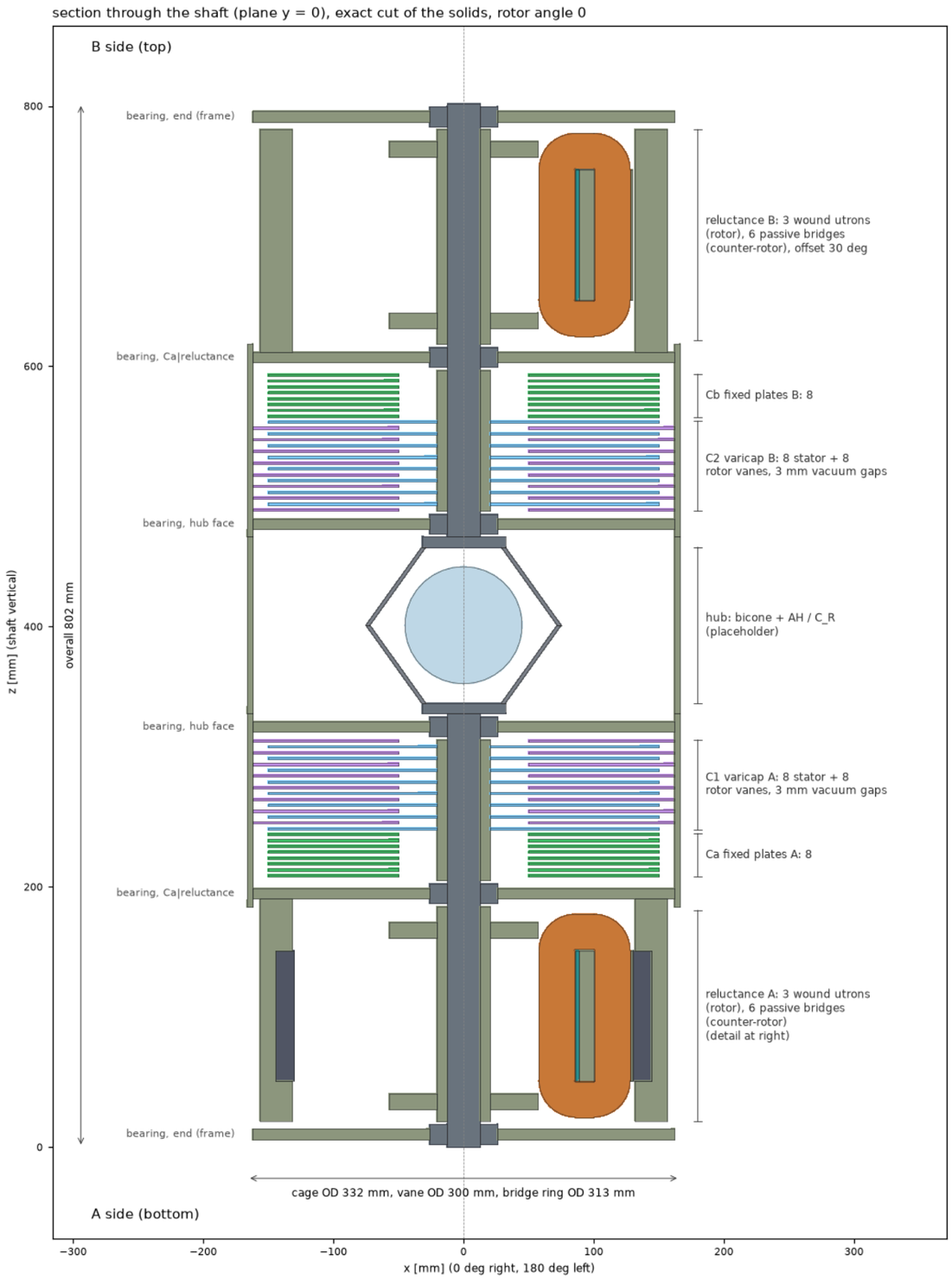


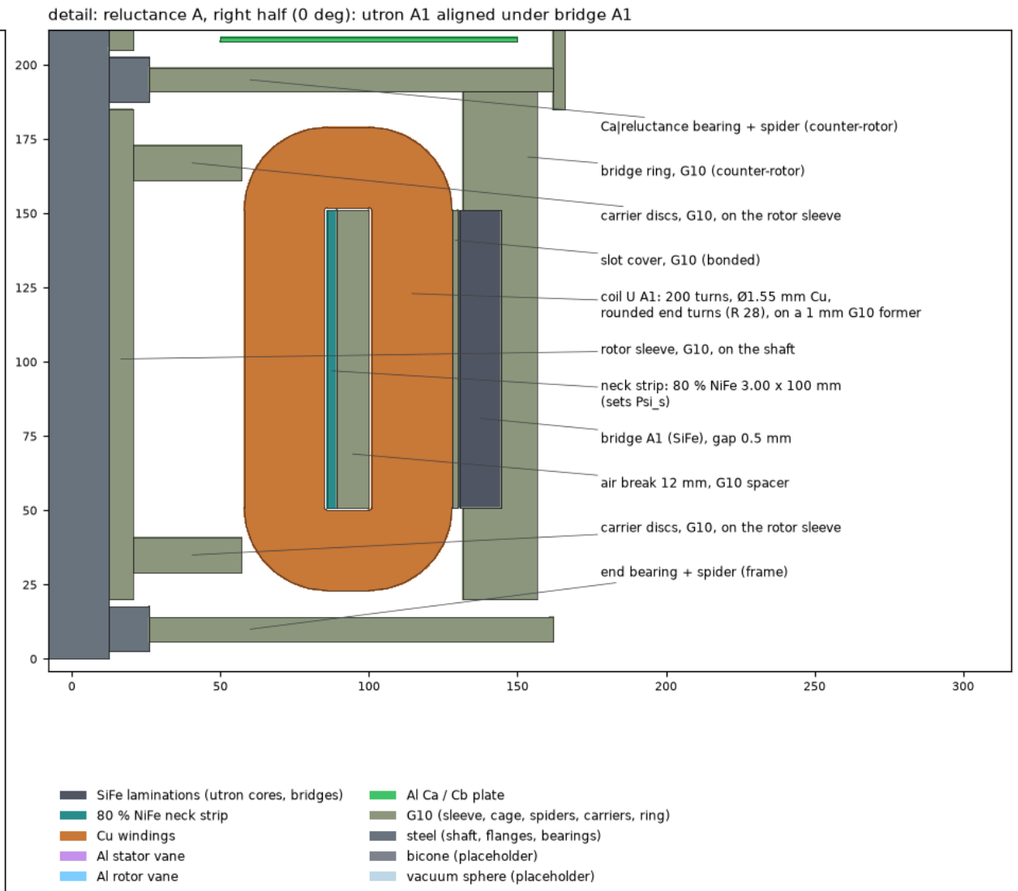
docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50-layout.step, rendered from the STEP (sim/step_to_glb.py + three.js)

The machine as modelled before the record: section

superseded by the record in solids: the old vacuum 8 + 8 stack, Ca / Cb on the counter-rotor, the 120 mm placeholder hub (802 mm tube); its reluctance sections are the record's

tube-r150-n8-wound-g0p5-6br: arrangement (g 0.5 / 6 bridges / 1200 rpm), 802 mm long





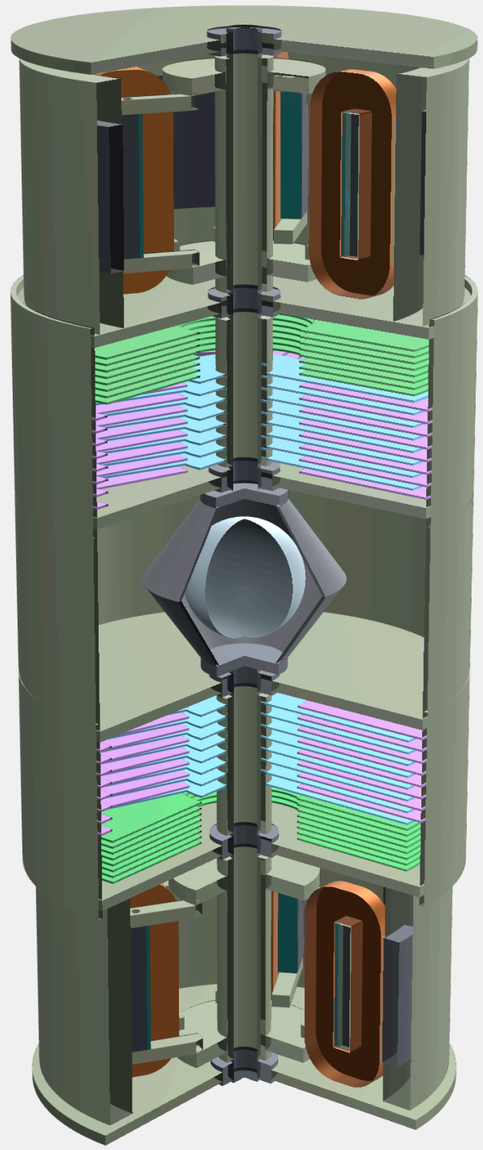
SiFe laminations (utron cores, bridges)
80 % NiFe neck strip
Cu windings
Al stator vane
Al rotor vane
Al Ca / Cb plate
G10 (sleeve, cage, spiders, carriers, ring)
steel (shaft, flanges, bearings)
bicone (placeholder)
vacuum sphere (placeholder)

Bodies. Rotor: shaft halves, flanges, bicone, rotor sleeve, rotor vanes, utrons + carrier discs.
Counter-rotor, geared 1 : -1 (gear or reversing belt, not drawn): stator cage (one tube across the hub), stator vanes, Ca / Cb plates, the hub-face and Ca|reluctance spiders, bridges + bridge rings.
Frame: the two end bearings and their spiders. The inner bearings run at the relative speed.
At rotor angle 0: utron A1 is aligned (gap 0.5 mm), utron B1 sits between two B bridges (antiphase).
This plane (v = 0) cuts the coil, the neck strip and the air-break spacer; the SiFe half-cores and the cheeks lie at |v| >= s / 2 (see the plan cuts and the utron detail drawing).

The machine as modelled before the record: quarter cutaway

as the section

Quarter cutaway at rotor angle 0 (rotor + counter-rotor geared 1 : −1; frame: the end bearings)

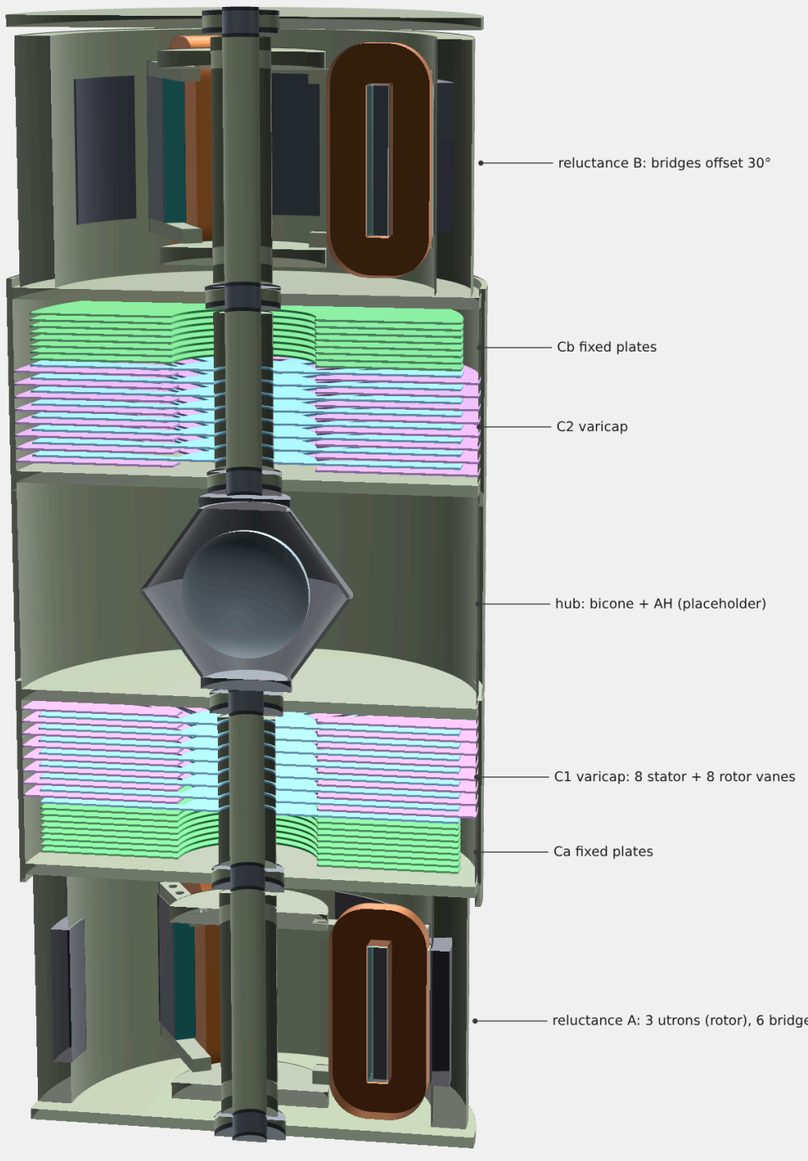


docs/geometry/tube/tube-r150-n8-wound-g0p5-6br.step, rendered from the STEP (sim/step_to_glb.py + three.js)

The machine as modelled before the record: half section

as the section

Half section (the near half removed): the machine from the front



docs/geometry/tube/tube-r150-n8-wound-g0p5-6br.step, rendered from the STEP (sim/step_to_glb.py + three.js)

Hub drive: the first two-pump schematic (2026-10-07)

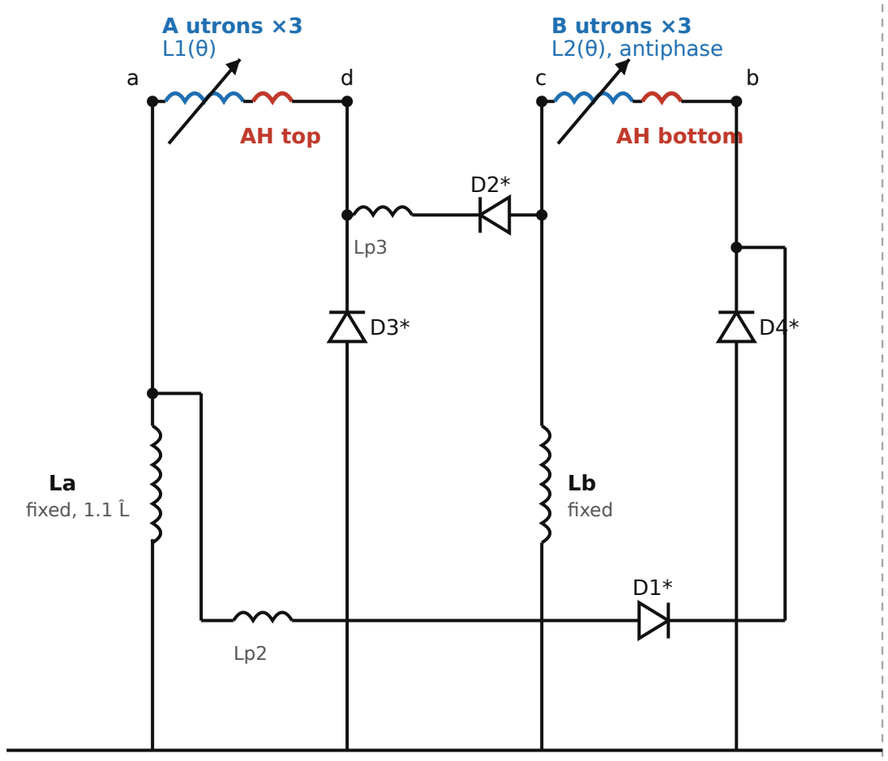
superseded by the rotor circuits: 300 rpm each way, 3 × 1150-turn toothed C-EM iron, the doubler on the stator vanes driving the bicone, a brush

Central cavity driven by two pumps — rotor and stator geared 1 : −1 (300 rpm each, 600 rpm relative)

(a) AH pair ← magnetic dual doubler on the rotor (wound utrons vs the stator's toothed C-EM iron) · (b) bicone ← electrostatic diode doubler. Only diodes switch; the gap in (b) is the optional dump test.

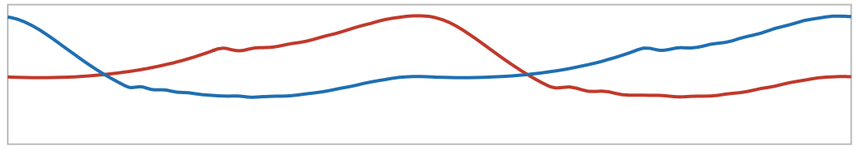
(a) MAGNETIC doubler → AH pair (pulsed, A / B in antiphase)

Diodes steer current between A and B: one group generates (L falling) while the other motors.



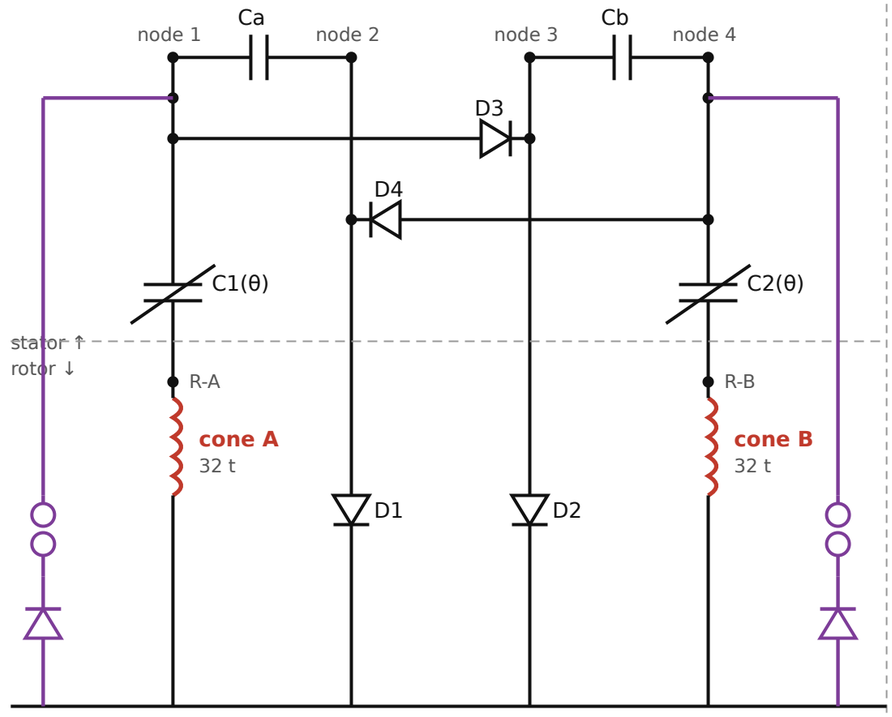
REF = rotor (shaft); everything in (a) turns with the rotor

|i| over one cycle (16.6 ms), 0 – 2.7 A: A group + AH top B group + AH bottom



(b) ELECTROSTATIC doubler → bicone cones

Limiter at nodes 1 / 4 (purple): an avalanche clamp (diodes only) OR a 20 kV spark gap with a quench diode (drawn), which dumps C1 / C2 through its cone.



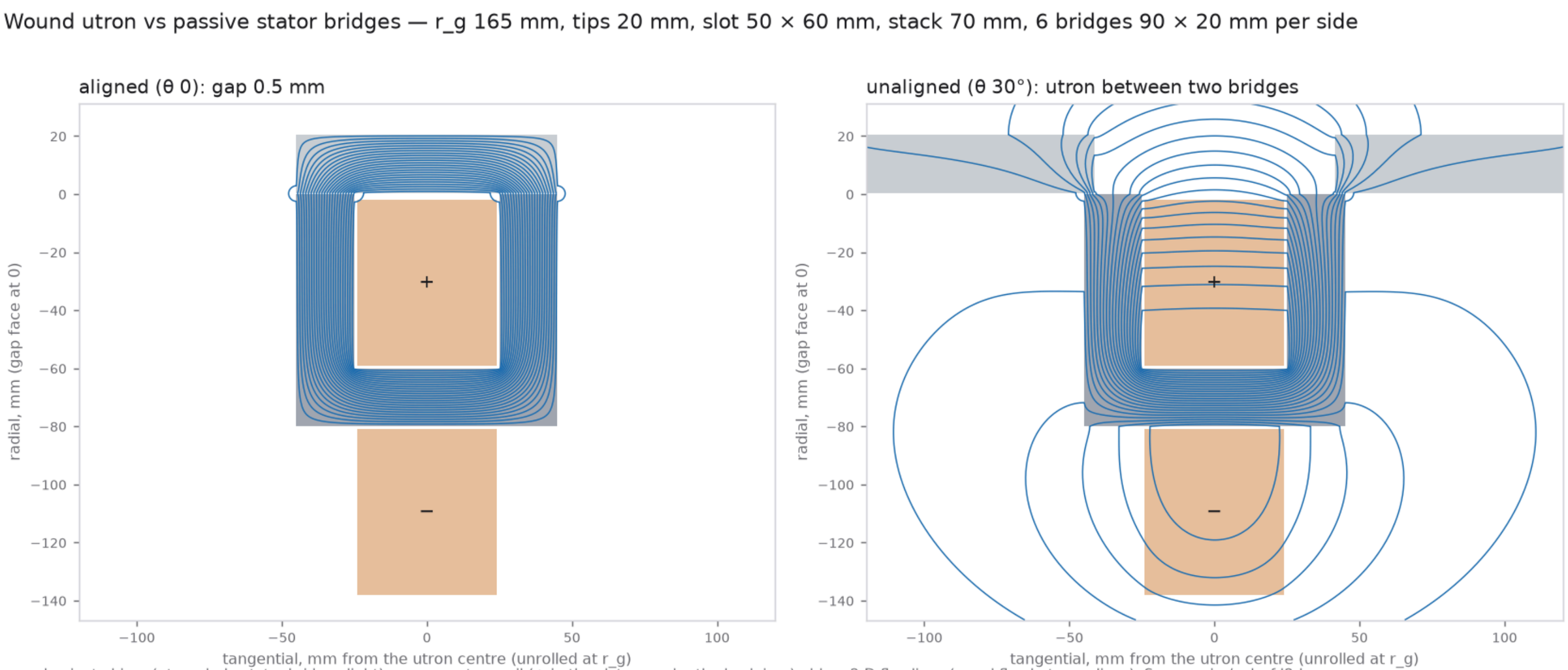
REF: shaft (rotor) ↔ stator reference through the shaft brush

diodes only: cone 1.7 mA rms (0.12 A-turns peak) — negligible
spark-gap dump: cone 21 A peak, 683 A-turns, ~60 dumps/s per side,
~54 mJ each, ring 1.8 MHz; pump belt 6.8 W at 600 rpm relative

(a) design point: L ratio 8 (toothed poles), τ = L/R 0.118 s, 3 coils × 1150 t per group, AH rewound to 160 t: utron current 1.0–2.7 A, AH 157–425 A-turns (rod limit ≈ 600).
Growth stops at iron saturation (B 1.35 T). Belt 144 W, copper 142 W on the rotor (currents scale with the core's saturation flux, power with its square).
Threshold: L ratio ≳ 3 and τ ≳ 0.08 s at 60 Hz; dual nodes carry 10 nF + 1 kΩ snubbers (winding capacitance) [IR].
(b) cones sit on the rotor side of C1 / C2 (R-A / R-B → cone → shaft): no rotating contact; the ES nodes live on the stator vanes.
Numbers: sim/magnetic_doubler_results.json, sim/bicone_drive_results.json · findings: sim/hub-drive-findings.md · generator: docs/make_schematic_hub.py

First pole pick: pole-pair flux at 0.5 mm

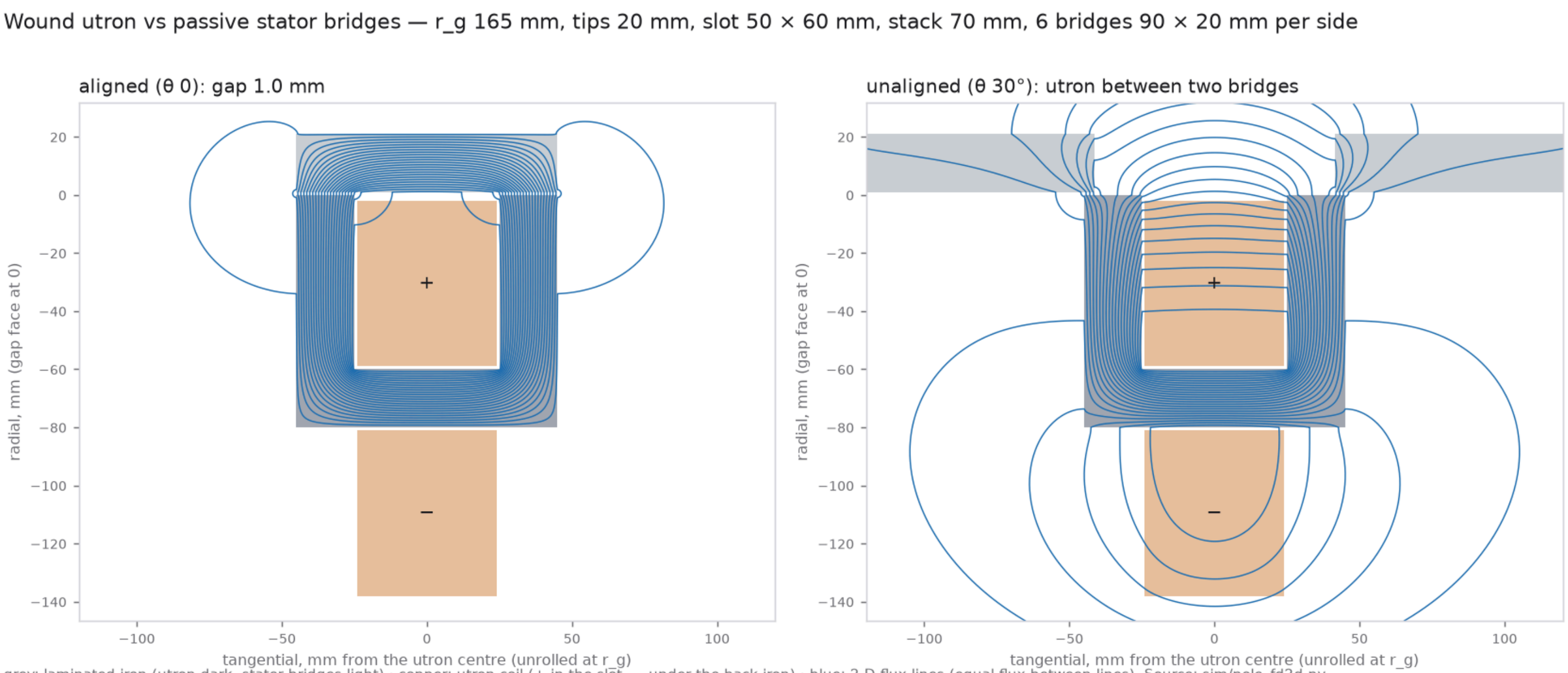
superseded geometry (r_g 165, slot 50 × 60, stack 70, bridges 90 × 20), before the mean-turn correction



grey: laminated iron (utron dark, stator bridges light) · copper: utron coil (+ in the slot, − under the back iron) · blue: 2-D flux lines (equal flux between lines). Source: sim/pole_fd2d.py

First pole pick: pole-pair flux at 1.0 mm

as the 0.5 mm sheet



grey: laminated iron (utron dark, stator bridges light) · copper: utron coil (+ in the slot, − under the back iron) · blue: 2-D flux lines (equal flux between lines). Source: sim/pole_fd2d.py

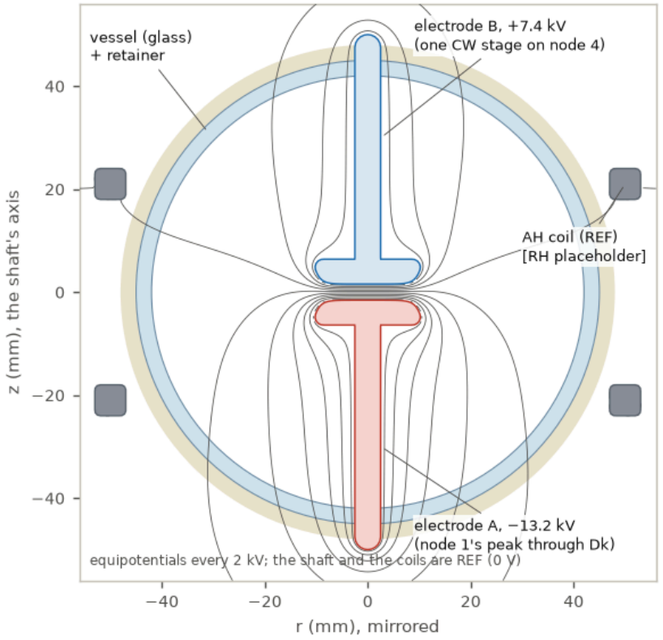
The DC pair inside the vacuum (not chosen)

Rogowski electrodes on the AH null in the placeholder hub, 65.7 kV/cm; superseded by the rings outside the glass

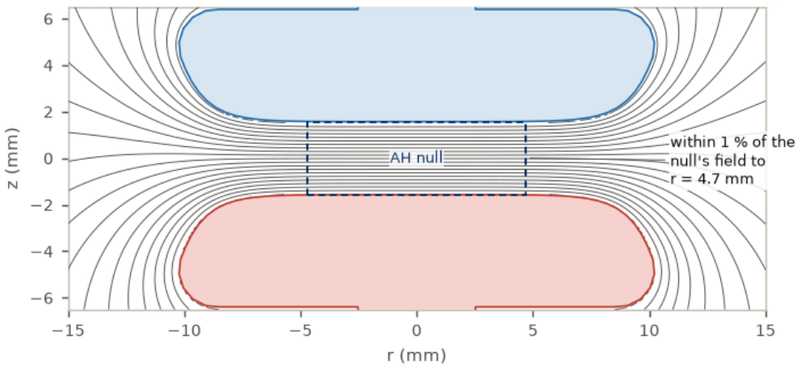
The strongest steady field at the AH null: two Rogowski electrodes in the vacuum, on the electrostatic pump's DC

Stack of record (3 mm, 6 + 6, 6 mm gaps, 13.1 kV clamp), 1200 rpm relative; held to the repo's vacuum rule, 6.67 kV/mm on any electrode surface (10 kV/mm over the 1.5 margin). Axisymmetric boundary elements (sim/core_null_field.py); the hub is the tube placeholder [RH].

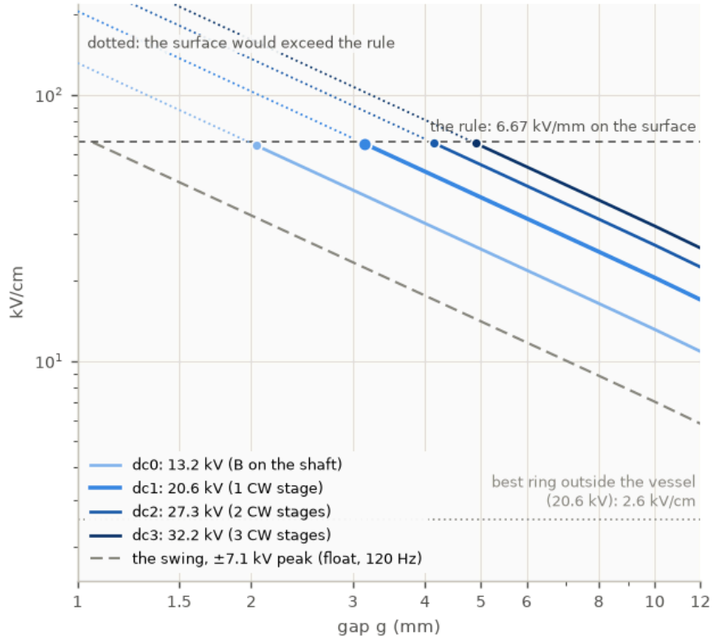
(a) the hub's section: two Rogowski electrodes on the AH null, dc1 at its design gap 3.14 mm



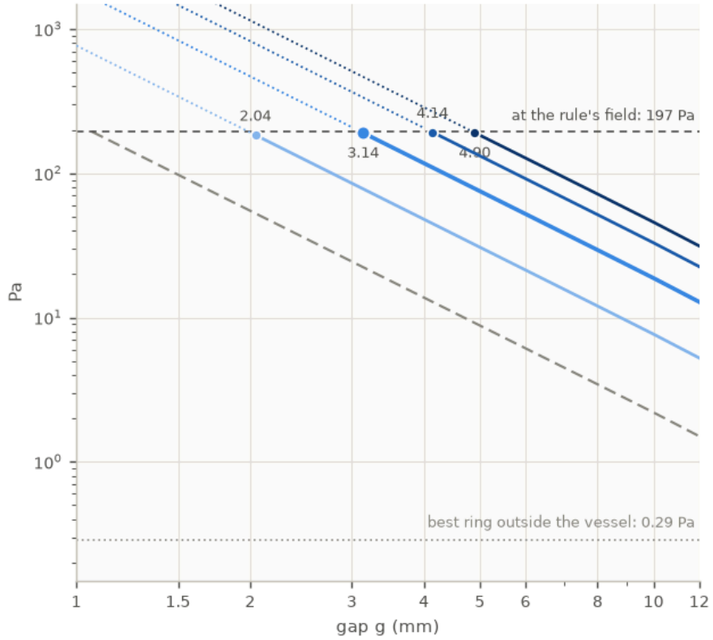
(b) the gap: 20.6 kV across 3.14 mm, 65.7 kV/cm and 191 Pa at the null; equipotentials every 1 kV; the surface peaks 1.5 % above the null, on the profile



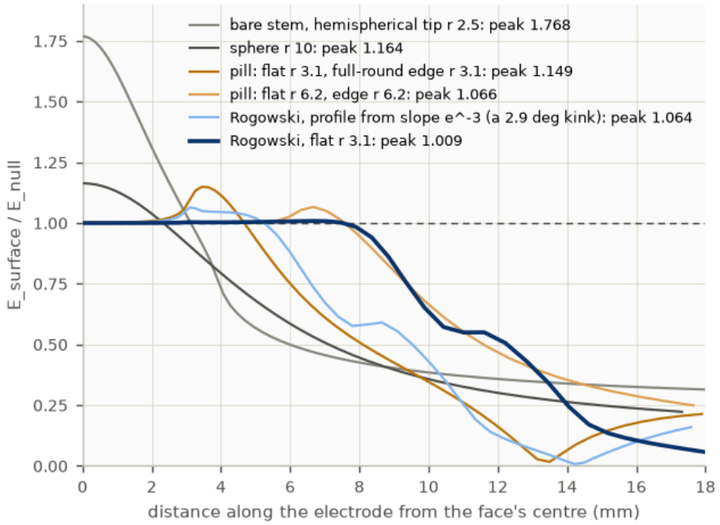
(c) the field at the null against the gap; dots: the smallest gap the rule allows



(d) the electrostatic pressure at the null, ½ ε0 E²



(e) the surface field along the electrode, over the null's (g 3.1 mm, 1 V across)



(f) the supplies: one field; the voltage buys room

	A / B	across	gap	E null	p	z start	ripple
	kV	kV	mm	kV/cm	Pa		V p-p, A / B
dc0	−13.2 / 0 (shaft)	13.2	2.04	64.5	184	1.237	103
dc1 ★	**−13.2 / +7.4**	**20.6**	**3.14**	**65.7**	**191**	**1.184**	**103 / 45**
dc2	−13.2 / +14.1	27.3	4.14	66.1	193	1.179	103 / 265
dc3	−13.2 / +19.0	32.2	4.90	65.8	191	1.178	103 / 703
swing	±3.5	±7.1	1.06	±66.3	0–195	1.231	AC 120 Hz

Each option is held to the same surface field, so each reaches about 65 kV/cm at the null. More multiplier stages widen the gap at that field (2.0 → 4.9 mm) and the radius within 1 % (3.1 → 7.3 mm); they do not raise the field.
No geometry beats a uniform gap: in the vacuum each component of E is harmonic, so the field at the null cannot exceed the field on the electrodes.
Rated higher (polished, conditioned, small area: the electrodes here are 13–32 mm across), they would carry more; the field at the null scales with the rated surface field, the pressure with its square.
z start: the pump's gain at start-up with 100 pF storage (the bare pump 1.310).
Ripple at the 10 GΩ placeholder leakage. ★ the default: one CW stage, Co / Dc / Dp / Cs.

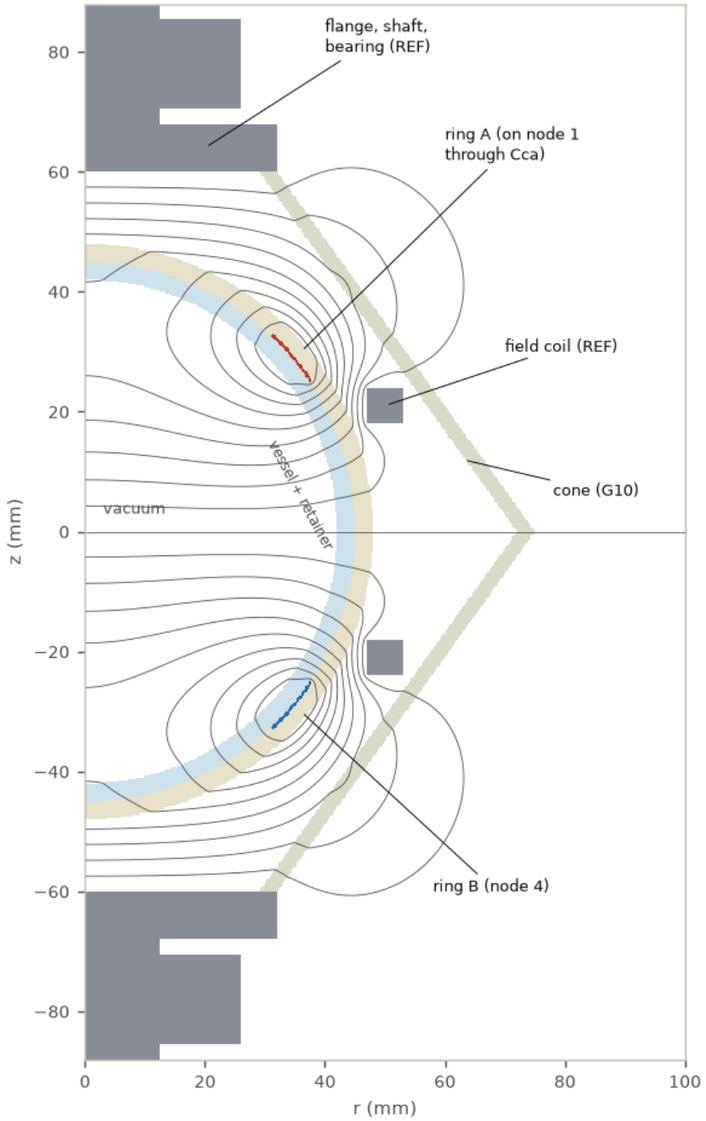
Rings on the placeholder vessel, AC-coupled

the swinging-rings study through 1 nF: field, strays, gain

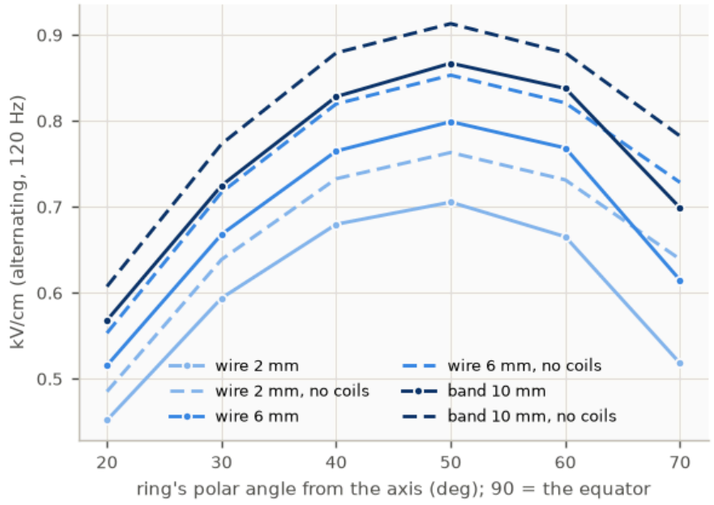
Two electrode rings on the core's vessel: the field at the centre, the strays they put on the pump, and the pump's gain

Stack of record (3 mm, 6 + 6, 6 mm gaps), each ring coupled through 1 nF to node 1 / 4; the hub's dimensions are the tube's placeholders in the designer's layer order (sim/core_rings.py).

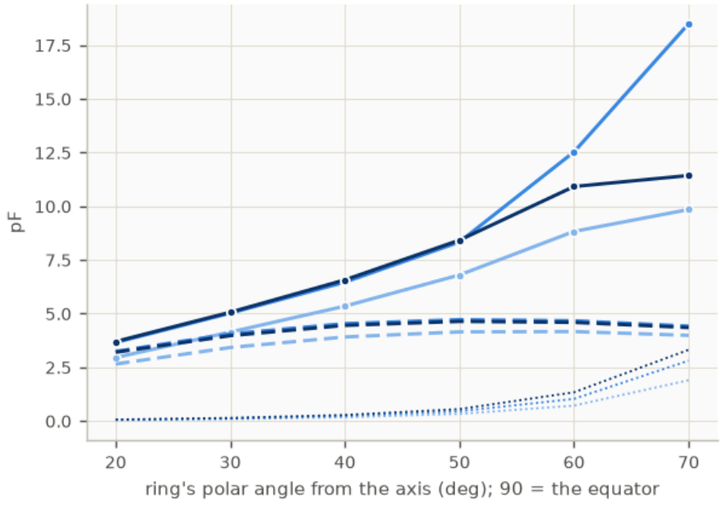
(a) the hub's half-section [RH placeholders]: a 10 mm band at 50°, equipotentials of A − B every 0.1



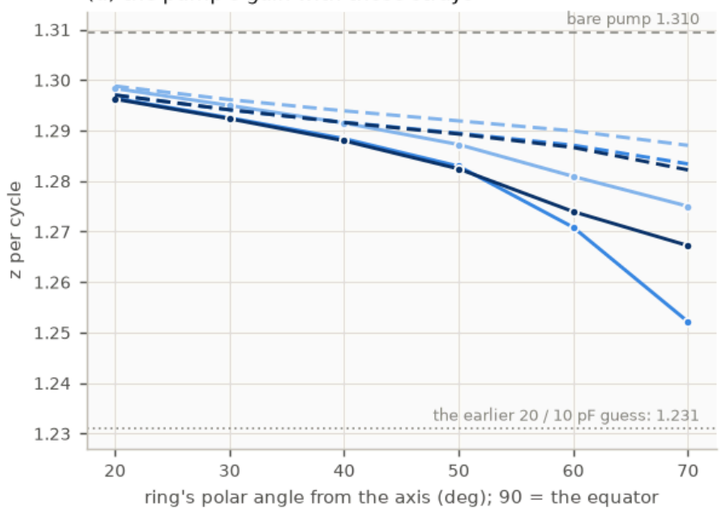
(b) the field at the centre, peak, at the pump's swing



(c) each ring's stray to the shaft side; dotted: ring to ring



(d) the pump's gain with those strays



(e) the pump's stray budget: z for any rings

to the shaft side ↓ / ring to ring →

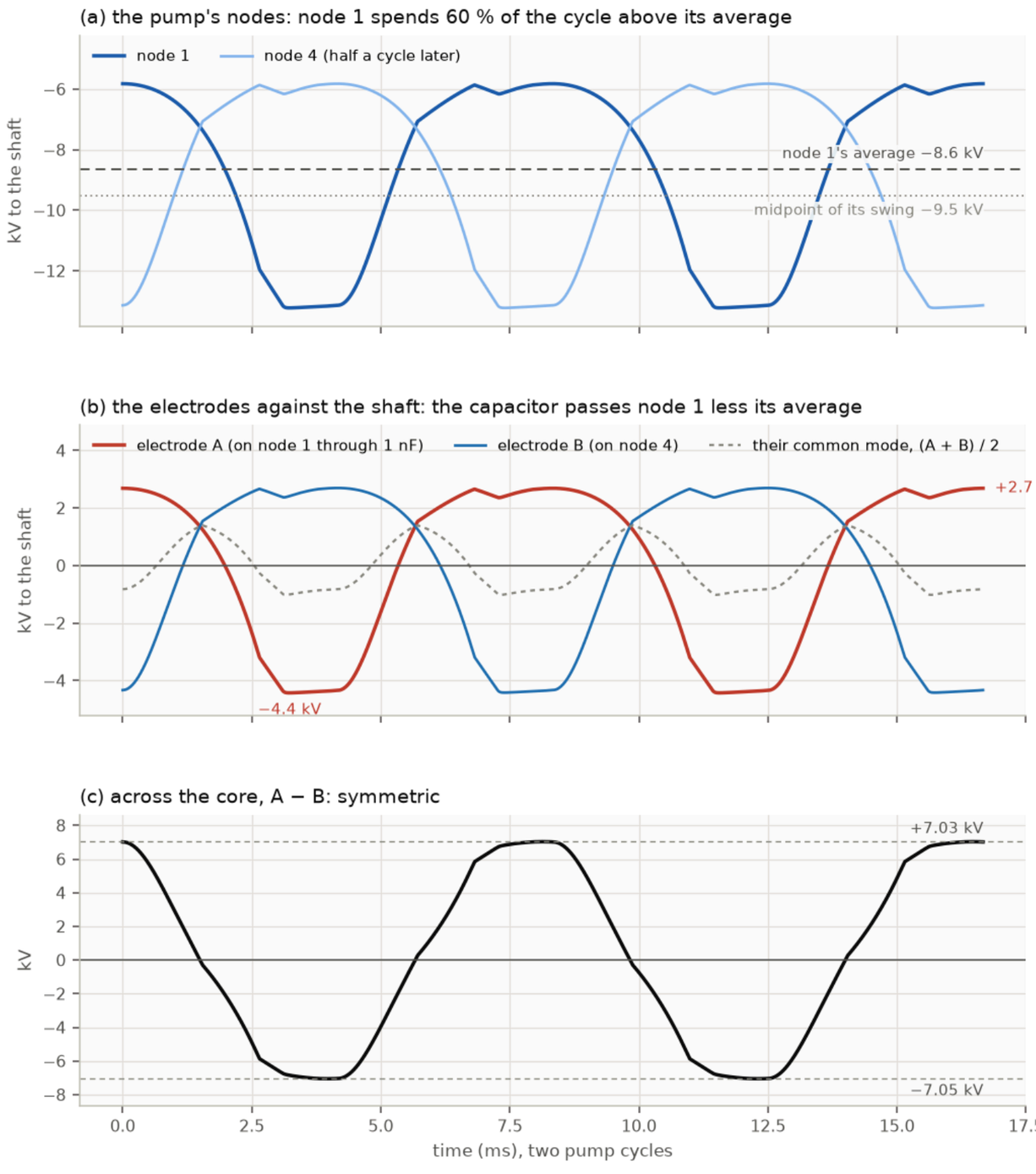
	0 pF	2 pF	5 pF	10 pF
0 pF	**1.307**	**1.301**	1.292	1.277
2 pF	**1.301**	1.295	1.286	1.272
5 pF	1.293	1.287	1.278	1.264
10 pF	1.280	1.274	1.266	1.253
20 pF	1.256	1.251	1.243	1.231

Bold: z ≥ 1.30. Each pF to the shaft side costs about 0.003 of z, each pF between the rings about 0.003 too.

cone electrodes on coupling capacitors swinging −4.4 / +2.7 kV at 120 Hz

The floating core electrodes: why each swings −4.4 / +2.7 kV against the shaft, while the field across the core is symmetric

Stack of record (3 mm, 6 + 6, 6 mm gaps, 13.1 kV clamp), 1200 rpm relative (120 Hz); each electrode coupled through 1 nF to node 1 / 4, its DC set by its leakage (sim/core_field.py).



Harmonic amplitudes, kV

× 120 Hz	1	2	3	4
node 1	3.84	1.19	0.10	0.22
electrode A	3.69	1.17	0.09	0.22
across the core	7.39	0.00	0.19	0.00
common mode	0.00	1.17	0.00	0.22

Node 1 is not a sine: it rests near its top for most of the cycle and dips briefly to the clamp at minimum C. Its average therefore sits nearer the top than the midpoint.

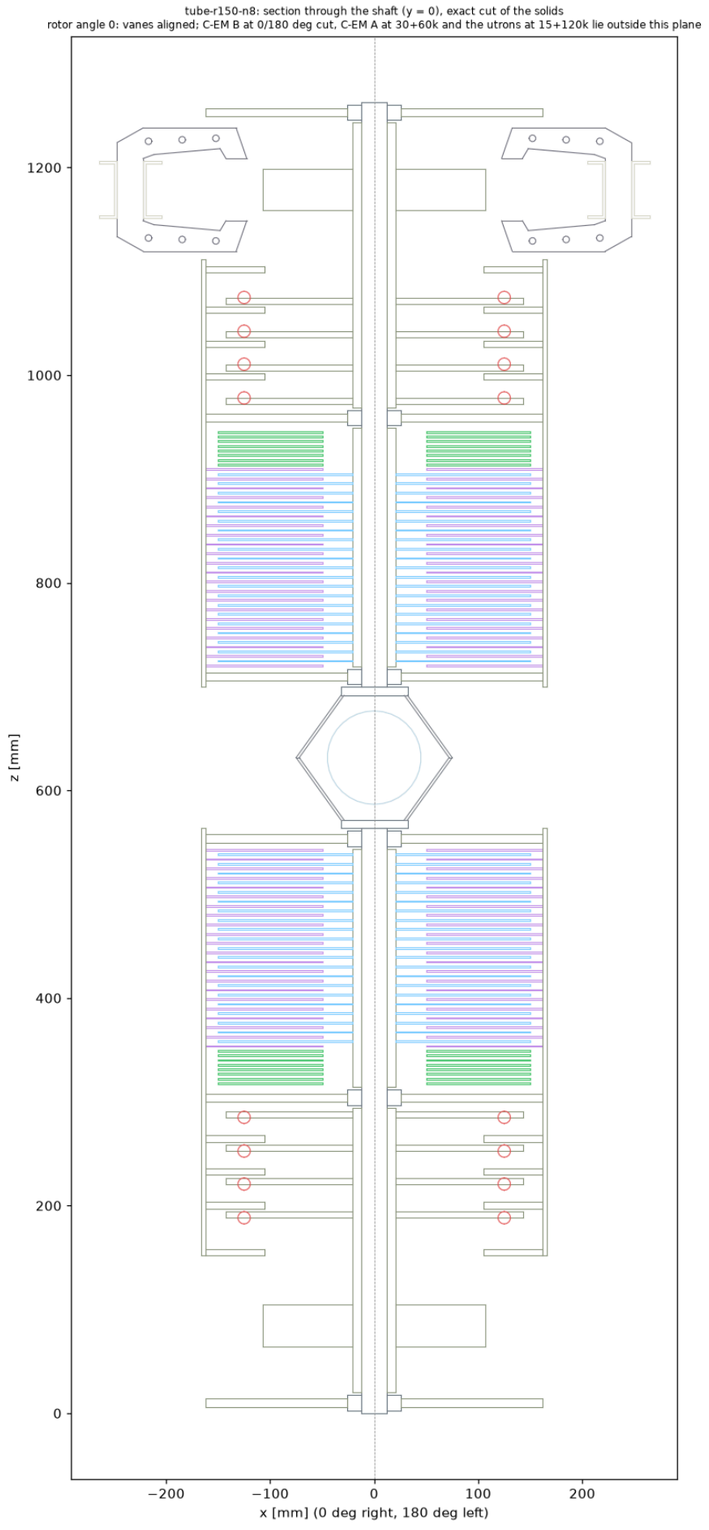
The coupling capacitor removes only that average, so each electrode swings more to the negative side than to the positive side: the 2nd harmonic.

Node 4 is node 1 half a cycle later. Half a cycle reverses the odd harmonics but leaves the even ones, so in A − B the 2nd harmonic cancels and the field is symmetric.

In (A + B) / 2 the fundamental cancels and the 2nd harmonic stays: both electrodes move together at 240 Hz against the shaft. That moves the core's potential, not its field.

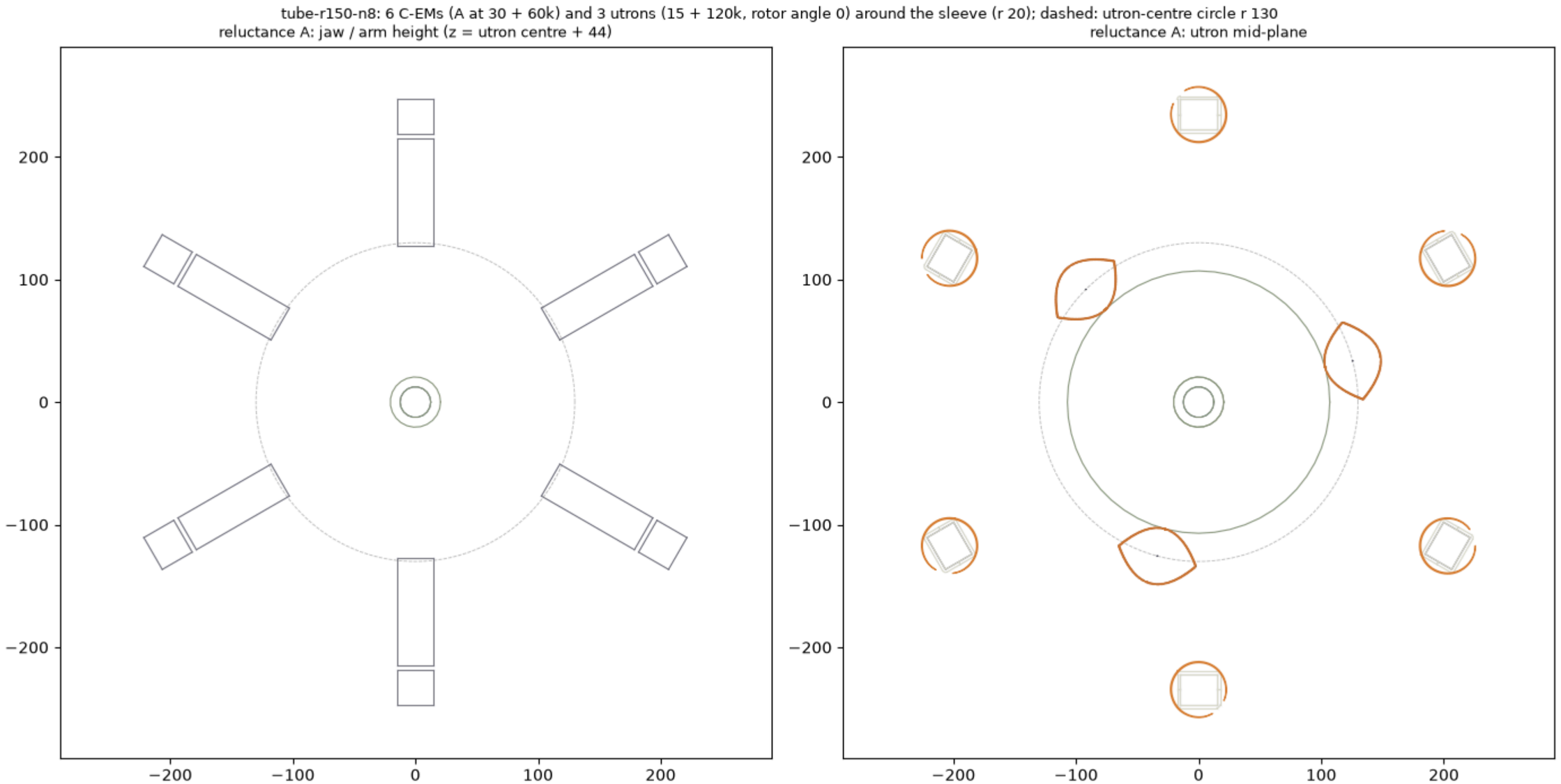
Earlier tube build: section

the spark-gap tube with clocking decks, C-EMs and the bicone, 1263 mm



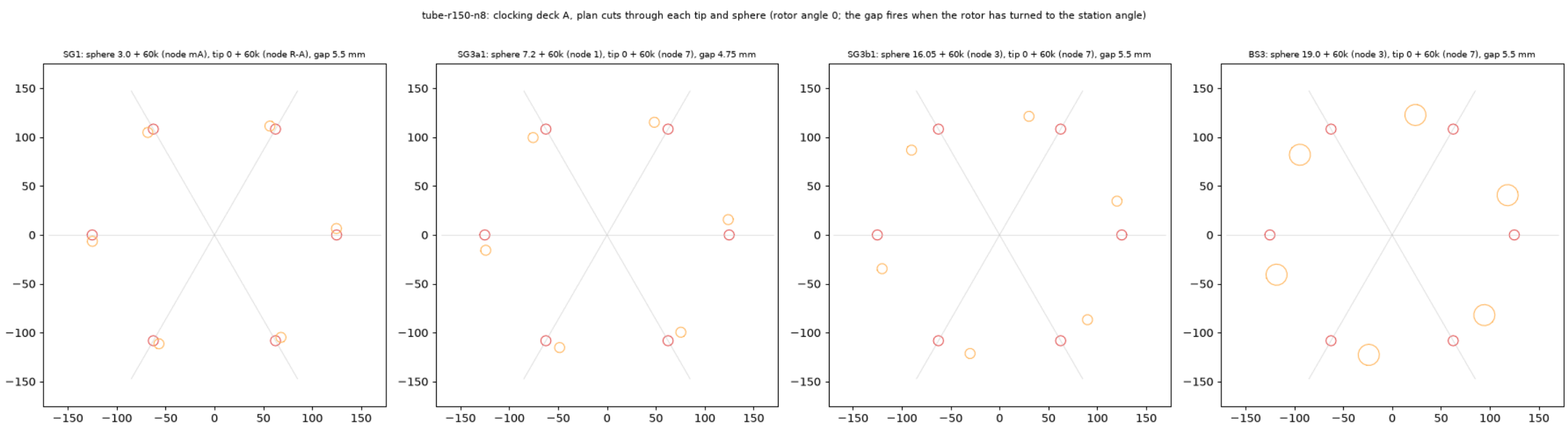
Earlier tube build: C-EM and utron plan

the designer's C-EM / utron sections



Earlier tube build: clocking plan

the spark-gap clocking deck

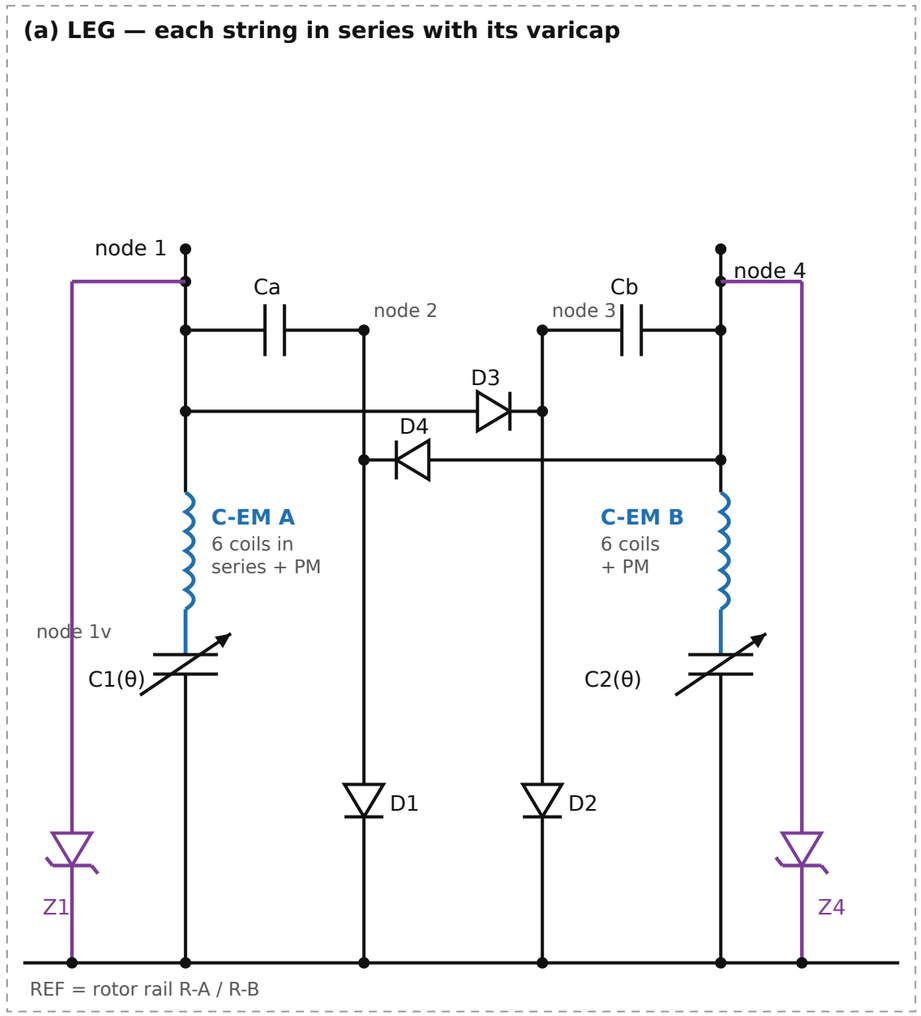


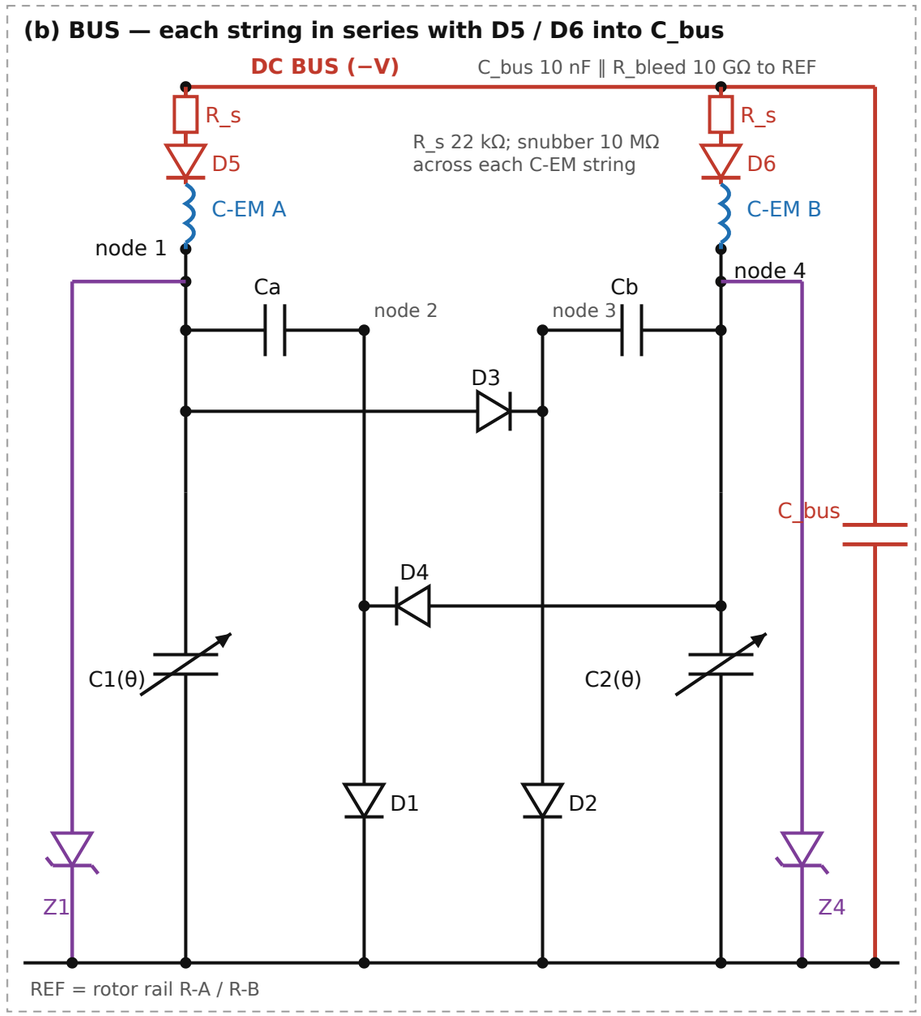
Switchless C-EMs on the diode core

leg against bus placement of the C-EMs

Diode machine (de Queiroz core, no flying bucket): C-EMs connected WITHOUT switches — two placements

black = core (solveDoubler4 netlist: D1 2→ref, D2 3→ref, D3 1→3, D4 4→2, anode→cathode) · blue = C-EM strings (PM utrons) · purple = avalanche clamp · red = bus parts
Negative polarity (all nodes ≤ 0). No transistor, thyristor, spark gap, commutator or controller anywhere: every one-way element is a diode. Wires that cross without a dot are not connected.





(a) design point k 20 (≈ 36920 turns per coil, string EMF 5.6 kV pk), flux extreme at plate alignment: 4.31 W to the shaft, 137 mN·m; belt 11.4 W, clamp 6.8 W, copper 0.29 W.
Stator: pump reaction 363 mN·m + drag 14.7 mN·m → net -241 mN·m (dragged along, as for every pump-fed motor). k 10: 2.1 W; k 40: copper and clamp take over.
(b) bus, k 1: -0.24 mN·m (once C_bus is charged only the 2 µA bleed flows through D5 / D6); k 20: −99 mN·m, a brake (the PM EMF drives 3.1 W into the snubbers); at two phases the pump collapses.
Without switches a DC bus cannot turn the coils (constant current, periodic flux → zero average torque) [OC].
Z1 / Z4: avalanche-diode strings (BV 20 kV, soft knee, ~100 kΩ): they set the operating voltage and take the surplus the coils don't.
Without them the pump runs away at every rewind (coil power ∝ V, pump surplus ∝ V²).
Coil timing comes from the plates alone: each coil must see 6 flux periods per rev (one per pump cycle), so 6 PM utrons per side.
The present 3 utrons per side give ≈ 0 average torque. Each string floats at node potential: coil insulation ≥ 20 kV.
ngspice, tube defaults at 300 rpm: near-ideal diodes, cosine varicaps, PM swing 0.27 mWb per coil [RH, gate G-SWING open].
Numbers: sim/switchless_cem_results.json · findings: sim/switchless-cem-findings.md · generator: docs/make_schematic_switchless.py.

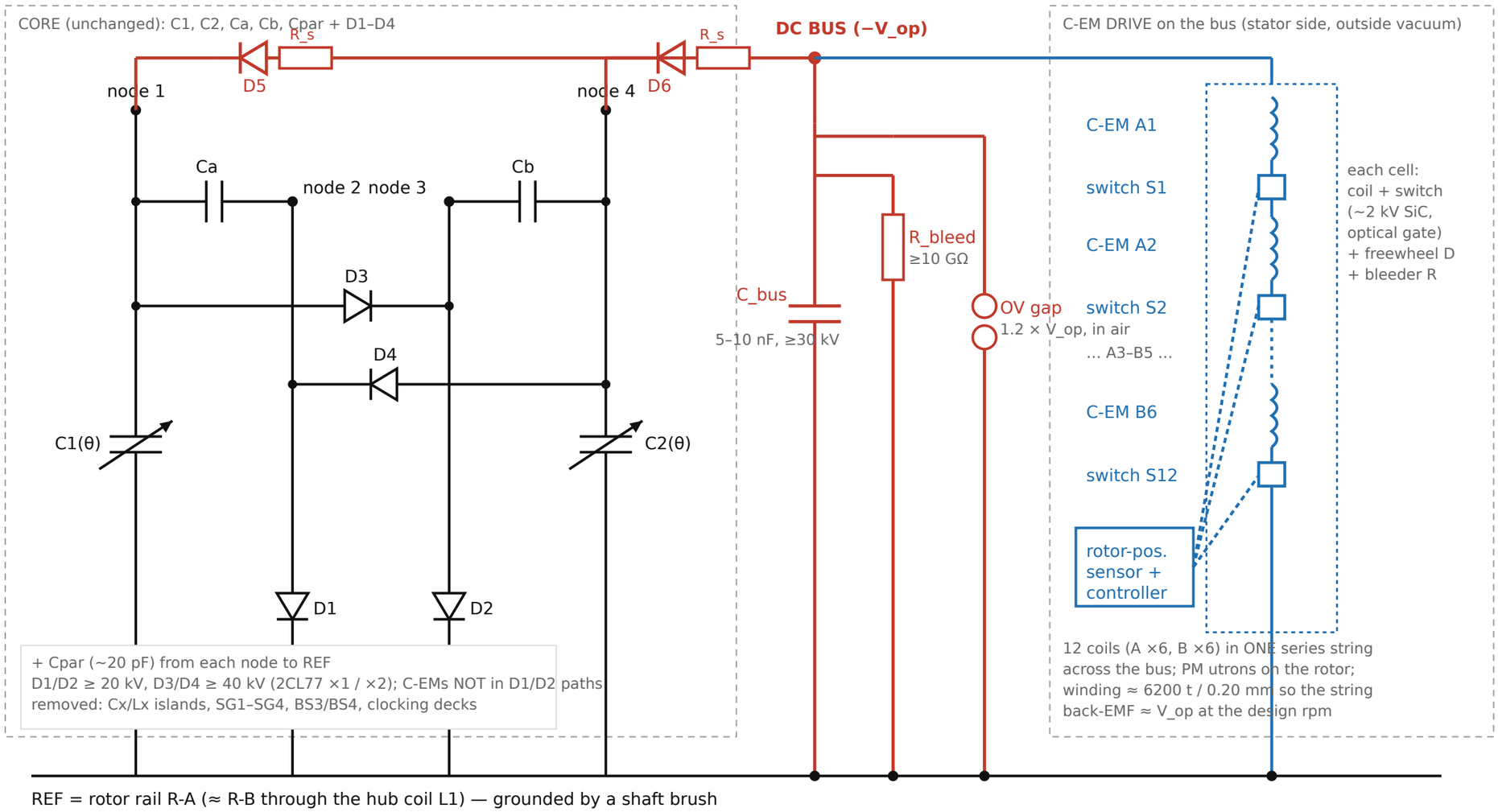
Diode core with a DC bus

marked SUPERSEDED in the file: a DC bus with a switched C-EM drive

Diode machine (de Queiroz core, no flying bucket) — DC bus and C-E

SUPERSEDED: this drive needs switches. See schematic-diode-core-switchl

black = existing core (solveDoubler4 topology) · red = NEW DC bus parts · blue = NEW C-EM drive (DC-tap ruling) · polarity as the engine's negative mode (all nodes ≤ 0)



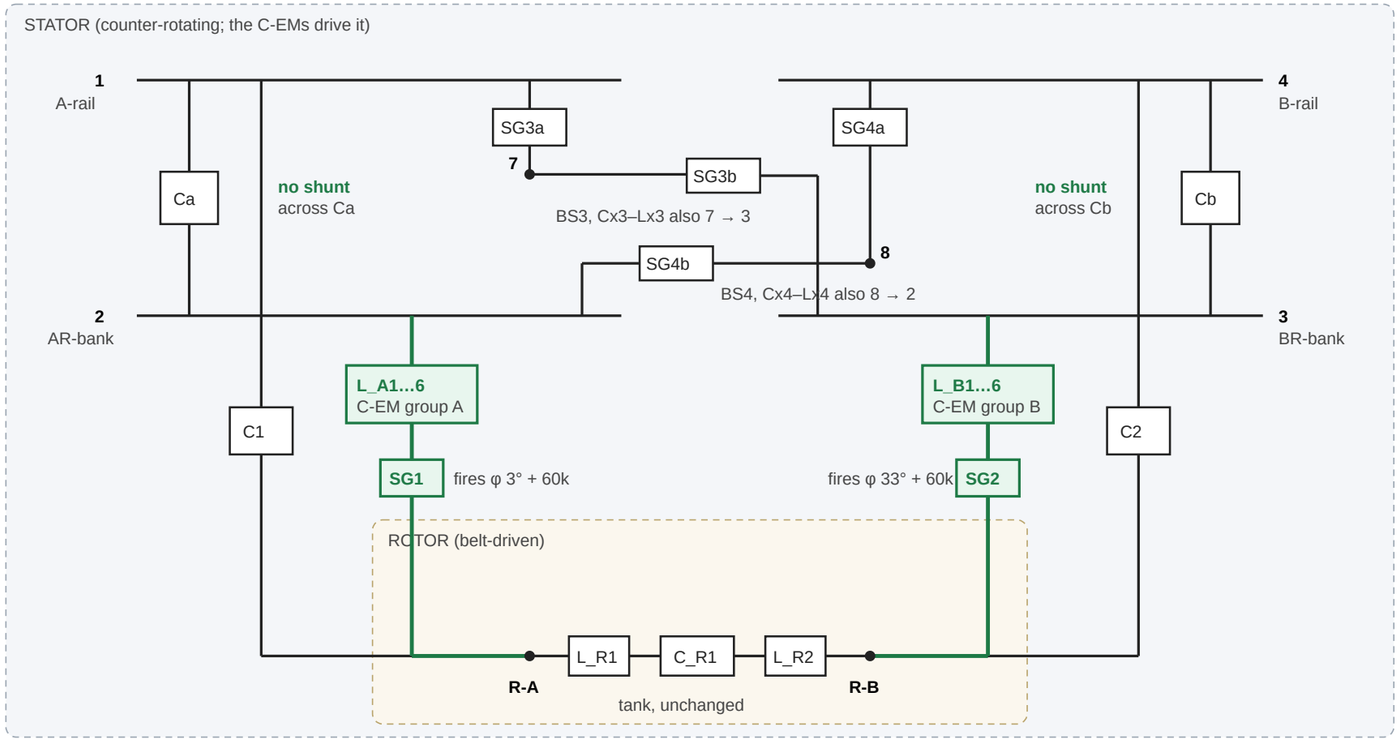
Bus voltage is set by the load: the drive draws only the surplus (voltage-priority control); without it the bus floats up to the OV gap. R_s 10–47 kΩ limits surge into D5/D6. D5/D6 reverse rating ≥ 2 × V_op (node swing + bus). Ripple at 30 cycles/s, 150 µA, 10 nF ≈ 500 V. Not yet engine-verified with the bus attached (gate G-LOAD).

C-EMs in the discharge path

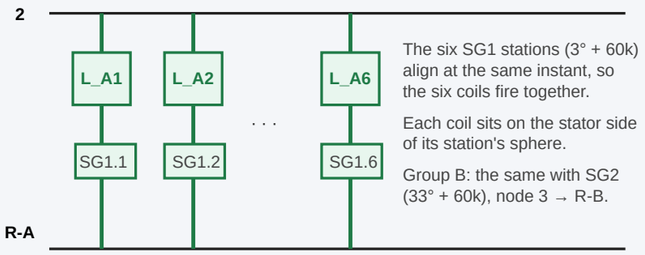
the C-EMs in series with the rail gaps SG1 / SG2

C-EMs in the discharge path (rail gaps SG1 / SG2) instead of across Ca / Cb

Edit of the netlist of record. Removed: the 12 shunt branches L_Ak + C_ARk (1 → 2) and C_BRk + L_Bk (3 → 4), and with them all 12 block caps (440 nF).
Added: nothing new. The same 12 coils, each in series with its own spark-gap station. No extra gaps, no buffer.



Per station: one coil in series with its own SG1 gap



The six SG1 stations (3° + 60k) align at the same instant, so the six coils fire together.

Each coil sits on the stator side of its station's sphere.

Group B: the same with SG2 (33° + 60k), node 3 → R-B.

Electrically: the gap in series with (six 0.64 H coils in parallel) = 0.107 H, 6.7 Ω.

Other series positions (not drawn)

- Load gaps SG3a (1 → 7) / SG4a (4 → 8), φ 7.2° / 37.2°
- Fire gaps SG3b (7 → 3) / SG4b (8 → 2), φ 16.05° / 46.05°
- In place of the island inductors Lx3 / Lx4

Why SG1 / SG2 first: they carry the rail transfers into the rotor halves, they are 30° apart like groups A and B, and they leave the islands' tuned Cx–Lx sink untouched. The other positions are worth one engine run each.

The fire gaps land just after an alignment (falling L → braking) unless the C-EMs are re-clocked.

What changes, in classic terms

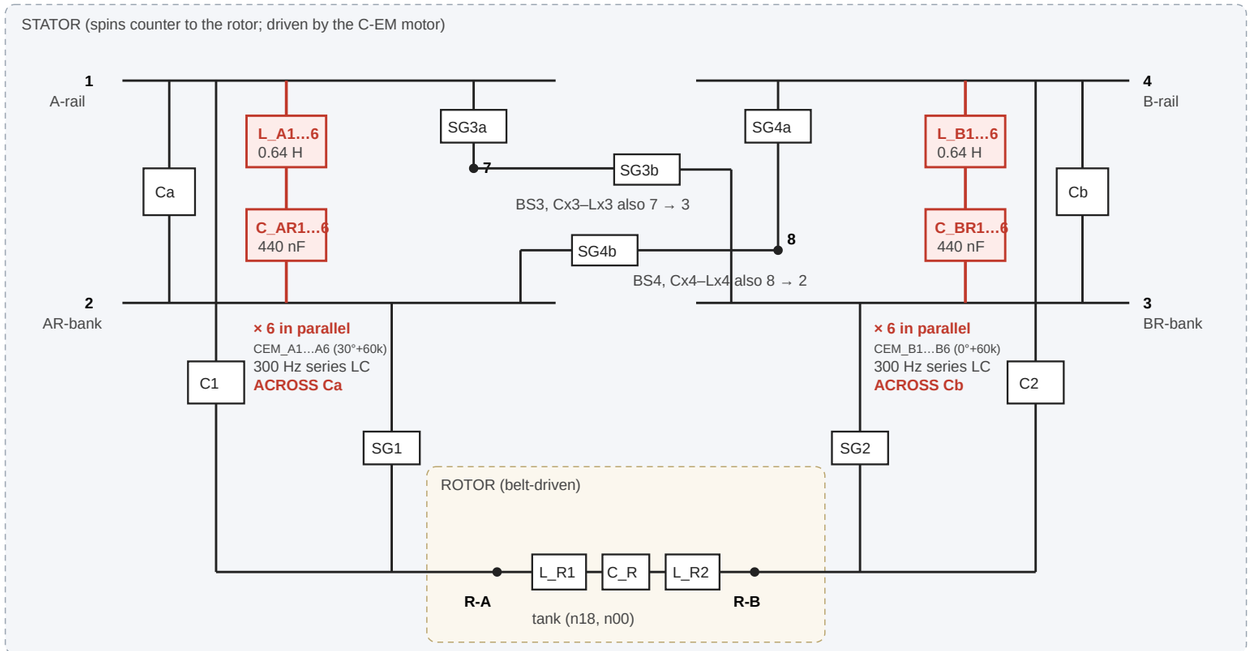
1. Ca and Cb carry no motor branch any more, so the Bennet ratio is not diluted. The 440 nF block caps disappear: the gap itself blocks DC.
2. Each rail transfer becomes a resonant half-cycle through the coils (inductive charging) instead of an abrupt spark equalisation. $\pi \cdot \sqrt{(0.107\ \text{H} \times \sim 200\ \text{pF})} \approx 15\ \mu\text{s}$, far inside the 233 µs between events. The gap opens at current zero, so the transfer is less lossy; it also changes how much charge each stroke moves, which is what the engine must check (pump z).
3. Torque = ½ i^2 dL/dθ on each pulse, so the pulse must land on rising L. As drawn, A aligns at φ 15° and SG1 fires at 3°, 12° early. Re-clock the C-EMs, per your offset ruling, so each group aligns 2–4° after its gap fires: both groups turn by about −9°, A to ≈ 21° + 60k and B to ≈ 51° + 60k, still 30° apart. The exact offset comes from the torque model.
4. The torque acts between the stator C-EMs and the rotor poles; the belt supplies the reaction. Each pulse scales with the transferred charge, so torque ∝ V^2.

C-EM motor placement

the C-EMs in the netlist of record; a buffer and angle-gated fire proposal

1. Where the C-EMs sit today (netlist of record, topology_edge_list.csv)

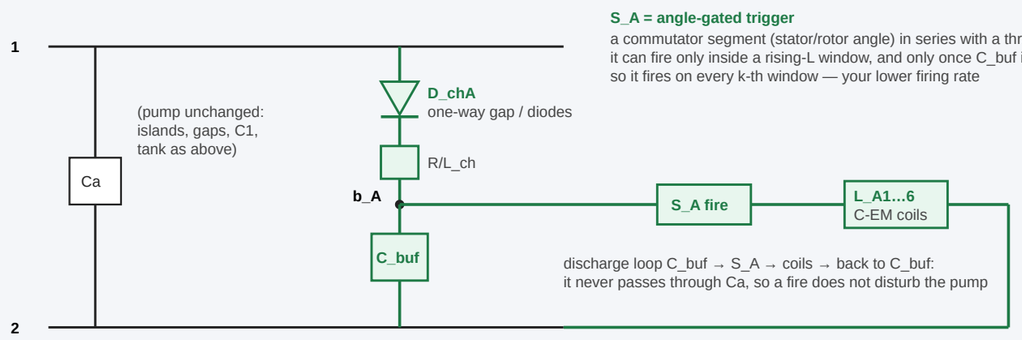
Stator body: nodes 1–4, Ca, Cb, islands 7/8, the 12 C-EM branches. Rotor body: R-A, R-B, the tank. Varicaps C1, C2 couple the two bodies.



Why it shorts the pump:
each branch puts 440 nF (≈ 1400 × Ca) behind a coil straight across Ca / Cb. Over a stroke that is extra transfer capacitance, so the Bennet ratio collapses (z → 1.000 in every detuning tried). The torque path and the pump's storage element are the same two nodes.

2. Your idea as a schematic edit: buffer cap, charged by the pump, fired into the C-EMs at a lower rate

Group A shown; group B mirrors it on nodes 4 / 3. The C-EM branches leave Ca. Only a one-way charging element touches the pump.



S_A = angle-gated trigger
a commutator segment (stator/rotor angle) in series with a threshold gap: it can fire only inside a rising-L window, and only once C_buf is at V_fire, so it fires on every k-th window — your lower firing rate

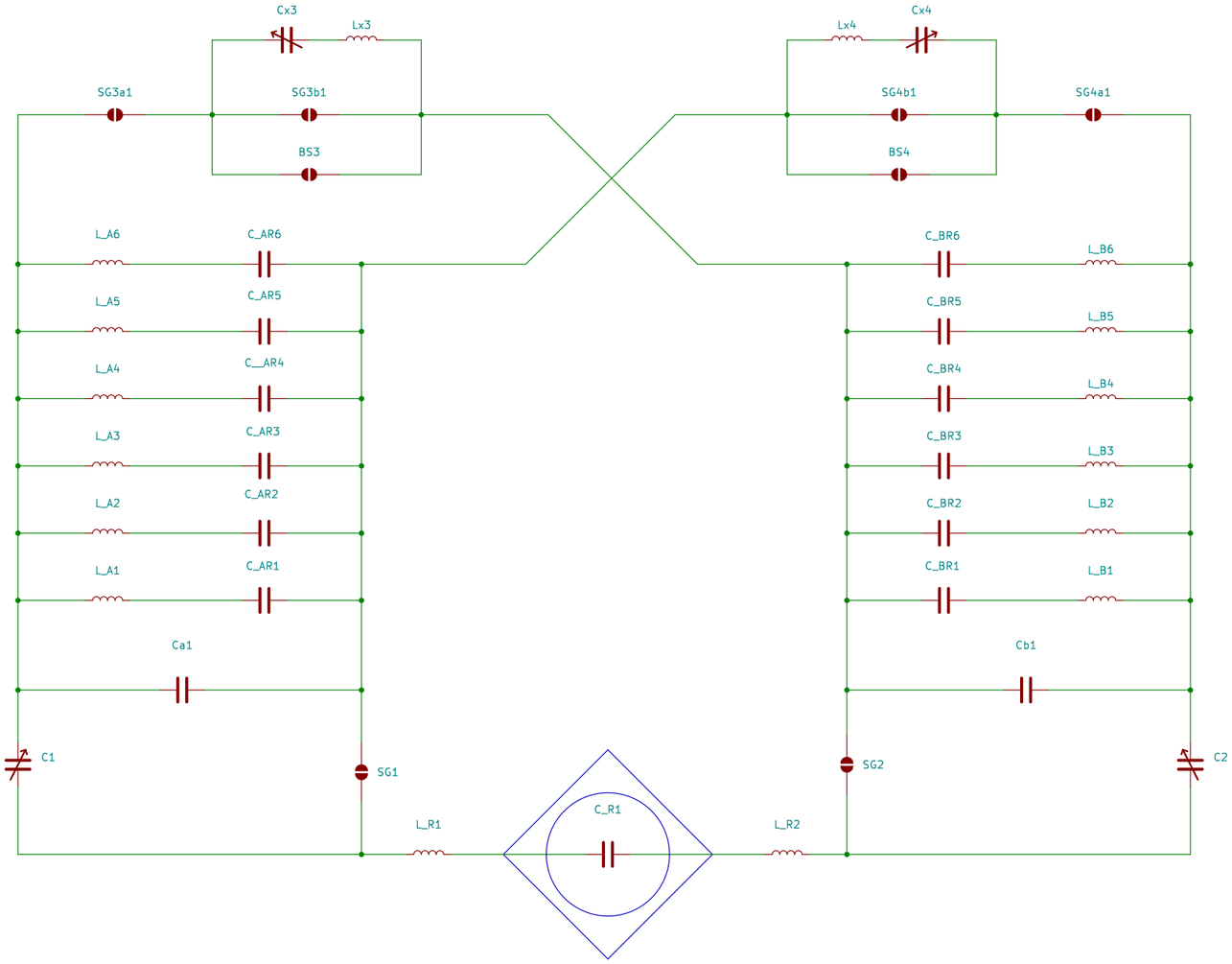
Charging: D_chA conducts only when V(1−2) exceeds V(b_A), so C_buf takes charge at the top of each stroke and fills over many strokes (R/L_ch limits each bite).
Fire: one half-cycle of C_buf with the coils, timed into the rising-L window, gives torque ½ i² dL/dθ. Group B: D_chB from node 4, C_buf between b_B and node 3.
Counter-rotation: torque acts between the stator C-EMs and the rotor poles; the belt supplies the reaction on the rotor.

Design relations to check (classic RLC + reluctance motor):
- pulse fits the window: π·√(L·C_buf) ≤ θ_window / ω_rel. 0.64 H with 10 nF gives 0.25 ms; a 5° window at 3000 rpm relative lasts 0.28 ms.

A slower stator lengthens the window, which allows a bigger C_buf, so more energy per fire.
- firing rate: f_fire ≈ I_charge / (C_buf·V_fire). Average motor power ≈ ½·C_buf·V_fire²·f_fire must stay below the pump's surplus.
- to check: D_ch clamps V(1−2) at V(b_A) while it conducts. This must hold the pump's growth (z) — the one thing the engine has to verify first.

Disc machine: the KiCad schematic of record

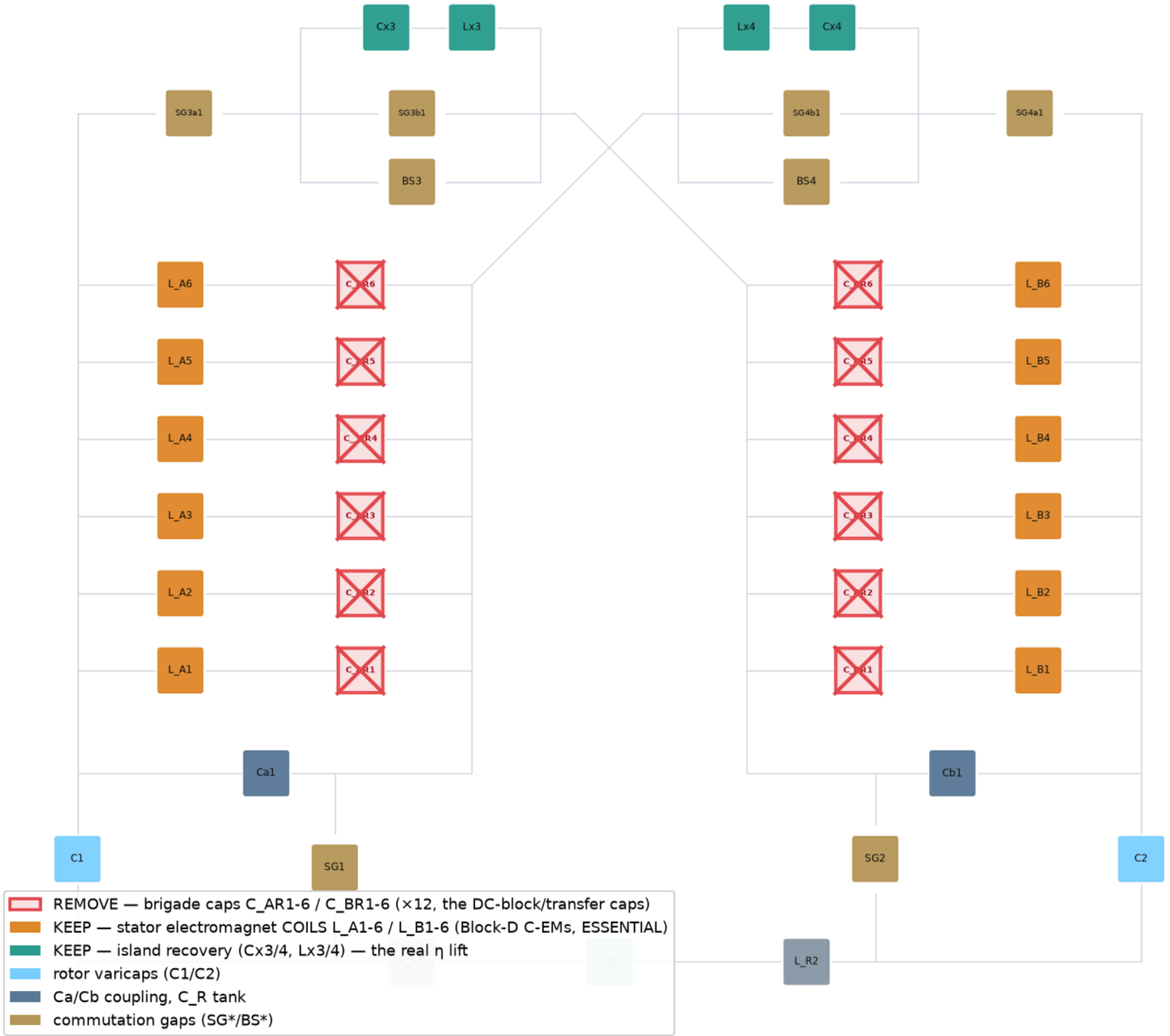
the designer's disc machine: 43 parts, 8 spark gaps (the source is the .kicad_sch beside it)



Disc machine: Ca / Cb simplification proposal

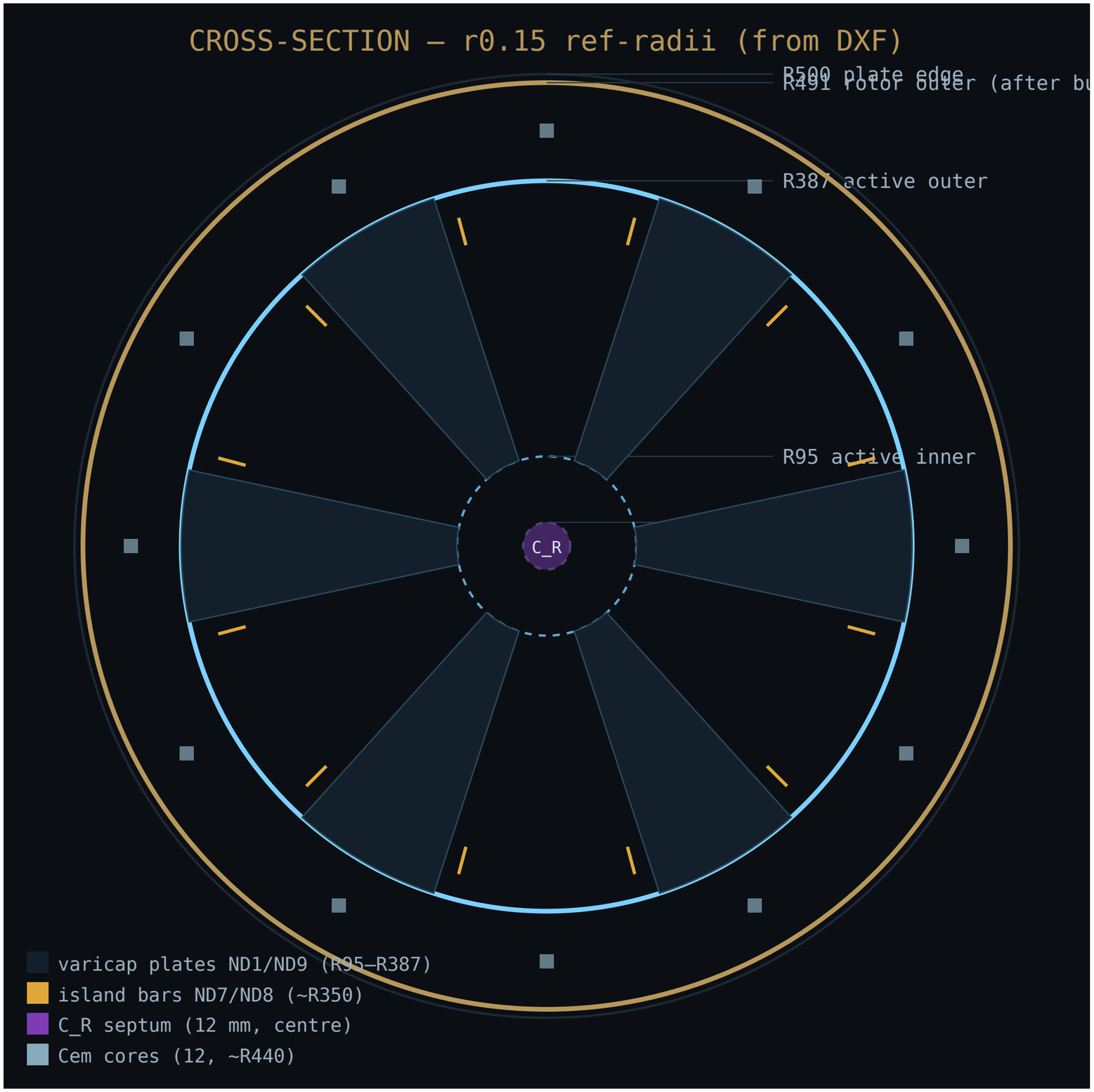
43 → 31 parts

DCCREG schematic — brigade simplification (corrected): remove the brigade CAPS C_AR/C_BR (red ×);
KEEP the stator electromagnet COILS L_A/L_B (orange, Block-D C-EMs) and the island Cx/Lx (teal). 43 → 31 components.



Disc machine: reference cross-section

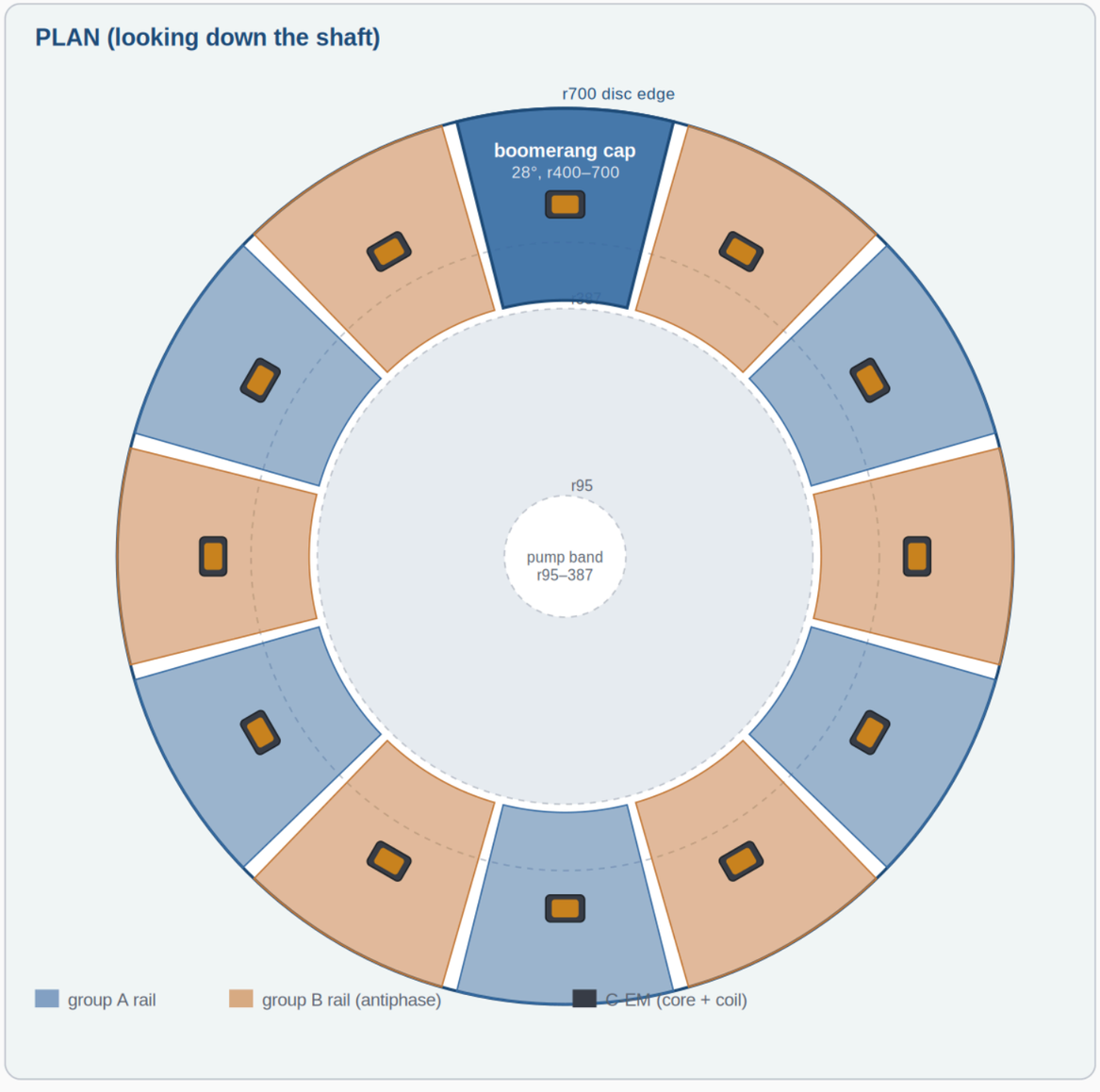
the disc's reference radii and named features from the r0.15 DXF

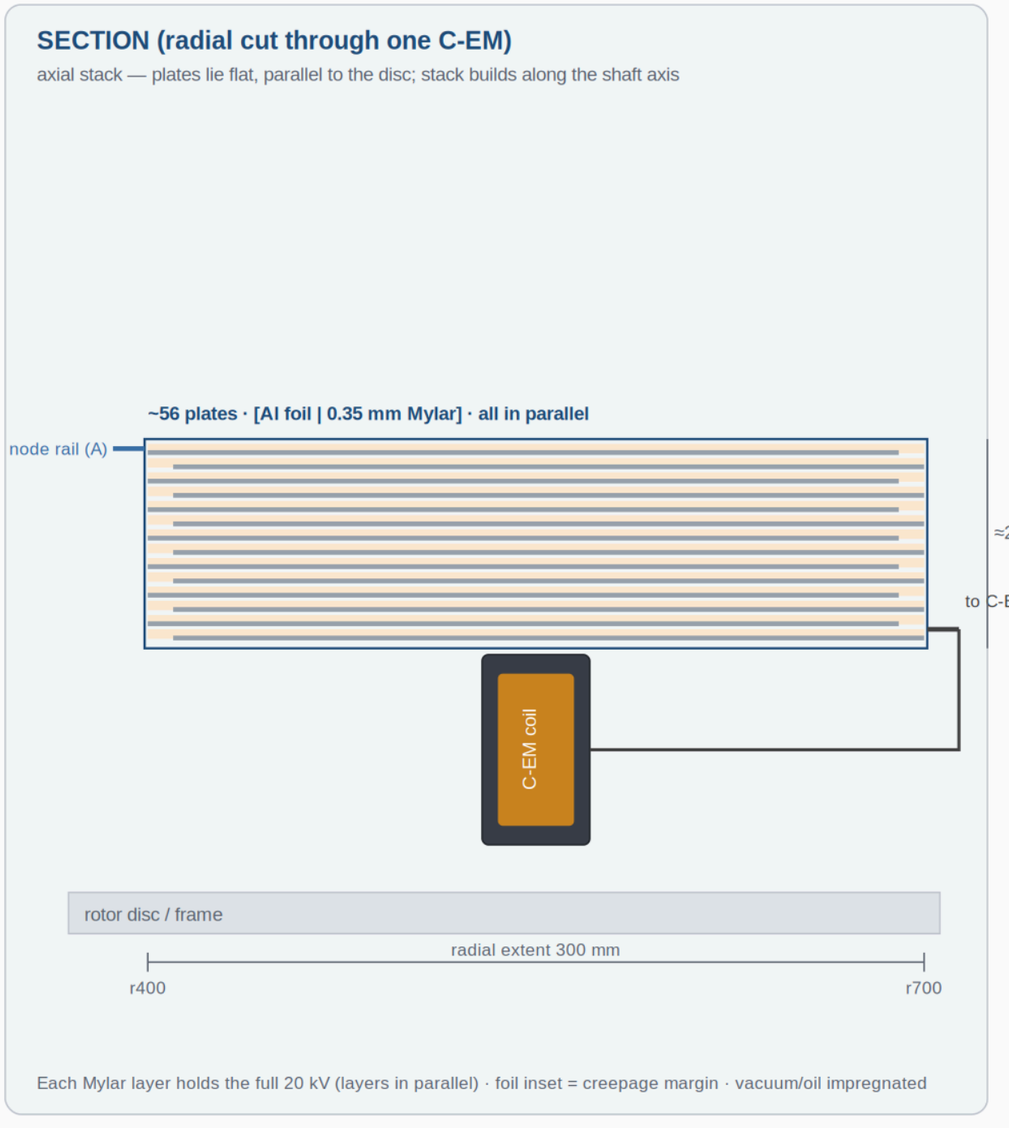


plan and section

Boomerang block-cap — one per C-EM, stacked axially

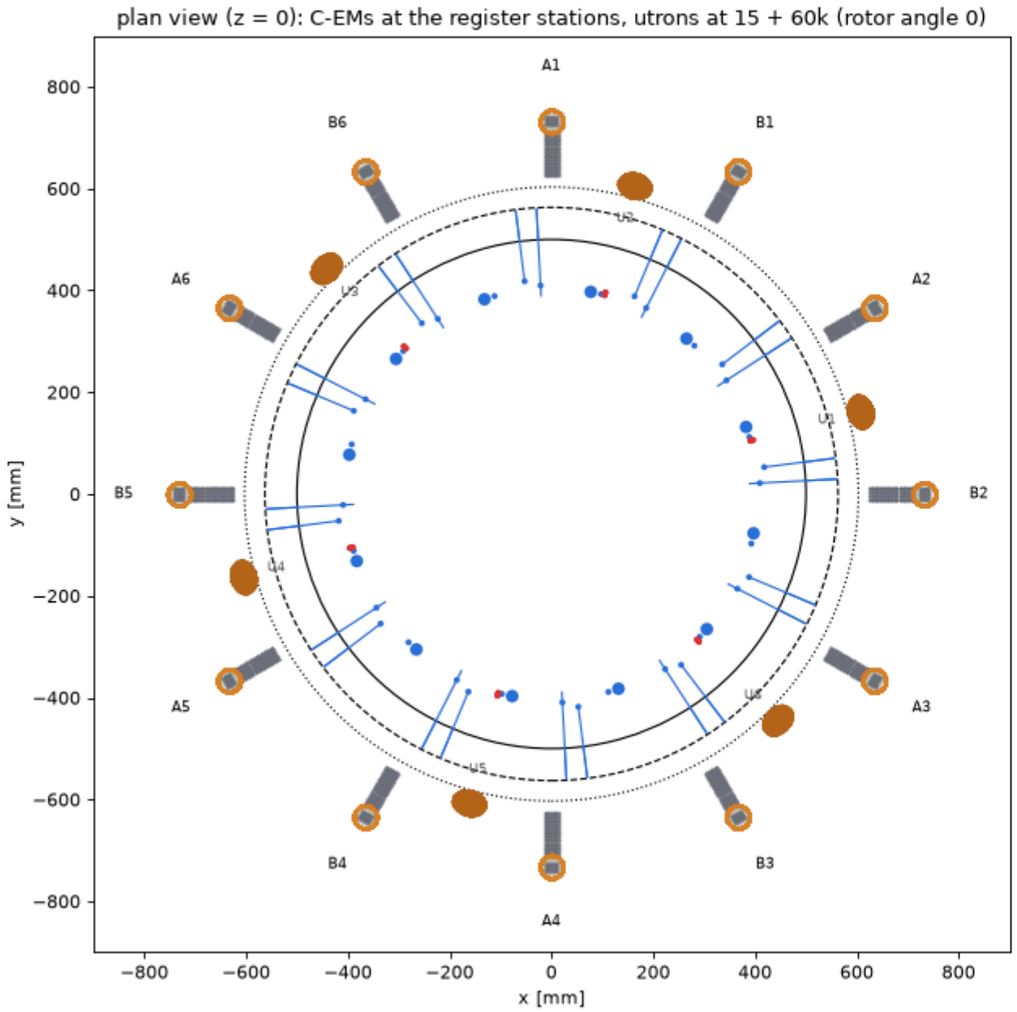
440 nF / 20 kV · annular-sector plates r400–700 mm · 28° arc · ~56 × (Al foil | 0.35 mm Mylar) · ~20 mm stack

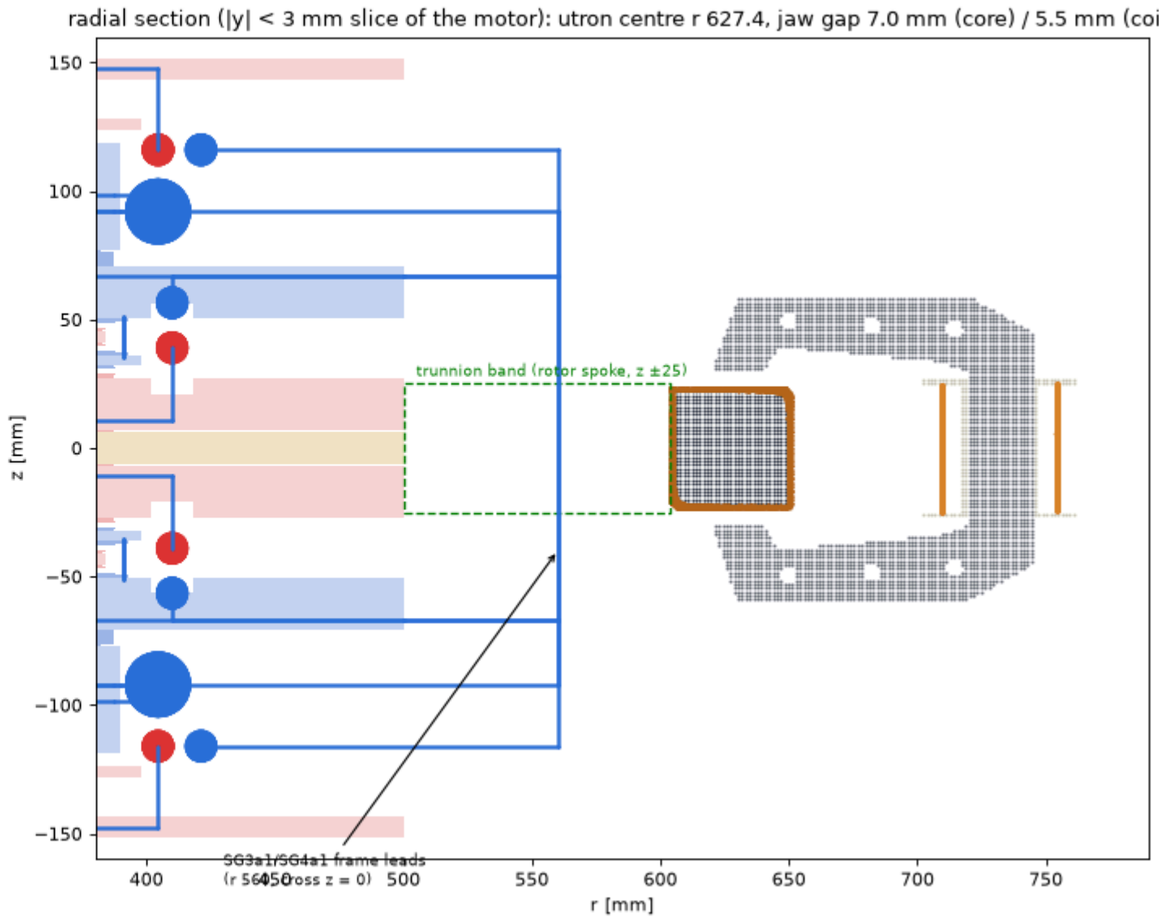




Disc machine: the placed C-EM motor

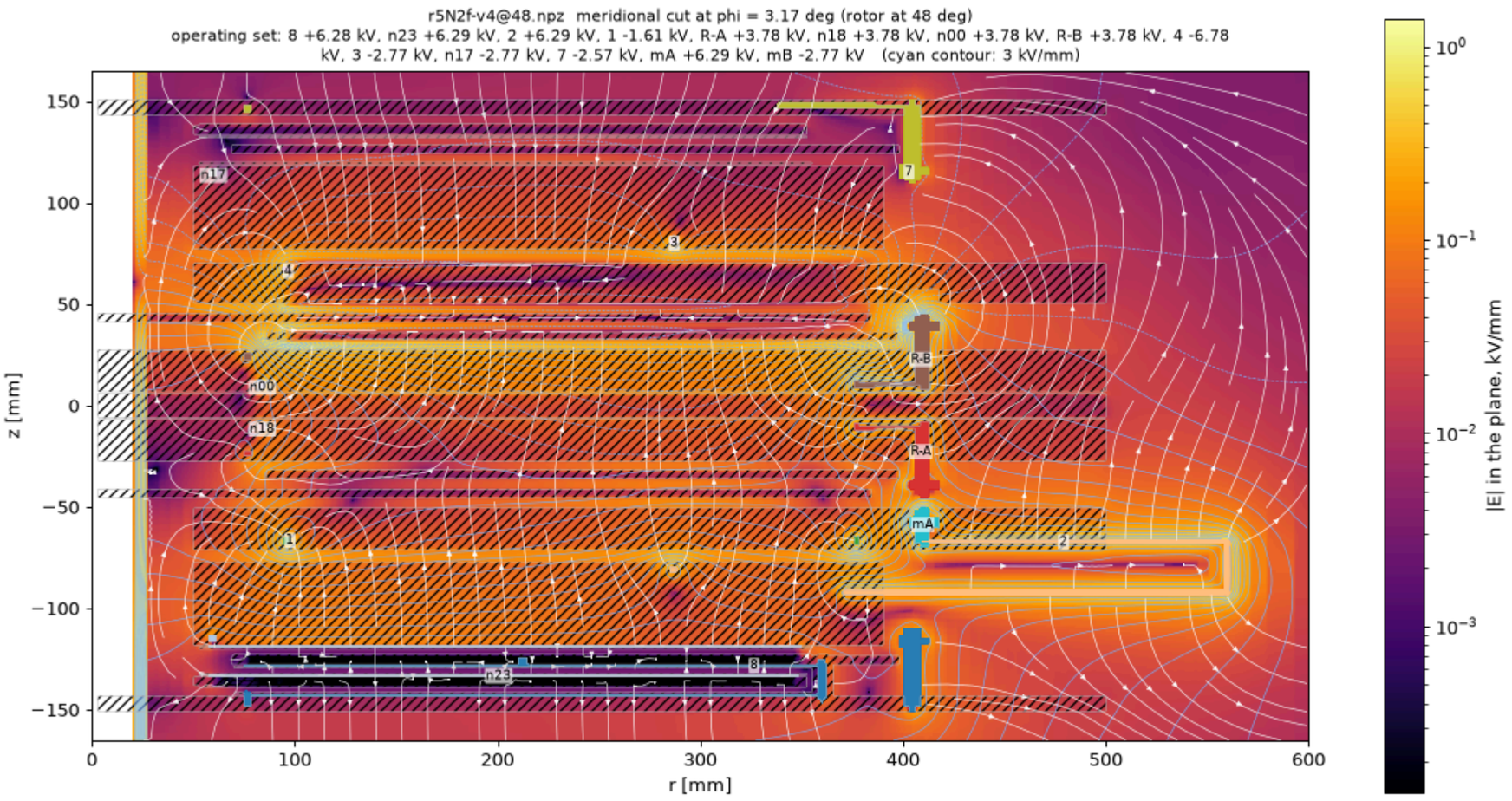
plan and radial section of the motor on the interleaved N = 2 disc





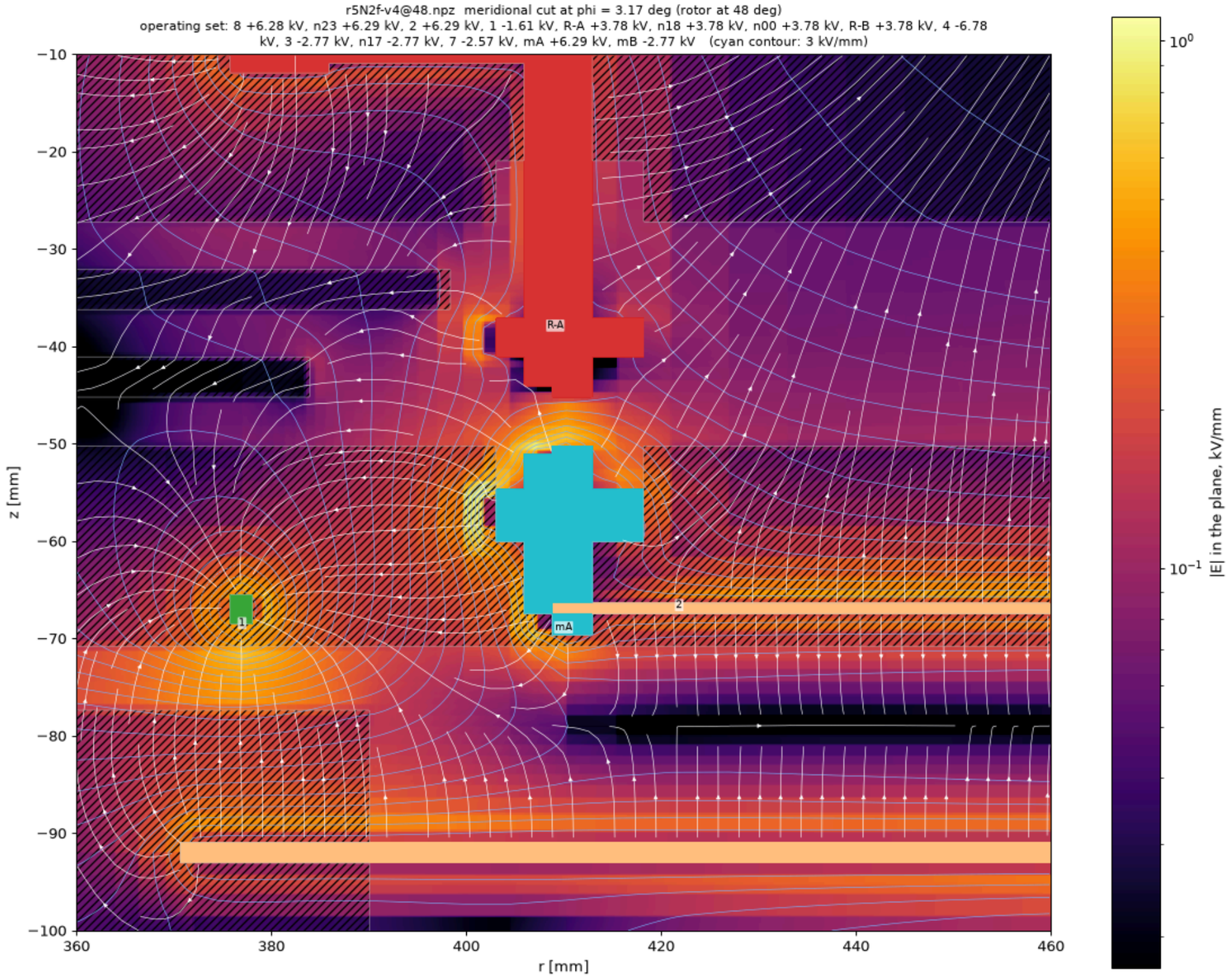
Disc machine: field cut at an SG1 station

meridional |E| before firing, v4 topology



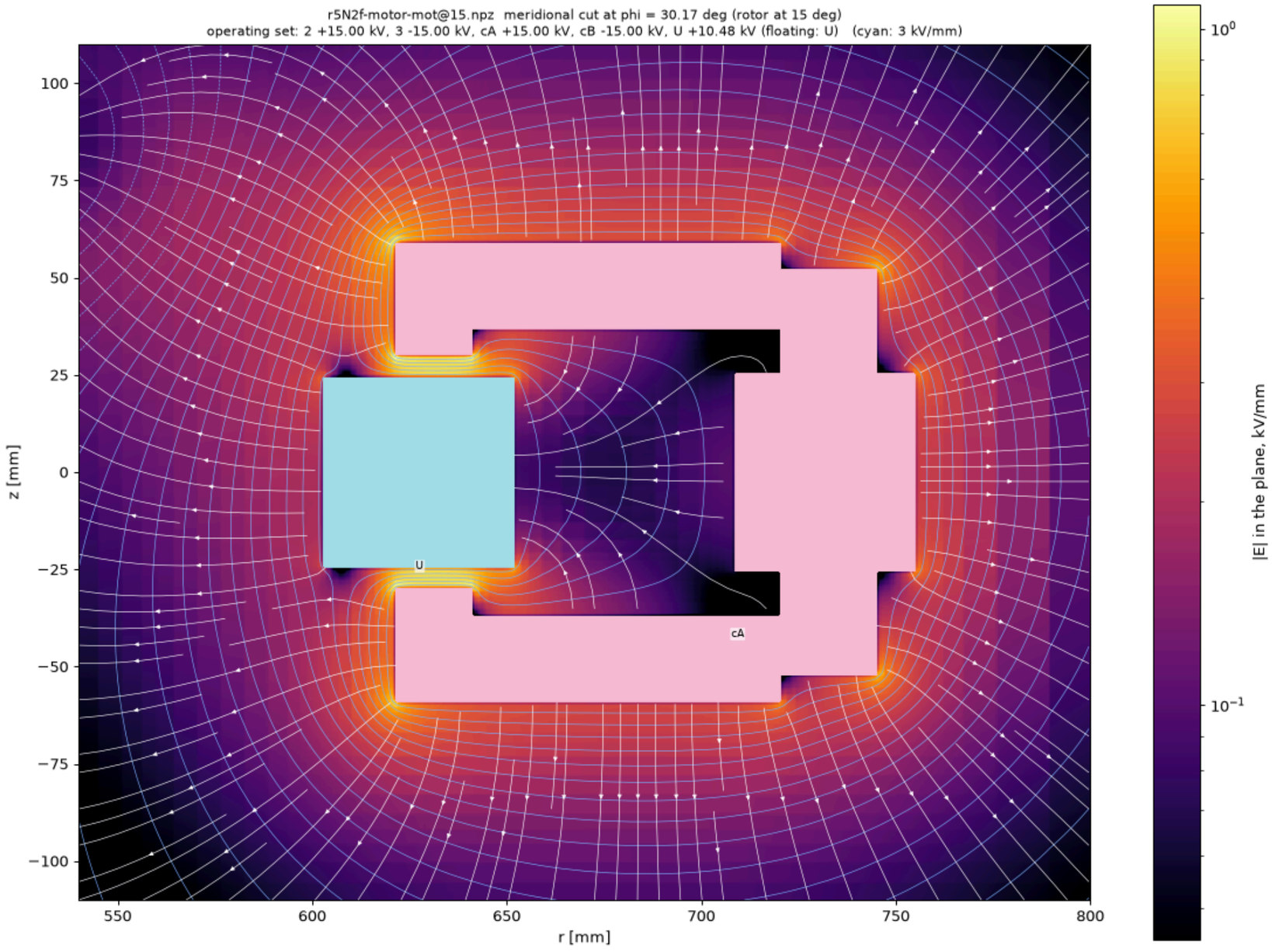
Disc machine: field cut at an SG1 station, zoom

as the station cut, magnified



Disc motor: field cut at ±15 kV

the motor model at rotor 15°



Disc motor: field cut, unit drive

the motor model at rotor 15°, unit drive on C_A

